Econometrics: Advanced Methods

Short lecture notes for Research Metrics 2 (MQE, LMU Munich), Winter 2026/27

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Preface

These notes accompany the course *Econometrics: Advanced Methods (Research Metrics 2)* in the Munich Graduate School of Economics MQE programme. They are deliberately short. Each chapter states the key definitions and results, works through one or two examples in detail, and ends with a computer tutorial in R. Proofs are sketched where the argument is instructive and otherwise referenced.

The notes complement the lecture slides; they do not replace the textbooks listed below.

What the course covers

The objective is to introduce fundamental estimation methods in econometrics. We spend a fair amount of time on the theoretical properties of the estimators and also discuss practical aspects of implementation. The course is not a course on causality or identification, and it is not an R course, although both appear throughout.

Chapter	Topic	Computer tutorial
1	Maximum likelihood	CT 1: logit by hand, PML vs. ML
2	Method of moments and GMM, including IV	CT 2: GMM estimation of an ATE
3	Simulation-based estimation	CT 3: Monte Carlo integration, Halton draws, quadrature
4	Nonparametric density and regression estimation	CT 4: kernel density, local linear regression
5	Introduction to statistical learning	CT 5: ridge, lasso, elastic net
6	Causal inference (optional reading, not exam-relevant)	
A	Large-sample asymptotics (assumed known; for reference)	
B	A short R primer	CT 0

Literature

- Hansen, B. E. (2022). *Econometrics*. Princeton University Press (B. E. Hansen 2022a); and (2022). *Probability and Statistics for Economists*. Princeton University Press (B. E. Hansen 2022b).
- Cameron, A. C. and Trivedi, P. K. (2005). *Microeconometrics: Methods and Applications*. Cambridge University Press (Cameron and Trivedi 2005).

- Wooldridge, J. M. (2010). *Econometric Analysis of Cross Section and Panel Data*, 2nd ed. MIT Press (Wooldridge 2010).
- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2023). *An Introduction to Statistical Learning*, 2nd ed. Springer (James et al. 2023).
- Casella, G. and Berger, R. L. (2002). *Statistical Inference*, 2nd ed. Duxbury (Casella and Berger 2002).

Chapter-specific readings are given at the start of each chapter.

Software

All tutorials use R. Install a recent version of R and an editor (RStudio or Positron), then install the packages used in the notes:

```
install.packages(c("sandwich", "lmtest", "MASS", "nleqslv",
                   "randtoolbox", "statmod", "glmnet", "haven"))
```

Every tutorial can be run from top to bottom. Data sets are in the `data/` folder of the course material on Moodle.

How to use the tutorials

Each tutorial first states the problems. Try them yourself before opening the solution boxes. Solutions are released after the tutorial session.

Conventions

Vectors and matrices are in bold, $\boldsymbol{\theta}$ denotes a generic parameter vector and $\boldsymbol{\theta}_0$ its true value. ln is the natural logarithm. $\xrightarrow{p}$ and $\xrightarrow{d}$ denote convergence in probability and in distribution.

Part I

Estimation principles

1 Maximum Likelihood

Readings

Wooldridge (2010), Ch. 13.1–13.7 and 13.11; Cameron and Trivedi (2005), Ch. 5.6–5.7; B. E. Hansen (2022b), Ch. 10; B. E. Hansen (2022a), Ch. 22, 25 and 26.10. Applications: Smith (2009) and Winkelmann (2004).

Maximum likelihood (ML) is the workhorse estimator for fully specified parametric models. This chapter introduces the likelihood principle, derives the large-sample properties of the MLE, discusses what happens when the assumed density is wrong, and closes with the classical trinity of tests.

1.1 Parametric models

We first consider unconditional distributions. A **parametric model** for a random variable Y is a complete probability function $f(y \mid \boldsymbol{\theta})$ that depends on an unknown parameter vector $\boldsymbol{\theta}$ in a **parameter space** Θ. We use the same notation for probability mass functions (discrete Y) and densities (continuous Y). For example, $Y \sim N(\mu, \sigma^2)$ has density $f(y \mid \mu, \sigma^2) = \sigma^{-1}\phi((y-\mu)/\sigma)$ with $\phi(z) = (2\pi)^{-1/2}\exp(-z^2/2)$, $\mu \in \mathbb{R}$ and $\sigma^2 > 0$. A parametric model need not be a textbook distribution; it can be built for a specific application.

Definition 1.1 (Model)**.** A model for a random sample is the assumption that $Y_1, \ldots, Y_n$ are i.i.d. with a known density (or mass) function $f(y \mid \boldsymbol{\theta})$ and unknown parameter $\boldsymbol{\theta} \in \Theta$.

Definition 1.2 (Correct specification)**.** A model is **correctly specified** if there exists a unique $\boldsymbol{\theta}_0 \in \Theta$ such that $f(y \mid \boldsymbol{\theta}_0) = f(y)$, the true density. $\boldsymbol{\theta}_0$ is the **true parameter value**. Uniqueness means that no other $\boldsymbol{\theta} \in \Theta$ satisfies $f(y \mid \boldsymbol{\theta}) = f(y \mid \boldsymbol{\theta}_0)$. A model is **misspecified** if no $\boldsymbol{\theta} \in \Theta$ satisfies $f(y \mid \boldsymbol{\theta}) = f(y)$.

Example 1.1 (Correct and incorrect models)**.** Let the true density be $f(y) = 2e^{-2y}$, $y > 0$.

- The exponential model $f(y \mid \lambda) = \lambda^{-1}e^{-y/\lambda}$ is correctly specified with $\lambda_0 = 1/2$.
- The gamma model $f(y \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1}e^{-\beta y}$ is correctly specified with $(\alpha_0, \beta_0) = (1, 2)$.
- The log-normal model $f(y \mid \theta, \nu) = (2\pi\nu)^{-1/2}y^{-1}\exp\{-(\ln y - \theta)^2/(2\nu)\}$ is misspecified: no (θ, ν) reproduces $2e^{-2y}$.

Uniqueness matters. A two-component normal mixture $p\,\phi_{\sigma_1}(y-\mu_1) + (1-p)\,\phi_{\sigma_2}(y-\mu_2)$ contains the standard normal $\phi(y)$, but at many parameter values: $(p, 0, 1, 0, 1)$ for any p, $(1, 0, 1, \mu_2, \sigma_2^2)$ for any (μ_2, σ_2^2), and so on. The model contains the truth, but $\boldsymbol{\theta}_0$ is not identified, so it is not correctly specified in the sense of Definition 1.2.

1.2 The likelihood function and the MLE

The **likelihood** is the joint density of the sample, evaluated at the observed data and viewed as a function of $\boldsymbol{\theta}$. Under independence it factorises, and under identical distributions all factors share the same form:

$$L_n(\boldsymbol{\theta}) = f(y_1, \dots, y_n \mid \boldsymbol{\theta}) = \prod_{i=1}^{n} f(y_i \mid \boldsymbol{\theta}), \qquad \ell_n(\boldsymbol{\theta}) \equiv \ln L_n(\boldsymbol{\theta}) = \sum_{i=1}^{n} \ln f(y_i \mid \boldsymbol{\theta}).$$

Definition 1.3 (Maximum likelihood estimator)**.** The MLE is the parameter value that maximises the likelihood,

$$\hat{\boldsymbol{\theta}}_n = \arg\max_{\boldsymbol{\theta} \in \Theta} L_n(\boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta} \in \Theta} \frac{1}{n} \sum_{i=1}^{n} \ln f(y_i \mid \boldsymbol{\theta}).$$

The second equality holds because $\ln(\cdot)$ is strictly increasing and scaling by $1/n$ does not change the maximiser.

The MLE is the parameter value under which the observed sample is most likely. Working with the average log-likelihood is convenient: it is a sample mean, so laws of large numbers and central limit theorems apply directly.

Example 1.2 (Bernoulli)**.** Let $Y_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\pi)$, so $f(y \mid \pi) = \pi^y (1-\pi)^{1-y}$ for $y \in \{0, 1\}$. Then

$$\ell_n(\pi) = \sum_{i=1}^{n} \left[y_i \ln \pi + (1 - y_i) \ln(1 - \pi) \right] = n \left[\bar{y} \ln \pi + (1 - \bar{y}) \ln(1 - \pi) \right].$$

The first-order condition $n\bar{y}/\pi - n(1-\bar{y})/(1-\pi) = 0$ gives $\hat{\pi} = \bar{y}$. Figure 1.1 plots the likelihood for two samples of size five.

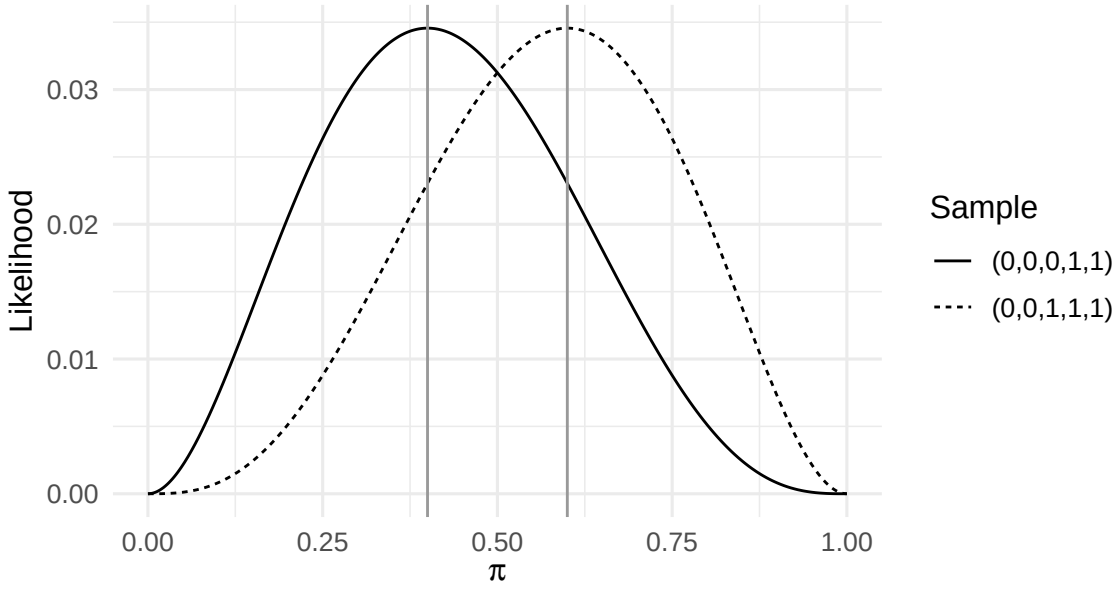


Figure 1.1: Bernoulli likelihood for the samples (0,0,0,1,1) and (0,0,1,1,1). The maximum is at the sample mean.

In practice we find the MLE in one of three ways:

1. **Solve the first-order condition** $\partial \ell_n(\boldsymbol{\theta})/\partial \boldsymbol{\theta} = \mathbf{0}$ and check the second-order condition. The FOC is necessary but not sufficient, and it only locates interior extrema; a maximum on the boundary of Θ need not satisfy it.
2. **Direct argument**, for instance by finding an upper bound of the likelihood that is attained at a unique point. The uniform distribution on $[0, \theta]$ is the standard case: the likelihood $\theta^{-n}\mathbf{1}\{\theta \geq \max_i x_i\}$ is maximised at the largest observation. Problem Set 1 treats a shifted exponential distribution, where the same argument applies.
3. **Numerical optimisation** when no closed form exists (logit, probit, Tobit, ...). We then have to worry about convergence to a local rather than a global maximum.

Example 1.3 (Exponential)**.** Let $f(y \mid \lambda) = \lambda^{-1} e^{-y/\lambda}$, $y > 0$. The log-likelihood and its derivatives are

$$\ell_n(\lambda) = -n \ln \lambda - \frac{n\bar{y}}{\lambda}, \qquad \frac{d\ell_n}{d\lambda} = -\frac{n}{\lambda} + \frac{n\bar{y}}{\lambda^2}, \qquad \frac{d^2\ell_n}{d\lambda^2} = \frac{n}{\lambda^2} - \frac{2n\bar{y}}{\lambda^3}.$$

The unique root of the FOC is $\hat{\lambda} = \bar{y}$, and $d^2\ell_n(\hat{\lambda})/d\lambda^2 = -n/\bar{y}^2 < 0$ confirms a maximum.

Theorem 1.1 (Invariance)**.** *If $\hat{\boldsymbol{\theta}}$ is the MLE of $\boldsymbol{\theta} \in \mathbb{R}^m$, then for any function $\boldsymbol{\beta} = h(\boldsymbol{\theta}) \in \mathbb{R}^l$ the MLE of $\boldsymbol{\beta}$ is $\hat{\boldsymbol{\beta}} = h(\hat{\boldsymbol{\theta}})$.*

In Example 1.3, reparametrise with the rate $\beta = 1/\lambda$. Then $\ell_n(\beta) = n \ln \beta - \beta n \bar{y}$, maximised at $\hat{\beta} = 1/\bar{y} = h(\hat{\lambda})$, as the theorem predicts. Most estimators do not share this property; an unbiased estimator of λ, for example, does not deliver an unbiased estimator of $1/\lambda$.

1.3 Score, Hessian and information

Assume $f(y \mid \boldsymbol{\theta})$ is twice differentiable in $\boldsymbol{\theta} \in \mathbb{R}^k$.

Definition 1.4 (Score and Hessian)**.** The **score** is the $k \times 1$ gradient of the log-likelihood and the **Hessian** is the $k \times k$ matrix of second derivatives:

$$S_n(\boldsymbol{\theta}) = \frac{\partial \ell_n(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \sum_{i=1}^{n} \underbrace{\frac{\partial \ln f(y_i \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}}_{S_i(\boldsymbol{\theta})}, \qquad H_n(\boldsymbol{\theta}) = \frac{\partial^2 \ell_n(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}'} = \sum_{i=1}^{n} \underbrace{\frac{\partial^2 \ln f(y_i \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}'}}_{H_i(\boldsymbol{\theta})}.$$

The score measures how sensitive the log-likelihood is to the parameters; at an interior MLE, $S_n(\hat{\boldsymbol{\theta}}) = \mathbf{0}$. The Hessian is symmetric and measures curvature. If $H_n(\boldsymbol{\theta})$ is negative definite for all $\boldsymbol{\theta}$, ℓ_n is strictly concave and a maximiser, if it exists, is unique. (The converse does not hold: a strictly concave function can have a singular Hessian at some points.) Both are sums over observations because the log-likelihood is.

Theorem 1.2 (Properties of the likelihood)**.** *Suppose the model is correctly specified, the support of Y does not depend on $\boldsymbol{\theta}$, and $\boldsymbol{\theta}_0$ is in the interior of Θ. Then*

(i) $\mathbb{E}[S_i(\boldsymbol{\theta}_0)] = \mathbf{0}$*;*

(ii) ***information matrix equality:*** $\mathbb{E}[S_i(\boldsymbol{\theta}_0) S_i(\boldsymbol{\theta}_0)'] = -\mathbb{E}[H_i(\boldsymbol{\theta}_0)] \equiv \mathcal{I}_1(\boldsymbol{\theta}_0)$.

$\mathcal{I}_1$ is the (Fisher) information of a single observation and $\mathcal{I}(\boldsymbol{\theta}_0) = n\mathcal{I}_1(\boldsymbol{\theta}_0)$ that of the sample. Because the score has mean zero, the left side of (ii) is the variance of the score. Part (i) follows from differentiating $\int f(y \mid \boldsymbol{\theta})\, dy = 1$ under the integral sign; part (ii) from differentiating once more. The support condition guarantees that this exchange of differentiation and integration is allowed. The uniform $U[0, \theta]$ violates it, which is why its MLE behaves non-standardly.

Proof of Theorem 1.2

Write $f = f(y \mid \boldsymbol{\theta})$ and use $\frac{\partial f}{\partial \boldsymbol{\theta}} = \frac{\partial \ln f}{\partial \boldsymbol{\theta}} f$.
(i) A density integrates to one for every $\boldsymbol{\theta}$: $\int f(y \mid \boldsymbol{\theta})\, dy = 1$. Differentiate with respect to $\boldsymbol{\theta}$ and exchange derivative and integral:

$$\mathbf{0} = \int \frac{\partial f}{\partial \boldsymbol{\theta}}\, dy = \int \frac{\partial \ln f}{\partial \boldsymbol{\theta}}\, f\, dy = \mathbb{E}\Big[\frac{\partial \ln f(y \mid \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\Big]. \qquad (*)$$

At $\boldsymbol{\theta} = \boldsymbol{\theta}_0$ the expectation is over the true distribution, so $\mathbb{E}[S_i(\boldsymbol{\theta}_0)] = \mathbf{0}$.
(ii) $(*)$ holds for every $\boldsymbol{\theta}$, so we can differentiate it with respect to $\boldsymbol{\theta}'$. By the product rule,

$$\mathbf{0} = \int \frac{\partial^2 \ln f}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}'}\, f\, dy + \int \frac{\partial \ln f}{\partial \boldsymbol{\theta}} \frac{\partial f}{\partial \boldsymbol{\theta}'}\, dy = \mathbb{E}\Big[\frac{\partial^2 \ln f}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}'}\Big] + \mathbb{E}\Big[\frac{\partial \ln f}{\partial \boldsymbol{\theta}} \frac{\partial \ln f}{\partial \boldsymbol{\theta}'}\Big],$$

using $\frac{\partial f}{\partial \boldsymbol{\theta}'} = \frac{\partial \ln f}{\partial \boldsymbol{\theta}'} f$ in the second term. Rearranging gives $\mathbb{E}[S_i S_i'] = -\mathbb{E}[H_i]$. □

Theorem 1.3 (Cramér–Rao lower bound)**.** *Under the same regularity conditions, any unbiased estimator $\tilde{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}_0$ satisfies* $\operatorname{Var}(\tilde{\boldsymbol{\theta}}) \succeq \mathcal{I}(\boldsymbol{\theta}_0)^{-1}$.

Example 1.4 (Bernoulli, continued)**.** For one observation, $S_i(\pi) = \dfrac{y_i - \pi}{\pi(1-\pi)}$ and $H_i(\pi) = -\dfrac{y_i}{\pi^2} - \dfrac{1 - y_i}{(1-\pi)^2}$. At π_0: $\mathbb{E}[S_i] = 0$, $\operatorname{Var}(S_i) = \dfrac{\pi_0(1-\pi_0)}{\pi_0^2(1-\pi_0)^2} = \dfrac{1}{\pi_0(1-\pi_0)}$, and $-\mathbb{E}[H_i] = \dfrac{1}{\pi_0} + \dfrac{1}{1-\pi_0} = \dfrac{1}{\pi_0(1-\pi_0)}$. The information equality holds and the CRLB is $\pi_0(1-\pi_0)/n$, which $\hat{\pi} = \bar{y}$ attains.

1.3.1 Conditional likelihood

Econometric models are usually about the distribution of y given covariates $\boldsymbol{x}$. Write the joint density as $f(y, \boldsymbol{x} \mid \boldsymbol{\delta}) = f(y \mid \boldsymbol{x}; \boldsymbol{\theta})\, f(\boldsymbol{x} \mid \boldsymbol{\gamma})$ with $\boldsymbol{\delta} = (\boldsymbol{\theta}', \boldsymbol{\gamma}')'$. If $\boldsymbol{\theta}$ and $\boldsymbol{\gamma}$ share no elements and are not linked by restrictions,

$$\ell_n(\boldsymbol{\delta}) = \sum_{i=1}^{n} \ln f(y_i \mid \boldsymbol{x}_i; \boldsymbol{\theta}) + \sum_{i=1}^{n} \ln f(\boldsymbol{x}_i \mid \boldsymbol{\gamma}),$$

and $\boldsymbol{\theta}$ can be estimated from the **conditional log-likelihood** $\sum_i \ln f(y_i \mid \boldsymbol{x}_i; \boldsymbol{\theta})$ alone, leaving the distribution of $\boldsymbol{x}$ unspecified. All results above carry over with expectations taken over the joint distribution of $(y, \boldsymbol{x})$.

1.4 Asymptotic properties of the MLE

Theorem 1.4 (Consistency (Wooldridge 2010, Thm. 13.1))**.** *Suppose (a) the conditional density $f(y \mid \boldsymbol{x}; \boldsymbol{\theta})$ is correctly specified; (b) $\boldsymbol{\theta}_0$ is globally identified: for every $\boldsymbol{\theta} \neq \boldsymbol{\theta}_0$ in Θ, $P\{f(y \mid \boldsymbol{x}; \boldsymbol{\theta}) \neq f(y \mid \boldsymbol{x}; \boldsymbol{\theta}_0)\} > 0$; (c) Θ is compact and the log-likelihood is continuous in $\boldsymbol{\theta}$ with a dominance condition. Then $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$.*

The idea: $\frac{1}{n}\ell_n(\boldsymbol{\theta}) \xrightarrow{p} \mathbb{E}[\ln f(y \mid \boldsymbol{x}; \boldsymbol{\theta})]$ uniformly in $\boldsymbol{\theta}$, and the limit is uniquely maximised at $\boldsymbol{\theta}_0$ (this follows from the KLIC inequality in Section 1.6). The maximiser of the sample objective then converges to the maximiser of the limit.

Theorem 1.5 (Asymptotic normality and efficiency)**.** *Under additional regularity conditions (interior $\boldsymbol{\theta}_0$, twice continuously differentiable log-density, nonsingular $\mathcal{I}_1(\boldsymbol{\theta}_0)$),*

$$\sqrt{n}\,(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0},\, \mathcal{I}_1(\boldsymbol{\theta}_0)^{-1}).$$

The MLE is asymptotically efficient: its asymptotic variance attains the Cramér–Rao bound, and no consistent and asymptotically normal estimator has a smaller asymptotic variance under the same assumptions.

Sketch. A mean-value expansion of the FOC around $\boldsymbol{\theta}_0$ gives $\mathbf{0} = S_n(\hat{\boldsymbol{\theta}}) = S_n(\boldsymbol{\theta}_0) + H_n(\bar{\boldsymbol{\theta}})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$, so

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = \Big[-\tfrac{1}{n} H_n(\bar{\boldsymbol{\theta}}) \Big]^{-1} \tfrac{1}{\sqrt{n}} S_n(\boldsymbol{\theta}_0).$$

Here $\bar{\boldsymbol{\theta}}$ lies between $\hat{\boldsymbol{\theta}}$ and $\boldsymbol{\theta}_0$, $\bar{\boldsymbol{\theta}} = \boldsymbol{\theta}_0 + \lambda(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$ with $0 \le \lambda \le 1$ (row by row), so $\bar{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}_0$ because $\hat{\boldsymbol{\theta}}$ is consistent. By the LLN and consistency, $-\frac{1}{n} H_n(\bar{\boldsymbol{\theta}}) \xrightarrow{p} \mathcal{I}_1(\boldsymbol{\theta}_0) \equiv \boldsymbol{A}_0$. By the CLT and Theorem 1.2, $\frac{1}{\sqrt{n}} S_n(\boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{B}_0)$ with $\boldsymbol{B}_0 = \mathbb{E}[S_i S_i']$. For a product of a matrix $\boldsymbol{H}_n \xrightarrow{p} \boldsymbol{H}$ and a vector $\boldsymbol{b}_n \xrightarrow{d} N(\boldsymbol{\mu}, \boldsymbol{B})$, Slutsky's theorem gives $\boldsymbol{H}_n \boldsymbol{b}_n \xrightarrow{d} N(\boldsymbol{H}\boldsymbol{\mu}, \boldsymbol{H}\boldsymbol{B}\boldsymbol{H}')$ (Appendix A). With $\boldsymbol{H}_n = [-\frac{1}{n} H_n(\bar{\boldsymbol{\theta}})]^{-1}$ this yields the general form $N(\mathbf{0}, \boldsymbol{A}_0^{-1} \boldsymbol{B}_0 \boldsymbol{A}_0^{-1})$, which collapses to $\mathcal{I}_1^{-1}$ because $\boldsymbol{A}_0 = \boldsymbol{B}_0$ under correct specification. Keep this sandwich form in mind; it reappears in Section 1.6 and in GMM.

1.5 Estimating the variance matrix

In finite samples we use $\operatorname{Var}(\hat{\boldsymbol{\theta}}) \approx \frac{1}{n} \mathcal{I}_1(\boldsymbol{\theta}_0)^{-1} = \mathcal{I}(\boldsymbol{\theta}_0)^{-1}$ and replace $\mathcal{I}_1$ by a consistent estimator. Three options are available, plus a fourth that does not rely on the information equality.

Table 1.1: Variance estimators after ML

	Estimator of $\operatorname{Var}(\hat{\boldsymbol{\theta}})$	Requires
(i) Expected Hessian	$\Big[-\sum_i \mathbb{E}\big(H_i(\boldsymbol{\theta}) \mid \boldsymbol{x}_i\big) \Big]^{-1}_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}$	closed form of $\mathbb{E}(H_i \mid \boldsymbol{x}_i)$
(ii) Observed Hessian	$\Big[-H_n(\hat{\boldsymbol{\theta}}) \Big]^{-1}$	second derivatives (analytic or numerical)
(iii) Outer product of gradients (OPG, BHHH)	$\Big[\sum_i S_i(\hat{\boldsymbol{\theta}}) S_i(\hat{\boldsymbol{\theta}})' \Big]^{-1}$	first derivatives only
(iv) Robust (sandwich)	$[-H_n(\hat{\boldsymbol{\theta}})]^{-1} \Big[\sum_i S_i S_i' \Big] [-H_n(\hat{\boldsymbol{\theta}})]^{-1}$	both

(i)–(iii) are all consistent under correct specification and typically differ in finite samples. The observed Hessian (ii) is the default in most software. The OPG (iii) is the cheapest but has the poorest finite-sample behaviour. The sandwich (iv) remains valid when the density is misspecified, as we see next.

For the Bernoulli model, all three of (i)–(iii) coincide at $\hat{\pi} = \bar{y}$ and equal $\bar{y}(1-\bar{y})/n$. Verifying this is a short exercise: use $\sum_i (y_i - \bar{y})^2 = n\bar{y}(1-\bar{y})$ for binary y_i.

1.5.1 Example: the normal linear model

Let $y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + \varepsilon_i$ with $\varepsilon_i \mid \boldsymbol{x}_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Then

$$\ell_n(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})^2.$$

Setting the derivatives to zero yields

$$\hat{\boldsymbol{\beta}}_{ML} = \Big(\sum_i \boldsymbol{x}_i\boldsymbol{x}_i'\Big)^{-1}\sum_i \boldsymbol{x}_i y_i, \qquad \hat{\sigma}^2_{ML} = \frac{1}{n}\sum_i (y_i - \boldsymbol{x}_i'\hat{\boldsymbol{\beta}})^2.$$

$\hat{\boldsymbol{\beta}}_{ML}$ is OLS. $\hat{\sigma}^2_{ML}$ lacks the degrees-of-freedom correction, so it is biased downward in finite samples, but the bias vanishes as $n \to \infty$.

1.6 Misspecification and pseudo-ML

What does the MLE estimate when the model is wrong? The answer uses a measure of distance between densities.

Definition 1.5 (Kullback–Leibler divergence)**.** For densities f and g, $\mathrm{KLIC}(f, g) = \int f(y)\ln\dfrac{f(y)}{g(y)}\,dy$.

KLIC is not symmetric, $\mathrm{KLIC}(f, g) \neq \mathrm{KLIC}(g, f)$ in general, so it is not a distance in the metric sense. Its key properties are $\mathrm{KLIC}(f, f) = 0$, $\mathrm{KLIC}(f, g) \geq 0$ and therefore $f = \arg\min_g \mathrm{KLIC}(f, g)$. The second follows from Jensen's inequality, since ln is concave:

$$-\mathrm{KLIC}(f, g) = \mathbb{E}_f\Big[\ln\frac{g(y)}{f(y)}\Big] \leq \ln\mathbb{E}_f\Big[\frac{g(y)}{f(y)}\Big] = \ln\int g(y)\,dy = \ln 1 = 0.$$

Applied to a parametric family $f_{\boldsymbol{\theta}} = f(\cdot \mid \boldsymbol{\theta})$: if the model is correct, $\boldsymbol{\theta}_0 = \arg\min_{\boldsymbol{\theta}} \mathrm{KLIC}(f, f_{\boldsymbol{\theta}})$.

When the model is misspecified, the parametric family may still be a good **approximating model**, for example a log-normal density for wages. The natural target is then the best-fitting member of the family.

Definition 1.6 (Pseudo-true parameter)**.** $\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}\in\Theta} \mathrm{KLIC}(f, f_{\boldsymbol{\theta}})$.

Since $\mathrm{KLIC}(f, f_{\boldsymbol{\theta}}) = \int f \ln f\,dy - \mathbb{E}[\ln f(y \mid \boldsymbol{\theta})]$ and only the second term depends on $\boldsymbol{\theta}$, $\boldsymbol{\theta}^*$ maximises $\mathbb{E}[\ln f(y \mid \boldsymbol{\theta})]$. The MLE maximises the sample analogue of this expectation. ML therefore has two roles: it estimates $\boldsymbol{\theta}_0$ when the model is correct and $\boldsymbol{\theta}^*$ when it is not. In the second case we call it the **pseudo-** or **quasi-MLE** (PMLE).

Theorem 1.6 (Asymptotic distribution of the PMLE (Huber 1967; White 1982))**.** *Under regularity conditions,* $\hat{\boldsymbol{\theta}}_{PMLE} \xrightarrow{p} \boldsymbol{\theta}^*$ *and*

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_{PMLE} - \boldsymbol{\theta}^*) \xrightarrow{d} N(\mathbf{0},\ \boldsymbol{A}^{*-1}\boldsymbol{B}^*\boldsymbol{A}^{*-1}), \qquad \boldsymbol{A}^* = -\mathbb{E}[H_i(\boldsymbol{\theta}^*)], \quad \boldsymbol{B}^* = \mathbb{E}[S_i(\boldsymbol{\theta}^*)S_i(\boldsymbol{\theta}^*)'].$$

The information equality fails, so $\boldsymbol{A}^* \neq \boldsymbol{B}^*$ and only the sandwich estimator (iv) in Table 1.1 is consistent. In general, $\boldsymbol{\theta}^* \neq \boldsymbol{\theta}_0$ and the PMLE is inconsistent for the parameter we care about. There is one important exception.

1.6.1 The linear exponential family

A density belongs to the **linear exponential family** (LEF) if it can be written

$$f(y \mid \mu) = \exp\{a(\mu) + b(y) + c(\mu)\, y\}, \qquad \mathbb{E}[y] = \mu, \quad \operatorname{Var}(y) = [c'(\mu)]^{-1}.$$

Table 1.2: Leading members of the linear exponential family

Distribution	$a(\mu)$, $b(y)$, $c(\mu)$	$\mathbb{E}[y]$	$\operatorname{Var}(y)$
Normal (σ^2 known)	$-\frac{\mu^2}{2\sigma^2} - \frac{1}{2}\ln(2\pi\sigma^2)$, $-\frac{y^2}{2\sigma^2}$, $\frac{\mu}{\sigma^2}$	μ	σ^2
Bernoulli	$\ln(1-\mu)$, 0, $\ln\frac{\mu}{1-\mu}$	μ	$\mu(1-\mu)$
Exponential	$-\ln\mu$, 0, $-1/\mu$	μ	μ^2
Poisson	$-\mu$, $-\ln y!$, $\ln\mu$	μ	μ

Theorem 1.7 (Consistency of LEF-based PMLE (Gouriéroux, Monfort, and Trognon 1984))**.** *If the assumed density is LEF and the conditional mean $\mathbb{E}[y \mid \boldsymbol{x}] = m(\boldsymbol{x}, \boldsymbol{\theta}_0)$ is correctly specified, the PMLE of $\boldsymbol{\theta}$ is consistent, whatever the true distribution of y given $\boldsymbol{x}$.*

The reason is that the LEF score is proportional to $y - m(\boldsymbol{x}, \boldsymbol{\theta})$, so its expectation is zero at $\boldsymbol{\theta}_0$ as soon as the mean is right. The practical consequence is important: Poisson regression consistently estimates an exponential conditional mean for *any* nonnegative outcome, count or not, provided inference uses the sandwich variance. Computer tutorial 1 illustrates this with overdispersed count data.

1.7 Hypothesis testing: the trinity

We test $H_0 : \boldsymbol{h}(\boldsymbol{\theta}_0) = \mathbf{0}$ against $H_1 : \boldsymbol{h}(\boldsymbol{\theta}_0) \neq \mathbf{0}$, where $\boldsymbol{h} : \mathbb{R}^k \to \mathbb{R}^q$ imposes $q \leq k$ restrictions with full-rank Jacobian $\boldsymbol{R}(\boldsymbol{\theta}) = \partial \boldsymbol{h}(\boldsymbol{\theta})/\partial \boldsymbol{\theta}'$. Linear restrictions are the special case $\boldsymbol{h}(\boldsymbol{\theta}) = \boldsymbol{R}\boldsymbol{\theta} - \boldsymbol{r}$. For example, with $k = 3$, $H_0 : \theta_1 = 1,\ \theta_2 - \theta_3 = 2$ has $q = 2$ and

$$\boldsymbol{R} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad \boldsymbol{r} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Let $\hat{\boldsymbol{\theta}}_U$ be the unrestricted MLE and $\hat{\boldsymbol{\theta}}_R$ the MLE subject to $\boldsymbol{h}(\boldsymbol{\theta}) = \mathbf{0}$. A restriction such as $\beta_1 = c\beta_2$ can often be imposed by reparametrisation: $x_1\beta_1 + x_2\beta_2 = (c\,x_1 + x_2)\beta_2$, so the restricted model has a single regressor.

Theorem 1.8 (LR, Wald and LM tests)**.** *Under H_0 and correct specification, each of the following converges in distribution to χ^2_q:*

$$\begin{aligned} LR &= 2\big[\ell_n(\hat{\boldsymbol{\theta}}_U) - \ell_n(\hat{\boldsymbol{\theta}}_R)\big], \\ W &= \boldsymbol{h}(\hat{\boldsymbol{\theta}}_U)' \Big[\boldsymbol{R}(\hat{\boldsymbol{\theta}}_U)\, \widehat{\operatorname{Var}}(\hat{\boldsymbol{\theta}}_U)\, \boldsymbol{R}(\hat{\boldsymbol{\theta}}_U)'\Big]^{-1} \boldsymbol{h}(\hat{\boldsymbol{\theta}}_U), \\ LM &= S_n(\hat{\boldsymbol{\theta}}_R)'\, \widehat{\operatorname{Var}}(S_n(\hat{\boldsymbol{\theta}}_R))^{-1} S_n(\hat{\boldsymbol{\theta}}_R), \qquad \widehat{\operatorname{Var}}(S_n(\hat{\boldsymbol{\theta}}_R)) = \mathcal{I}(\hat{\boldsymbol{\theta}}_R) \text{ or } -H_n(\hat{\boldsymbol{\theta}}_R). \end{aligned}$$

The three statistics are asymptotically equivalent under H_0 and under local alternatives.

The intuition differs (Figure 1.2):

- **LR**: if the restriction is valid, imposing it should not reduce the maximised log-likelihood by much.
- **Wald**: if the restriction is valid, $\boldsymbol{h}(\hat{\boldsymbol{\theta}}_U)$ should be close to zero, because $\hat{\boldsymbol{\theta}}_U$ is consistent.
- **LM (score)**: if the restriction is valid, the slope of the log-likelihood at $\hat{\boldsymbol{\theta}}_R$ should be close to zero, because $\hat{\boldsymbol{\theta}}_R$ is then close to the unrestricted maximum.

For a single coefficient, $H_0 : \theta_j = 0$, the Wald statistic is the square of the usual t-ratio $\hat{\theta}_j / \mathrm{se}(\hat{\theta}_j)$.

i Why χ^2? Wald and LR derivations

Wald. Let $\sqrt{n}(\hat{\boldsymbol{\theta}}_U - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Omega})$ with $\boldsymbol{\Omega} = \mathcal{I}_1(\boldsymbol{\theta}_0)^{-1}$. A mean-value expansion gives $\boldsymbol{h}(\hat{\boldsymbol{\theta}}_U) = \boldsymbol{h}(\boldsymbol{\theta}_0) + \boldsymbol{R}(\boldsymbol{\theta}^*)(\hat{\boldsymbol{\theta}}_U - \boldsymbol{\theta}_0)$ with $\boldsymbol{\theta}^*$ between $\hat{\boldsymbol{\theta}}_U$ and $\boldsymbol{\theta}_0$. Under H_0, $\boldsymbol{h}(\boldsymbol{\theta}_0) = \mathbf{0}$, so by the delta method

$$\sqrt{n}\,\boldsymbol{h}(\hat{\boldsymbol{\theta}}_U) \xrightarrow{d} N\big(\mathbf{0},\ \boldsymbol{R}(\boldsymbol{\theta}_0)\boldsymbol{\Omega}\boldsymbol{R}(\boldsymbol{\theta}_0)'\big),$$

and the quadratic form $n\,\boldsymbol{h}(\hat{\boldsymbol{\theta}}_U)'\big[\boldsymbol{R}\boldsymbol{\Omega}\boldsymbol{R}'\big]^{-1}\boldsymbol{h}(\hat{\boldsymbol{\theta}}_U) \xrightarrow{d} \chi^2_q$. Replacing $\boldsymbol{R}(\boldsymbol{\theta}_0)$ by $\boldsymbol{R}(\hat{\boldsymbol{\theta}}_U)$ and $\boldsymbol{\Omega}$ by the consistent estimator $n\widehat{\mathrm{Var}}(\hat{\boldsymbol{\theta}}_U)$ does not change the limit; this gives W.
LR (scalar θ, $H_0 : \theta = \theta_0$). A second-order expansion of $\ell_n(\theta_0)$ around $\hat{\theta}$, using $S_n(\hat{\theta}) = 0$:

$$\ell_n(\theta_0) = \ell_n(\hat{\theta}) + \tfrac{1}{2}H_n(\theta^*)(\theta_0 - \hat{\theta})^2 \quad \Longrightarrow \quad LR = 2\big[\ell_n(\hat{\theta}) - \ell_n(\theta_0)\big] = \Big[-\tfrac{1}{n}H_n(\theta^*)\Big]\Big[\sqrt{n}(\hat{\theta} - \theta_0)\Big]^2.$$

The first factor converges to $\mathcal{I}_1(\theta_0)$ and $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \mathcal{I}_1^{-1})$, so $LR \xrightarrow{d} \mathcal{I}_1 \cdot \mathcal{I}_1^{-1}\chi^2_1 = \chi^2_1$. The same argument shows $LR - W \xrightarrow{p} 0$.

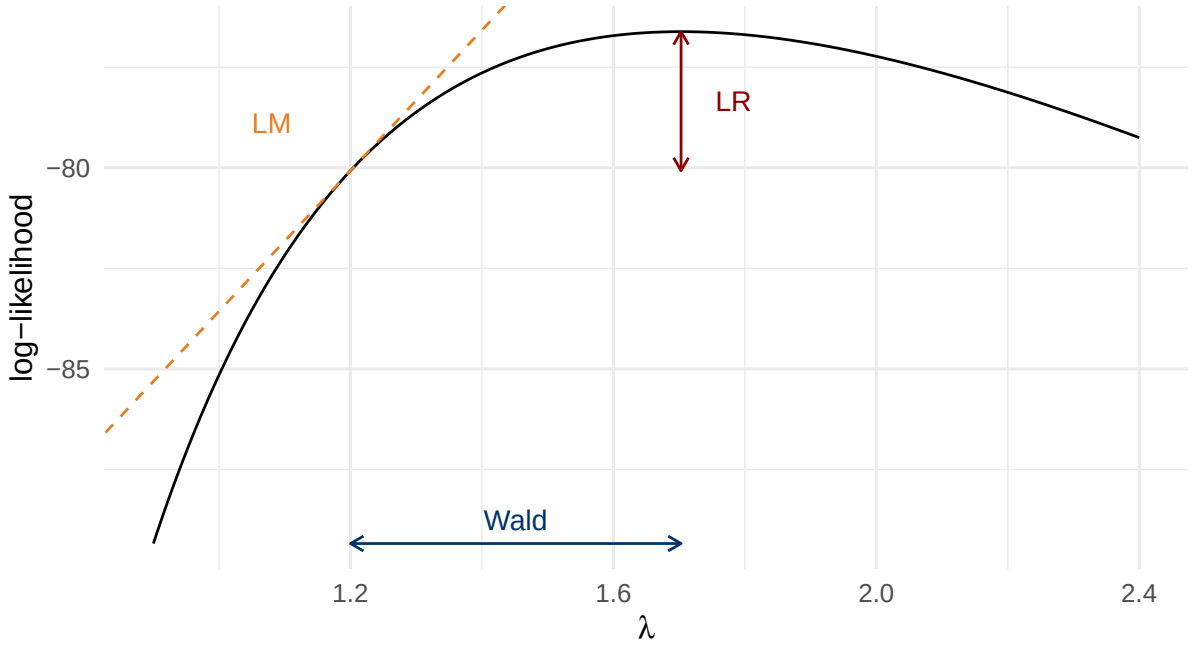


Figure 1.2: The trinity for the exponential model, testing $H_0 : \lambda = 1.2$ with n = 50 simulated observations. LR compares heights, Wald compares horizontal distance, LM compares the slope at the restricted value.

The three statistics for this sample:

```r
# y: 50 draws from an exponential with lambda = 1.5 (set.seed(42)); H0: lambda
↪  = 1.2
info <- function(l) n / l^2                  # Fisher information of the sample
LR <- 2 * (ll(l_hat) - ll(l0))
W  <- (l_hat - l0)^2 * info(l_hat)           # uses Var(l_hat) = l_hat^2 / n
LM <- slope^2 / info(l0)                     # score and information at l0
round(c(LR = LR, Wald = W, LM = LM, crit_5pct = qchisq(0.95, 1)), 3)
```

```
#>        LR      Wald        LM crit_5pct
#>     6.907     4.361     8.782     3.841
```

They differ in finite samples but lead to the same decision here. Which to use in practice?

1. **Computation.** LR needs both models, Wald only the unrestricted one, LM only the restricted one. LM is attractive when the restricted model is much simpler, e.g. when the restriction makes a nonlinear model linear.
2. **Invariance.** Only LR is invariant to how the model and the restriction are parametrised. The Wald statistic for $H_0 : \theta = 1$ and for the equivalent $H_0 : 1/\theta = 1$ can differ substantially in small samples (Cameron and Trivedi 2005, Ch. 7).
3. **Robustness.** Under misspecification, LR and the information-based LM are no longer χ^2. The Wald test with the sandwich variance remains valid, which is why robust Wald tests dominate applied work.

1.8 Computer tutorial 1

Learning goals. Write a log-likelihood as an R function and maximise it with `optim`; compute Hessian, OPG and sandwich variance estimators by hand and check them against `glm` and `sandwich`; see in a Monte Carlo study why pseudo-ML needs robust standard errors.

Packages. `sandwich`, `lmtest`, `MASS`. **Data.** `mroz.csv` (Mroz 1987), 753 married women in 1975.

1.8.1 Problem 1: Logit model for labour force participation

Variable	Description
`inlf`	= 1 if in labour force, 1975
`hours`	hours worked, 1975
`kidslt6`	number of kids under 6
`kidsge6`	number of kids aged 6–18
`age`	woman's age in years
`educ`	years of schooling
`wage`	estimated hourly wage from earnings (0 if not working)
`faminc`	family income, 1975
`exper`	actual labour market experience
`expersq`	experience squared

a) Create non-wife income in thousand dollars, `nwifeinc` = (`faminc` − `wage` × `hours`)/1000.

b) Estimate the linear probability model (LPM)

$$\texttt{inlf} = \beta_0 + \beta_1 \texttt{nwifeinc} + \beta_2 \texttt{educ} + \beta_3 \texttt{exper} + \beta_4 \texttt{expersq} + \beta_5 \texttt{age} + \beta_6 \texttt{kidslt6} + \beta_7 \texttt{kidsge6} + \varepsilon.$$

Why are heteroskedasticity-robust standard errors required here? How many fitted values lie outside $[0, 1]$?

c) Write down the logit model and its log-likelihood formally. Derive the score and the Hessian of observation i.

d) Estimate the logit model with the regressors from (b) in two ways.

1. Code the **negative** log-likelihood as an R function `negll(b, y, X)` and minimise it with `optim(..., method = "BFGS", hessian = TRUE)`, using the LPM coefficients as starting values. Supply the analytical gradient as well.
2. Use `glm(..., family = binomial(link = "logit"))`.

e) Compute the variance matrix of $\hat{\boldsymbol{\beta}}$ in four ways: (i) from the Hessian returned by `optim`; (ii) the inverse of the negative analytical Hessian; (iii) the inverse of the OPG; (iv) the sandwich. Compare with `vcov()` and `sandwich()` applied to the `glm` object.

f) Compute the average partial effects (APEs) of `educ` and `kidslt6` from the logit model and compare them with the LPM coefficients.

1.8.2 Problem 2: PML vs. ML, a Monte Carlo study

The goal is to see how ML behaves when the likelihood is misspecified but the conditional mean is right (see "Misspecification and pseudo-ML" in Chapter 1 of the lecture notes).

Correct density.

a) Simulate $n = 1000$ observations from $y_i \mid \boldsymbol{x}_i \sim \text{Poisson}(\lambda_i)$ with $\lambda_i = \exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})$, $(\beta_0, \beta_1, \beta_2) = (0.2, 0.1, 0)$ and x_1, x_2 independent $N(0, 1)$.

b) Estimate $\boldsymbol{\beta}$ with `glm(..., family = poisson)`.

c) Using the score and Hessian of the Poisson model from Problem Set 1, Problem 3, compute Hessian-based and sandwich standard errors by hand. Check them against `glm` and `coeftest(..., vcov = sandwich)`.

Misspecified density.

d) Now draw y_i from a negative binomial distribution with the same mean $\mu_i = \exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i})$ and variance $\text{Var}(y_i \mid \boldsymbol{x}_i) = \mu_i + \mu_i^2/\theta$ with $\theta = 3$.

e) Estimate the mean function by (i) Poisson PML with Hessian-based variance, (ii) Poisson PML with sandwich variance, (iii) negative binomial ML (`MASS::glm.nb`) with Hessian-based variance.

f) Repeat (d)–(e) for 1000 Monte Carlo samples and fill in the table:

	Poisson PML (Hessian)	Poisson PML (sandwich)	NegBin ML (Hessian)
MC mean of $\hat{\beta}_2$			
MC mean of s.e.$(\hat{\beta}_2)$			

	Poisson PML (Hessian)	Poisson PML (sandwich)	NegBin ML (Hessian)
MC std. dev. of $\hat{\beta}_2$			
Rejection rate of $H_0 : \beta_2 = 0$ at 5%			

g) Discuss your results.

1.9 Exercises

Problem Set 1 contains the analytical exercises for this chapter:

- **Problem 1**: a regular one-parameter model from start to finish (score, information, bias and the Cramér–Rao bound, asymptotics, invariance) and the three tests, including the non-invariance of the Wald test.
- **Problem 2**: the shifted exponential distribution as a non-regular case, with an n-rate, non-normal limit.
- **Problem 3**: Poisson regression, its score and Hessian; with a binary regressor in closed form, with Poisson-based and robust standard errors under overdispersion. Computer tutorial 1, Problem 2 builds on it.
- **Problem 4**: canonical links in the linear exponential family.

2 Method of Moments and GMM

Readings

Cameron and Trivedi (2005), Ch. 6.1–6.5; B. E. Hansen (2022a), Ch. 12 (instrumental variables) and Ch. 13 (GMM) and 17.36–17.42 (dynamic panels); B. E. Hansen (2022b), Ch. 11 (method of moments); Wooldridge (2010), Ch. 8 and 14. Potential outcomes and weighting (Section 2.9): Imbens and Wooldridge (2009), Sections 1–5; Wooldridge (2010), Ch. 21.1–21.3. Applications: Arellano and Bond (1991) and Imai and Ratkovic (2014).

Maximum likelihood requires the full distribution of the data. Many economic models deliver less: a set of expectations that should be zero at the true parameter value. The method of moments (MM) and its generalization, the generalized method of moments (GMM), turn such moment conditions directly into estimators. OLS, IV, 2SLS and ML are all special cases, and the GMM framework gives a single recipe for their asymptotic distribution and for efficient estimation when there are more conditions than parameters (L. P. Hansen 1982).

2.1 Moment conditions

Let $\boldsymbol{w}_i$ collect all observed variables for unit i (outcomes, regressors, instruments) and let $\boldsymbol{\theta} \in \Theta \subset \mathbb{R}^k$.

Definition 2.1 (Moment function)**.** A $q \times 1$ vector function $\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})$ with $q \geq k$ is a **moment function** if

$$\mathbb{E}\big[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)\big] = \mathbf{0}.$$

This equation is a (population) **moment condition** or moment restriction.

A moment function depends only on the data and the parameters, it has at least as many elements as there are parameters, and its expectation is zero at the true value. Three examples recur throughout the chapter.

Example 2.1 (Familiar estimators as moment estimators)**.**

1. **Mean.** $\psi(y_i, \mu) = y_i - \mu$. The sample analogue $\frac{1}{n}\sum_i (y_i - \hat{\mu}) = 0$ gives $\hat{\mu} = \bar{y}$.
2. **OLS.** In $y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + u_i$, the assumption $\mathbb{E}[u_i \mid \boldsymbol{x}_i] = 0$ implies the k conditions $\mathbb{E}[\boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})] = \mathbf{0}$. Solving the sample analogue gives $\hat{\boldsymbol{\beta}} = \big(\sum_i \boldsymbol{x}_i\boldsymbol{x}_i'\big)^{-1}\sum_i \boldsymbol{x}_i y_i$.
3. **Maximum likelihood.** Under correct specification the score has mean zero (Theorem 1.2). With $\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}) = \partial \ln f(y_i \mid \boldsymbol{x}_i; \boldsymbol{\theta})/\partial\boldsymbol{\theta}$, the MM estimator is the root of the score, i.e. the MLE.

2.2 The method of moments estimator

Definition 2.2 (Method of moments estimator)**.** If $q = k$, the MM estimator $\hat{\boldsymbol{\theta}}$ solves the k sample moment conditions

$$\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) \equiv \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{\psi}(\boldsymbol{w}_i, \hat{\boldsymbol{\theta}}) = \mathbf{0}.$$

MM estimators are consistent whenever a law of large numbers makes the sample moments converge to the population moments and $\boldsymbol{\theta}_0$ is the unique solution of $\mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})] = \mathbf{0}$. For OLS,

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + \Big(\tfrac{1}{n}\textstyle\sum_i \boldsymbol{x}_i\boldsymbol{x}_i'\Big)^{-1}\tfrac{1}{n}\textstyle\sum_i \boldsymbol{x}_i u_i \;\xrightarrow{p}\; \boldsymbol{\beta} + \mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}\mathbb{E}[\boldsymbol{x}_i u_i] = \boldsymbol{\beta},$$

provided $\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']$ is nonsingular.

Theorem 2.1 (Asymptotic distribution of the MM estimator)**.** *Under regularity conditions and i.i.d. sampling,*

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0},\; \boldsymbol{A}_0^{-1}\boldsymbol{V}_0\boldsymbol{A}_0^{-1\prime}), \qquad \boldsymbol{A}_0 = \mathbb{E}\Big[\frac{\partial \boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}'}\Big], \quad \boldsymbol{V}_0 = \mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)'].$$

Sketch. A mean-value expansion of $\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) = \mathbf{0}$ around $\boldsymbol{\theta}_0$ gives $\mathbf{0} = \bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0) + \boldsymbol{A}_n(\bar{\boldsymbol{\theta}})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$ with $\boldsymbol{A}_n = \frac{1}{n}\sum_i \partial\boldsymbol{\psi}_i/\partial\boldsymbol{\theta}'$. Hence $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = -\boldsymbol{A}_n^{-1}\sqrt{n}\,\bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0)$. The LLN gives $\boldsymbol{A}_n \xrightarrow{p} \boldsymbol{A}_0$, the CLT gives $\sqrt{n}\,\bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{V}_0)$, and Slutsky does the rest. This is the same argument as for the MLE in Chapter 1; the MLE simply has $\boldsymbol{A}_0 = -\mathcal{I}_1$ and $\boldsymbol{V}_0 = \mathcal{I}_1$.

The variance is estimated by the sandwich $\hat{\boldsymbol{A}}^{-1}\hat{\boldsymbol{V}}\hat{\boldsymbol{A}}^{-1\prime}$ with $\hat{\boldsymbol{A}} = \frac{1}{n}\sum_i \partial\boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})/\partial\boldsymbol{\theta}'$ and $\hat{\boldsymbol{V}} = \frac{1}{n}\sum_i \boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})\boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})'$.

Example 2.2 (OLS under heteroskedasticity)**.** With $\boldsymbol{\psi} = \boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})$ we get $\boldsymbol{A}_0 = -\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']$ and $\boldsymbol{V}_0 = \mathbb{E}[u_i^2\boldsymbol{x}_i\boldsymbol{x}_i']$, so

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} N(\mathbf{0},\; \mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}\mathbb{E}[u_i^2\boldsymbol{x}_i\boldsymbol{x}_i']\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']^{-1}).$$

White's heteroskedasticity-robust variance estimator is the MM sandwich for OLS. Likewise, the PML sandwich in Section 1.6 is the MM sandwich with the score as moment function.

2.3 The generalized method of moments

When $q > k$ the model is **overidentified**: in general no $\boldsymbol{\theta}$ sets all q sample moments to zero at once. GMM instead makes them as small as possible in a quadratic metric.

Example 2.3 (Two sets of conditions for one regression)**.** If, in addition to $\mathbb{E}[u_i \mid \boldsymbol{x}_i] = 0$, the errors are conditionally symmetric, then $\mathbb{E}[u_i^3 \mid \boldsymbol{x}_i] = 0$ and

$$\mathbb{E}\begin{pmatrix} \boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta}) \\ \boldsymbol{x}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})^3 \end{pmatrix} = \mathbf{0},$$

which gives $2k$ conditions for k parameters.

Definition 2.3 (GMM estimator)**.** Let $\boldsymbol{W}_n$ be a $q \times q$ symmetric positive definite weighting matrix, possibly data dependent. The GMM estimator is

$$\hat{\boldsymbol{\theta}}(\boldsymbol{W}_n) = \arg\min_{\boldsymbol{\theta}\in\Theta} \; Q_n(\boldsymbol{\theta}), \qquad Q_n(\boldsymbol{\theta}) = \bar{\boldsymbol{g}}_n(\boldsymbol{\theta})' \, \boldsymbol{W}_n \, \bar{\boldsymbol{g}}_n(\boldsymbol{\theta}).$$

The first-order conditions are

$$\boldsymbol{A}_n(\hat{\boldsymbol{\theta}})' \, \boldsymbol{W}_n \, \bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) = \boldsymbol{0}, \qquad \boldsymbol{A}_n(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n} \frac{\partial \boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}'},$$

i.e. GMM sets k linear combinations of the q sample moments to zero. If $q = k$, $\boldsymbol{A}_n$ is square and (generically) invertible, so the FOC reduce to $\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) = \boldsymbol{0}$: in the just-identified case GMM is MM, whatever $\boldsymbol{W}_n$.

2.4 Asymptotic properties of GMM

Theorem 2.2 (Consistency and asymptotic normality of GMM)**.** *Assume i.i.d. data and*

(i) ***identification:*** $\mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})] = \boldsymbol{0}$ *if and only if* $\boldsymbol{\theta} = \boldsymbol{\theta}_0$*;*

(ii) $\boldsymbol{A}_0 = \mathbb{E}[\partial \boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)/\partial \boldsymbol{\theta}']$ *is finite with full column rank* k*;*

(iii) $\boldsymbol{W}_n \xrightarrow{p} \boldsymbol{W}_0$*, symmetric positive definite;*

(iv) $\sqrt{n}\,\bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0) \xrightarrow{d} N(\boldsymbol{0}, \boldsymbol{V}_0)$*,* $\boldsymbol{V}_0 = \mathbb{E}[\boldsymbol{\psi}_i \boldsymbol{\psi}_i']$ *positive definite;*

plus standard regularity conditions (compact Θ*, uniform convergence, smoothness). Then* $\hat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}_0$ *and*

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} N\Big(\boldsymbol{0}, \; (\boldsymbol{A}_0'\boldsymbol{W}_0\boldsymbol{A}_0)^{-1}\boldsymbol{A}_0'\boldsymbol{W}_0\boldsymbol{V}_0\boldsymbol{W}_0\boldsymbol{A}_0(\boldsymbol{A}_0'\boldsymbol{W}_0\boldsymbol{A}_0)^{-1}\Big).$$

The proof follows the MM sketch: expand the FOC around $\boldsymbol{\theta}_0$ and solve, $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \approx -(\boldsymbol{A}_0'\boldsymbol{W}_0\boldsymbol{A}_0)^{-1}\boldsymbol{A}_0'\boldsymbol{W}_0\sqrt{n}\,\bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0)$. Two features are worth stressing. First, **the weighting matrix does not affect consistency**, only the asymptotic variance. Second, condition (ii) is the local version of identification: if $\boldsymbol{A}_0$ has deficient rank, the moments do not pin down all parameters, which is the essence of weak or irrelevant instruments.

The variance is estimated by replacing $\boldsymbol{A}_0, \boldsymbol{V}_0, \boldsymbol{W}_0$ by $\hat{\boldsymbol{A}} = \boldsymbol{A}_n(\hat{\boldsymbol{\theta}})$, $\hat{\boldsymbol{V}} = \frac{1}{n}\sum_i \boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})\boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}})'$ and $\boldsymbol{W}_n$, and dividing by n.

2.5 Efficient GMM

Theorem 2.3 (Optimal weighting matrix)**.** *Among all choices of* $\boldsymbol{W}_0$*,* $\boldsymbol{W}_0^* = \boldsymbol{V}_0^{-1}$ *minimizes the asymptotic variance (in the matrix sense of Chapter 1). The minimum is*

$$\operatorname{Avar}\left(\hat{\boldsymbol{\theta}}(\boldsymbol{V}_0^{-1})\right) = \left(\boldsymbol{A}_0'\boldsymbol{V}_0^{-1}\boldsymbol{A}_0\right)^{-1}.$$

Proof. Write $\boldsymbol{B} = (\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A})^{-1}\boldsymbol{A}'\boldsymbol{V}^{-1}$ and $\boldsymbol{C} = (\boldsymbol{A}'\boldsymbol{W}\boldsymbol{A})^{-1}\boldsymbol{A}'\boldsymbol{W} - \boldsymbol{B}$ (dropping subscripts). The asymptotic variance with weight $\boldsymbol{W}$ is $(\boldsymbol{B}+\boldsymbol{C})\boldsymbol{V}(\boldsymbol{B}+\boldsymbol{C})'$. The cross term vanishes because $\boldsymbol{C}\boldsymbol{V}\boldsymbol{B}' = [(\boldsymbol{A}'\boldsymbol{W}\boldsymbol{A})^{-1}\boldsymbol{A}'\boldsymbol{W}\boldsymbol{A} - (\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A})^{-1}\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A}](\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A})^{-1} = \boldsymbol{0}$. Hence the variance equals $(\boldsymbol{A}'\boldsymbol{V}^{-1}\boldsymbol{A})^{-1} + \boldsymbol{C}\boldsymbol{V}\boldsymbol{C}'$, and $\boldsymbol{C}\boldsymbol{V}\boldsymbol{C}'$ is positive semidefinite. □

A fact that is useful for such comparisons: for symmetric positive definite $\boldsymbol{P}$ and $\boldsymbol{Q}$, $\boldsymbol{P} - \boldsymbol{Q}$ is positive semidefinite if and only if $\boldsymbol{Q}^{-1} - \boldsymbol{P}^{-1}$ is. Comparing variances is therefore equivalent to comparing information (precision) matrices.

Intuitively, moments that are measured with more noise get less weight. "Optimal" is relative to a given set of moment conditions: poorly chosen moments remain inefficient even with the best weight. Adding valid moment conditions never increases the asymptotic variance of efficient GMM.

$\boldsymbol{V}_0$ is unknown, so efficient GMM is implemented in steps:

1. Estimate $\hat{\boldsymbol{\theta}}^{(1)}$ with a convenient weight, e.g. $\boldsymbol{W}_n = \boldsymbol{I}$ or, in linear IV, $(\boldsymbol{Z}'\boldsymbol{Z}/n)^{-1}$. It is consistent but inefficient.
2. Compute $\hat{\boldsymbol{V}} = \frac{1}{n}\sum_i \boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}}^{(1)})\boldsymbol{\psi}_i(\hat{\boldsymbol{\theta}}^{(1)})'$ and re-estimate with $\boldsymbol{W}_n = \hat{\boldsymbol{V}}^{-1}$. This **two-step GMM** estimator is asymptotically efficient.

Repeating step 2 until convergence gives **iterated GMM**; letting the weight depend on $\boldsymbol{\theta}$ inside the objective, $Q_n(\boldsymbol{\theta}) = \bar{\boldsymbol{g}}_n(\boldsymbol{\theta})'\hat{\boldsymbol{V}}(\boldsymbol{\theta})^{-1}\bar{\boldsymbol{g}}_n(\boldsymbol{\theta})$, gives the **continuously updated estimator** (CUE). All three are asymptotically equivalent. In finite samples, estimating $\boldsymbol{V}_0$ introduces bias, especially with many moment conditions (Altonji and Segal 1996), so efficient GMM need not beat a simple one-step estimator in practice.

2.6 Testing overidentifying restrictions

GMM consistency rests on $\mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)] = \boldsymbol{0}$. With $q > k$ there is information left to check it: at the efficient estimate the sample moments should be close to zero.

Theorem 2.4 (Hansen's J test (L. P. Hansen 1982))**.** *Let $\hat{\boldsymbol{\theta}}$ be the efficient GMM estimator and $\hat{\boldsymbol{V}}$ a consistent estimator of $\boldsymbol{V}_0$. Under $H_0 : \mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)] = \boldsymbol{0}$,*

$$J = n\,\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}})'\,\hat{\boldsymbol{V}}^{-1}\,\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) \;\xrightarrow{d}\; \chi^2_{q-k}.$$

Three remarks. The factor n matters: without it the statistic converges to zero. The degrees of freedom are the number of overidentifying restrictions, $q - k$; in a just-identified model $J = 0$ by construction and nothing can be tested. Finally, the test has power only against violations that cannot be absorbed by $\boldsymbol{\theta}$: if all instruments are invalid in the same direction, J may not reject. A rejection says that *some* moment conditions fail, not which.

2.7 Conditional moments and optimal instruments

Models are often stated as a conditional restriction $\mathbb{E}[\rho(\boldsymbol{w}_i, \boldsymbol{\theta}_0) \mid \boldsymbol{z}_i] = 0$, e.g. $\rho = y_i - \boldsymbol{x}_i'\boldsymbol{\beta}$ with exogenous $\boldsymbol{z}_i$. By the law of iterated expectations, *any* function $\boldsymbol{D}(\boldsymbol{z}_i)$ yields valid unconditional moments $\mathbb{E}[\boldsymbol{D}(\boldsymbol{z}_i)\rho(\boldsymbol{w}_i, \boldsymbol{\theta}_0)] = \boldsymbol{0}$, so there are infinitely many GMM estimators. The asymptotically best choice, the **optimal instrument**, is

$$\boldsymbol{D}^*(\boldsymbol{z}_i) = \mathbb{E}\Big[\frac{\partial \rho(\boldsymbol{w}_i, \boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}} \,\Big|\, \boldsymbol{z}_i\Big] \Big/ \operatorname{Var}\left(\rho(\boldsymbol{w}_i, \boldsymbol{\theta}_0) \mid \boldsymbol{z}_i\right).$$

In the linear model with exogenous regressors ($\boldsymbol{z}_i = \boldsymbol{x}_i$) this is $-\boldsymbol{x}_i/\sigma_i^2$: the efficient moment estimator is weighted least squares (GLS). With endogenous regressors it is $-\mathbb{E}[\boldsymbol{x}_i \mid \boldsymbol{z}_i]/\sigma_i^2$, which motivates 2SLS: the first stage approximates $\mathbb{E}[\boldsymbol{x}_i \mid \boldsymbol{z}_i]$ by a linear projection.

2.8 Linear instrumental variables as GMM

Consider $y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + u_i$ with $\boldsymbol{x}_i$ of dimension k and some regressors correlated with u_i (omitted variables, measurement error, simultaneity). OLS is inconsistent. Let $\boldsymbol{z}_i$ be an $l \times 1$ vector of instruments, containing the exogenous regressors, with

- **exogeneity:** $\mathbb{E}[\boldsymbol{z}_i u_i] = \boldsymbol{0}$;
- **relevance:** $\operatorname{rank} \mathbb{E}[\boldsymbol{z}_i \boldsymbol{x}_i'] = k$ (requires $l \geq k$), and $\mathbb{E}[\boldsymbol{z}_i \boldsymbol{z}_i']$ nonsingular.

The moment function is $\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\beta}) = \boldsymbol{z}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})$, with sample analogue $\frac{1}{n}\boldsymbol{Z}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$ in matrix notation. Minimizing $\frac{1}{n^2}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$ gives the closed form

$$\hat{\boldsymbol{\beta}}(\boldsymbol{W}_n) = \left(\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{X}\right)^{-1}\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{y}.$$

Here $\boldsymbol{A}_0 = -\mathbb{E}[\boldsymbol{z}_i \boldsymbol{x}_i']$ and $\boldsymbol{V}_0 = \mathbb{E}[u_i^2 \boldsymbol{z}_i \boldsymbol{z}_i']$. The familiar estimators are special cases.

Table 2.1: Linear GMM estimators

Case	Weight $\boldsymbol{W}_n$	Estimator
Just identified, $l = k$	any	IV: $\hat{\boldsymbol{\beta}}_{IV} = (\boldsymbol{Z}'\boldsymbol{X})^{-1}\boldsymbol{Z}'\boldsymbol{y}$
Overidentified	$(\boldsymbol{Z}'\boldsymbol{Z})^{-1}$	2SLS: $\left[\boldsymbol{X}'\boldsymbol{Z}(\boldsymbol{Z}'\boldsymbol{Z})^{-1}\boldsymbol{Z}'\boldsymbol{X}\right]^{-1}\boldsymbol{X}'\boldsymbol{Z}(\boldsymbol{Z}'\boldsymbol{Z})^{-1}\boldsymbol{Z}'\boldsymbol{y}$
Overidentified, homoskedastic	$(\hat{\sigma}^2\boldsymbol{Z}'\boldsymbol{Z}/n)^{-1}$ (optimal)	2SLS
Overidentified, heteroskedastic	$\left(\frac{1}{n}\sum_i \hat{u}_i^2 \boldsymbol{z}_i \boldsymbol{z}_i'\right)^{-1}$ (optimal)	two-step efficient GMM

2SLS is the GMM estimator with $\boldsymbol{W}_n = (\boldsymbol{Z}'\boldsymbol{Z})^{-1}$; under homoskedasticity, $\boldsymbol{V}_0 = \sigma^2\mathbb{E}[\boldsymbol{z}_i\boldsymbol{z}_i']$ and this weight is optimal. Under heteroskedasticity both 2SLS and two-step GMM are consistent, GMM is more efficient, and **2SLS standard errors must be heteroskedasticity robust**:

$$\widehat{\operatorname{Var}}(\hat{\boldsymbol{\beta}}) = \left(\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{X}\right)^{-1}\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\Big(\sum_i \hat{u}_i^2 \boldsymbol{z}_i\boldsymbol{z}_i'\Big)\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{X}\left(\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}_n\boldsymbol{Z}'\boldsymbol{X}\right)^{-1}.$$

For two-step GMM this simplifies to $\left[\boldsymbol{X}'\boldsymbol{Z}\left(\sum_i \hat{u}_i^2 \boldsymbol{z}_i\boldsymbol{z}_i'\right)^{-1}\boldsymbol{Z}'\boldsymbol{X}\right]^{-1}$.

Weak instruments. If the instruments are only weakly correlated with the endogenous regressors, $\boldsymbol{A}_0$ is close to rank deficient. IV and 2SLS are then biased towards OLS and their normal approximation is poor, even in large samples (Staiger and Stock 1997). Always report the first-stage regression and the F statistic for the excluded instruments; a common rule of thumb flags $F < 10$ as weak.

2.8.1 Illustration: 2SLS versus efficient GMM

The simulation below has one endogenous regressor, three instruments and errors whose variance depends on the first instrument. Both estimators are consistent, but efficient GMM puts less weight on the noisy moment $\mathbb{E}[z_{1i}u_i] = 0$.

```r
set.seed(2027)
one_sample <- function(n = 500) {
  Z <- cbind(1, matrix(rnorm(3 * n), n))              # constant + 3
↪  instruments
  v <- rnorm(n)
  x <- drop(Z[, 2:4] %*% c(0.4, 0.4, 0.4)) + v
  u <- (0.5 * v + sqrt(0.75) * rnorm(n)) * exp(Z[, 2]) # endogenous,
↪  heteroskedastic
  y <- 1 + x + u
  X <- cbind(1, x)
  gmm <- function(W) solve(t(X) %*% Z %*% W %*% t(Z) %*% X, t(X) %*% Z %*% W
↪  %*% t(Z) %*% y)
  b_2sls <- gmm(solve(crossprod(Z)))                   # step 1: 2SLS
  S      <- crossprod(Z * drop(y - X %*% b_2sls)) / n
  b_gmm  <- gmm(solve(S))                              # step 2: efficient GMM
  e      <- drop(y - X %*% b_gmm)
  gbar   <- colMeans(Z * e)
  J      <- n * drop(t(gbar) %*% solve(crossprod(Z * e) / n) %*% gbar)
  c(b_2sls = b_2sls[2], b_gmm = b_gmm[2], J_reject = J > qchisq(0.95, df = 2))
}
mc <- t(replicate(1000, one_sample()))
round(c(mean_2sls = mean(mc[, 1]), mean_gmm = mean(mc[, 2]),
        sd_2sls = sd(mc[, 1]), sd_gmm = sd(mc[, 2]),
        J_rejection_rate = mean(mc[, 3])), 3)
```

```
#>        mean_2sls         mean_gmm          sd_2sls           sd_gmm
#>            1.016            1.002            0.273            0.199
#> J_rejection_rate
#>            0.039
```

Both estimators are centred on the true slope of one. The standard deviation of efficient GMM is about a quarter smaller than that of 2SLS. The J test, with $q - k = 4 - 2 = 2$ degrees of freedom, rejects the true moment conditions at roughly the nominal 5% rate.

2.9 Two-step estimators as joint GMM

Many estimators proceed in two steps: a first step estimates a nuisance parameter $\boldsymbol{\alpha}$ (a propensity score, a first-stage regression, a selection equation), and a second step estimates the parameter of interest τ given $\hat{\boldsymbol{\alpha}}$. Treating $\hat{\boldsymbol{\alpha}}$ as known in the second step gives wrong standard errors. Stacking both steps into one MM problem gives the correct ones automatically. We first introduce the leading example of this course, the average treatment effect estimated by inverse probability weighting, and then derive the general result.

2.9.1 Background: potential outcomes and the average treatment effect

Potential outcomes. Let $D_i \in \{0, 1\}$ indicate whether unit i receives a treatment (a training programme, smoking during pregnancy, a policy). Each unit has two *potential outcomes*: Y_{1i} if

treated and Y_{0i} if not (Rubin 1974). The individual effect is $Y_{1i} - Y_{0i}$. We observe only one of the two,

$$Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i},$$

so individual effects are never observed. We therefore target averages, most often the **average treatment effect**

$$\tau = \mathbb{E}[Y_{1i} - Y_{0i}] = \mu_1 - \mu_0, \qquad \mu_d = \mathbb{E}[Y_{di}],$$

or the average treatment effect on the treated, $\tau_T = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1]$. The notation implicitly assumes that a unit's outcome does not depend on the treatment of other units.

Why a simple comparison fails. The difference in observed means decomposes as

$$\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] = \underbrace{\mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1]}_{\tau_T} + \underbrace{\mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]}_{\text{selection bias}}.$$

If treated and untreated units differ in their untreated outcomes, e.g. mothers who smoke also differ in age, education and marital status, the comparison is biased. Under random assignment D_i is independent of (Y_{0i}, Y_{1i}), the selection bias is zero and $\tau_T = \tau$.

Selection on observables. Without randomization, suppose we observe covariates $\boldsymbol{x}_i$ that capture all differences between the groups that are relevant for the potential outcomes:

- *Unconfoundedness:* $(Y_{0i}, Y_{1i}) \perp D_i \mid \boldsymbol{x}_i$. Within cells of $\boldsymbol{x}_i$, treatment is as good as random.
- *Overlap:* $0 < p(\boldsymbol{x}) < 1$ for all $\boldsymbol{x}$ in the support, where $p(\boldsymbol{x}_i) = P(D_i = 1 \mid \boldsymbol{x}_i)$ is the **propensity score**.

Unconfoundedness is an identifying assumption and cannot be tested with the data; overlap can be checked by comparing the estimated propensity scores of treated and controls.

Identification by weighting. Under these two assumptions

$$\mu_1 = \mathbb{E}\Big[\frac{D_i Y_i}{p(\boldsymbol{x}_i)}\Big], \qquad \mu_0 = \mathbb{E}\Big[\frac{(1 - D_i) Y_i}{1 - p(\boldsymbol{x}_i)}\Big], \qquad \tau = \mathbb{E}\Big[\frac{D_i Y_i}{p(\boldsymbol{x}_i)} - \frac{(1 - D_i) Y_i}{1 - p(\boldsymbol{x}_i)}\Big]. \tag{2.1}$$

The proof uses $D_i Y_i = D_i Y_{1i}$, the law of iterated expectations and unconfoundedness; it is Problem Set 2, Problem 4(a). The intuition: a treated unit with propensity score 0.2 represents five units in the population, of which only one was treated. Weighting each treated unit by $1/p(\boldsymbol{x}_i)$ and each control by $1/(1 - p(\boldsymbol{x}_i))$ creates a pseudo-population in which treatment no longer depends on $\boldsymbol{x}_i$. This idea goes back to survey sampling (Horvitz and Thompson 1952); Rosenbaum and Rubin (1983) showed that the propensity score is the key balancing score.

Estimation. Replacing expectations by sample means gives the **inverse probability weighting (IPW)** estimator

$$\hat{\tau}_{IPW} = \frac{1}{n} \sum_{i=1}^{n} \Big[\frac{D_i Y_i}{\hat{p}(\boldsymbol{x}_i)} - \frac{(1 - D_i) Y_i}{1 - \hat{p}(\boldsymbol{x}_i)}\Big].$$

In finite samples the weights $D_i / \hat{p}(\boldsymbol{x}_i)$ need not average to one. The **normalized** version divides by the sum of the weights within each group,

$$\hat{\tau}_{norm} = \frac{\sum_i D_i Y_i / \hat{p}(\boldsymbol{x}_i)}{\sum_i D_i / \hat{p}(\boldsymbol{x}_i)} - \frac{\sum_i (1 - D_i) Y_i / (1 - \hat{p}(\boldsymbol{x}_i))}{\sum_i (1 - D_i) / (1 - \hat{p}(\boldsymbol{x}_i))},$$

and is usually more stable. With limited overlap some $\hat{p}(\boldsymbol{x}_i)$ are close to 0 or 1, a few observations get very large weights, and both estimators become imprecise. Regression adjustment, $\tau = \mathbb{E}[\mathbb{E}(Y_i \mid D_i = 1, \boldsymbol{x}_i) - \mathbb{E}(Y_i \mid D_i = 0, \boldsymbol{x}_i)]$, and doubly robust estimators that combine both models are the main alternatives; Imbens and Wooldridge (2009) survey them and the optional Chapter 6 discusses them together with matching, instrumental variables and regression discontinuity.

Why this is a two-step problem. The propensity score is unknown. In practice it is modelled parametrically, typically as a logit $p(\boldsymbol{x}_i;\boldsymbol{\alpha}) = \Lambda(\boldsymbol{x}_i'\boldsymbol{\alpha})$, and $\boldsymbol{\alpha}$ is estimated by ML in a first step. The ML first-order condition is a moment condition (Example 2.1), so both steps are MM problems:

$$\boldsymbol{\psi}_1(\boldsymbol{w}_i,\boldsymbol{\alpha}) = (D_i - \Lambda(\boldsymbol{x}_i'\boldsymbol{\alpha}))\boldsymbol{x}_i, \qquad \psi_2(\boldsymbol{w}_i,\boldsymbol{\alpha},\tau) = \frac{D_iY_i}{\Lambda(\boldsymbol{x}_i'\boldsymbol{\alpha})} - \frac{(1-D_i)Y_i}{1-\Lambda(\boldsymbol{x}_i'\boldsymbol{\alpha})} - \tau.$$

The question is how the estimation error in $\hat{\boldsymbol{\alpha}}$ affects the distribution of $\hat{\tau}$.

2.9.2 Stacking the two steps

Let $\boldsymbol{\psi}_1(\boldsymbol{w}_i,\boldsymbol{\alpha})$ (k_1 conditions) define the first step and $\psi_2(\boldsymbol{w}_i,\boldsymbol{\alpha},\tau)$ the second. The stacked moment function

$$\boldsymbol{\psi}(\boldsymbol{w}_i,\boldsymbol{\theta}) = \begin{pmatrix} \boldsymbol{\psi}_1(\boldsymbol{w}_i,\boldsymbol{\alpha}) \\ \psi_2(\boldsymbol{w}_i,\boldsymbol{\alpha},\tau) \end{pmatrix}, \qquad \boldsymbol{\theta} = (\boldsymbol{\alpha}',\tau)',$$

is just identified, and $\boldsymbol{A}_0$ is block lower triangular:

$$\boldsymbol{A}_0 = \begin{pmatrix} \boldsymbol{A}_{11} & \boldsymbol{0} \\ \boldsymbol{A}_{21} & A_{22} \end{pmatrix}, \quad \boldsymbol{A}_{11} = \mathbb{E}\Big[\frac{\partial \boldsymbol{\psi}_1}{\partial \boldsymbol{\alpha}'}\Big],\ \boldsymbol{A}_{21} = \mathbb{E}\Big[\frac{\partial \psi_2}{\partial \boldsymbol{\alpha}'}\Big],\ A_{22} = \mathbb{E}\Big[\frac{\partial \psi_2}{\partial \tau}\Big].$$

Applying Theorem 2.1 and reading off the last row gives the influence representation

$$\sqrt{n}(\hat{\tau} - \tau_0) = -\frac{1}{A_{22}}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\Big[\psi_{2i} - \boldsymbol{A}_{21}\boldsymbol{A}_{11}^{-1}\boldsymbol{\psi}_{1i}\Big] + o_p(1).$$

The term $\boldsymbol{A}_{21}\boldsymbol{A}_{11}^{-1}\boldsymbol{\psi}_{1i}$ is the **correction for the estimated first step**. It vanishes only if $\boldsymbol{A}_{21} = \boldsymbol{0}$, i.e. if the second-step moment does not depend on $\boldsymbol{\alpha}$ in expectation. Solving the stacked system jointly gives the same point estimates as the sequential procedure, because the first block does not involve τ; only the standard errors change.

For IPW, $A_{22} = -1$ and $\boldsymbol{A}_{21} \neq \boldsymbol{0}$. A surprising but general feature: when the first step is efficient ML of a correctly specified propensity score model, the correction *reduces* the variance, so using the estimated propensity score is more efficient than using the true one (Hirano, Imbens, and Ridder 2003). The naive standard error that treats $\hat{p}(\boldsymbol{x}_i)$ as known is therefore conservative for IPW, but in other two-step problems it can be too small.

```
set.seed(2027)
one_draw <- function(n = 1000) {
  x  <- rnorm(n)
  p  <- plogis(-0.3 + 0.8 * x)            # true propensity score
  d  <- rbinom(n, 1, p)
  y  <- 1 + 2 * x + d * 1 + rnorm(n)       # true ATE = 1
  ph <- fitted(glm(d ~ x, family = binomial))
  ipw <- function(p) mean(d * y / p - (1 - d) * y / (1 - p))
  c(true_p = ipw(p), estimated_p = ipw(ph))
}
sim <- replicate(2000, one_draw())
round(rbind(mean = rowMeans(sim), sd = apply(sim, 1, sd)), 3)
```

```
#>      true_p estimated_p
#> mean  1.000       1.002
#> sd    0.183       0.086
```

Both versions are centred on the true ATE of one, and the estimator with the *estimated* propensity score has the clearly smaller standard deviation.

Problem Set 2, Problem 4 derives the moment functions and the blocks of $\boldsymbol{A}_0$ for the average treatment effect on the treated, and Computer Tutorial 2 estimates the model jointly, computes the stacked sandwich variance and compares it with the bootstrap.

2.10 Further applications

GMM is the natural framework whenever theory delivers orthogonality conditions rather than a full likelihood. Two examples from the slides:

- **Dynamic panel data.** In $y_{it} = \gamma y_{i,t-1} + \boldsymbol{x}_{it}'\boldsymbol{\beta} + \eta_i + \varepsilon_{it}$, first differencing removes η_i but makes $\Delta y_{i,t-1}$ correlated with $\Delta\varepsilon_{it}$. Lags $y_{i,t-2}, y_{i,t-3}, \ldots$ are valid instruments, and their number grows with t. Arellano and Bond (1991) use all of them in an efficient GMM estimator.
- **Covariate balancing propensity score.** Imai and Ratkovic (2014) estimate the propensity score by GMM, combining the logit score with moment conditions that require the weighted covariate means to balance between treated and controls. The overidentified version allows a J test of the propensity score model.

2.11 Computer tutorial 2

Learning goals. Estimate a two-step estimator as one joint MM problem with `nleqslv`; compute its sandwich variance by hand and compare with the bootstrap; implement 2SLS, efficient two-step GMM and the J test in matrix form and check them against `AER::ivreg`.

Packages. `nleqslv`, `boot`, `sandwich`, `lmtest`, `AER`. **Data.** `cattaneo2.csv` (Cattaneo 2010), 4642 births; `card1995.csv` (Card 1995), 3613 young men from the NLS 1976 cross-section.

2.11.1 Problem 1: The effect of smoking on birth weight (IPW as joint MM)

This problem uses the IPW results of Section 2.9 of the lecture notes and Problem Set 2, Problem 4(a).

Variable	Description
`bweight`	infant birth weight (grams)
`mbsmoke`	= 1 if the mother smoked during pregnancy
`mmarried`	= 1 if the mother is married
`mage`	mother's age (years)
`medu`	mother's education (years)
`fbaby`	= 1 if first baby

We want the average treatment effect τ of `mbsmoke` on `bweight`. The propensity score is modelled as a logit,

$$P(\texttt{mbsmoke} = 1 \mid X_i) = \Lambda\big(\alpha_0 + \alpha_1 \texttt{mmarried} + \alpha_2 \texttt{mage} + \alpha_3 \texttt{mage}^2 + \alpha_4 \texttt{fbaby} + \alpha_5 \texttt{medu}\big).$$

a) Compute the raw difference in mean birth weight between smokers and non-smokers. Why is it unlikely to be the ATE?

b) Write the stacked moment function for $\boldsymbol{\theta} = (\boldsymbol{\alpha}', \tau)'$ from Section 2.9.2 of the lecture notes as an R function that returns the $k+1$ sample moments, and solve the system with `nleqslv`. Compare $\hat{\boldsymbol{\alpha}}$ with `glm` and $\hat{\tau}$ with the IPW formula evaluated at the `glm` propensity score. Why do they coincide?

c) Estimate the asymptotic variance of $\hat{\boldsymbol{\theta}}$ with the MM sandwich $\hat{\boldsymbol{A}}^{-1}\hat{\boldsymbol{V}}\hat{\boldsymbol{A}}^{-1\prime}/n$, computing $\hat{\boldsymbol{A}}$ analytically. Report the standard error of $\hat{\tau}$ and compare it with the "naive" standard error that treats the propensity score as known.

d) Compute the ATE with normalized weights (weights that sum to one within each treatment group).

e) Estimate the standard errors of both ATE estimators by the nonparametric bootstrap (500 replications). Remember that each bootstrap replication has to re-estimate the propensity score.

2.11.2 Problem 2: Returns to schooling with college proximity (IV and GMM)

Variable	Description
`lwage76`	log hourly wage, 1976
`ed76`	years of education, 1976
`age76`	age in 1976
`black`, `reg76r`, `smsa76r`	= 1 if black; lives in the South; lives in an urban area (1976)
`nearc4a`, `nearc4b`	= 1 if grew up near a 4-year public / private college

Following Card (1995) and B. E. Hansen (2022a, Ch. 12), consider

$$\log w_i = \beta_0 + \beta_1\,\texttt{educ}_i + \beta_2\,\texttt{exper}_i + \beta_3\,\texttt{exper}_i^2/100 + \beta_4\texttt{black}_i + \beta_5\texttt{south}_i + \beta_6\texttt{urban}_i + u_i,$$

with potential experience `exper` = `age` − `educ` − 6. Education is endogenous (ability is omitted), and so is experience, because it is a function of education. The excluded instruments are `nearc4a`, `nearc4b`, `age76` and `age76^2/100`.

a) Construct `exper` and `exper2`, drop observations with missing values and estimate the equation by OLS.

b) Estimate the first stage for `ed76`. Test the joint significance of `nearc4a` and `nearc4b` with a heteroskedasticity-robust F test. Are the instruments strong?

c) Count the moment conditions q and parameters k. Compute 2SLS with the GMM formula and $\boldsymbol{W}_n = (\boldsymbol{Z}'\boldsymbol{Z})^{-1}$, with heteroskedasticity-robust standard errors. Check your results with `AER::ivreg` and `sandwich`.

d) Compute the two-step efficient GMM estimator and its standard errors.

e) Compute Hansen's J statistic and its p-value. What does it test here?

f) Compare the estimated returns to schooling across OLS, 2SLS and GMM.

2.12 Exercises

Problem Set 2 contains the analytical exercises for this chapter:

- **Problem 1**: two MM estimators of an exponential mean, their asymptotic distributions, efficient GMM combining both moments, and why the identity weighting matrix depends on the units of measurement.
- **Problem 2**: ML, MM and GMM for the beta distribution.
- **Problem 3**: one endogenous regressor and two instruments, by hand: IV, GMM as a weighted average of the IV estimators, 2SLS, efficient GMM, robust standard errors and the J test.
- **Problem 4**: identification of the ATE and the ATT by inverse probability weighting, the ATT as a joint MM problem, and covariate balancing. Computer tutorial 2, Problem 1 implements the ATE version.

Part II

Computation and flexible models

3 Simulation-Based Estimation

> **Readings**
>
> Cameron and Trivedi (2005), Ch. 12 (simulation-based methods) and Ch. 11 (bootstrap); B. E. Hansen (2022a), Ch. 10 (resampling); Train (2009), Ch. 9 and 10.

Many likelihood functions and moment conditions contain integrals without a closed form: random effects in nonlinear panel models, random coefficients, unobserved heterogeneity, multinomial probit choice probabilities. This chapter shows how to approximate such integrals, deterministically by quadrature or stochastically by simulation, and what happens to the statistical properties of an estimator when the exact integral is replaced by an approximation. The last section uses the same simulation idea for inference: the bootstrap.

3.1 Integrals in econometric models

Two examples recur.

Random effects. In a panel with individual effect $c_i \sim g(c \mid \boldsymbol{\eta})$, the likelihood contribution of individual i is

$$f(\boldsymbol{y}_i \mid \boldsymbol{X}_i; \boldsymbol{\beta}, \boldsymbol{\eta}) = \int f(\boldsymbol{y}_i \mid \boldsymbol{X}_i, \boldsymbol{\beta}, c)\, g(c \mid \boldsymbol{\eta})\, dc,$$

a one-dimensional integral for every i and every trial value of the parameters.

Discrete choice. In a multinomial probit with m alternatives the choice probability is an $(m-1)$-dimensional integral of a multivariate normal density over a rectangle.

In both cases the object we need is an **expectation**,

$$I = \mathbb{E}[h(X)] = \int h(x)\, g(x)\, dx, \qquad X \sim g, \tag{3.1}$$

where h and g are known for given parameter values. A plain integral $\int_a^b f(x)\, dx$ is the special case $h = (b-a)f$ and g the uniform density on $[a, b]$.

3.2 Deterministic integration: quadrature

Quadrature approximates an integral by a weighted sum of function values at chosen **nodes** x_j with **weights** ω_j,

$$\hat{I} = \sum_{j=1}^{m} \omega_j\, f(x_j).$$

Newton–Cotes rules use equally spaced nodes on $[a, b]$. The midpoint rule evaluates f at the midpoints of n intervals with weights $(b-a)/n$; the trapezoidal rule averages the function values at the two ends of each interval; Simpson's rule fits a quadratic through three neighbouring

points. Their error shrinks at a polynomial rate in n (e.g. n^{-2} for the midpoint rule if f'' is bounded).

Gaussian quadrature chooses nodes and weights optimally for a given weight function $w(x)$:

$$\int_c^d w(x)\, r(x)\, dx \approx \sum_{j=1}^{m} \omega_j\, r(x_j).$$

With m nodes the rule is **exact if r is a polynomial of degree $2m-1$ or less**. Nodes and weights are tabulated and available in software (`statmod::gauss.quad` in R). The common cases are Gauss–Legendre ($w=1$ on $[-1,1]$), Gauss–Laguerre ($w=e^{-x}$ on $(0,\infty)$) and **Gauss–Hermite** ($w=e^{-x^2}$ on $\mathbb{R}$).

Example 3.1 (Gauss–Hermite for a normal expectation)**.** Let $X \sim N(\mu,\sigma^2)$. The substitution $x=\mu+\sqrt{2}\sigma z$ gives

$$\mathbb{E}[h(X)] = \int h(x)\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/(2\sigma^2)}dx = \frac{1}{\sqrt{\pi}}\int e^{-z^2}h(\mu+\sqrt{2}\sigma z)\,dz \approx \frac{1}{\sqrt{\pi}}\sum_{j=1}^{m}\omega_j\, h(\mu+\sqrt{2}\sigma z_j).$$

In the random effects probit with $c_i \sim N(0,\sigma_c^2)$ this gives the likelihood contribution $\ell_i \approx \pi^{-1/2}\sum_j \omega_j f(\boldsymbol{y}_i \mid \boldsymbol{X}_i, \boldsymbol{\beta}, \sqrt{2}\sigma_c z_j)$, which is how `xtprobit` in Stata computes it.

Table 3.1: Gauss<U+2013>Hermite nodes and weights for $w(z)=e^{-z^2}$ (only $z_j \geq 0$; the rule is symmetric).

m	node	weight
2	0.707107	0.886227
3	0.000000	1.181636
3	1.224745	0.295409
4	0.524648	0.804914
4	1.650680	0.081313
5	0.000000	0.945309
5	0.958572	0.393619
5	2.020183	0.019953

Choosing m. There is a trade-off between speed and accuracy. In practice one re-estimates with more nodes and checks that the results do not change (Stata's `quadchk`).

Many dimensions. A d-dimensional integral can be handled by a product rule, applying a one-dimensional rule in each coordinate. This needs m^d function evaluations: with $m=10$ nodes and $d=5$ dimensions already 10^5 per observation. This **curse of dimensionality** is the main reason to simulate.

3.3 Monte Carlo integration

The integral in Equation 3.1 is a population mean. Draw $x_1,\dots,x_S$ independently from g and use the sample mean:

$$\hat{I}_S = \frac{1}{S}\sum_{s=1}^{S} h(x_s).$$

Theorem 3.1 (Direct Monte Carlo integration)**.** *If* $\mathbb{E}|h(X)| < \infty$*, then* $\hat{I}_S$ *is unbiased,* $\mathbb{E}[\hat{I}_S] = I$*, and* $\hat{I}_S \xrightarrow{p} I$ *as* $S \to \infty$*. If moreover* $\sigma_h^2 = \mathrm{Var}(h(X)) < \infty$*,*

$$\mathrm{Var}(\hat{I}_S) = \frac{\sigma_h^2}{S}, \qquad \sqrt{S}(\hat{I}_S - I) \xrightarrow{d} N(0, \sigma_h^2).$$

These are the WLLN and the CLT applied to the draws. Three consequences:

- The **simulation standard error** $\hat{\sigma}_h/\sqrt{S}$, with $\hat{\sigma}_h$ the standard deviation of the $h(x_s)$, is computed from the same draws. Report it.
- The error shrinks at rate $S^{-1/2}$ **whatever the dimension of** X. This is why simulation wins in high dimensions, and loses to quadrature in one or two smooth dimensions.
- Halving the error requires four times as many draws.

Draws from g are usually generated from uniforms by the **inverse transform**: if $U \sim U(0,1)$ and G is the cdf of X, then $G^{-1}(U) \sim G$.

Example 3.2 (A normal expectation with a known answer)**.** Let $X \sim N(0,1)$ and $I = \mathbb{E}[\Phi(0.5 + X)]$. Since $\Phi(0.5 + X) = P(Z \le 0.5 + X \mid X)$ for an independent $Z \sim N(0,1)$, $I = P(Z - X \le 0.5) = \Phi(0.5/\sqrt{2}) = 0.638163$. Figure 3.1 compares Monte Carlo, Halton draws (Section 3.3.2) and Gauss–Hermite quadrature. Computer Tutorial 3 asks you to reproduce it.

3.3.1 Variance reduction

The variance σ_h^2/S can be reduced without more draws by using draws that are not independent or not from g.

Importance sampling. Rewrite the integral with a proposal density p that is positive wherever $hg \neq 0$:

$$I = \int \frac{h(x)g(x)}{p(x)}\, p(x)\, dx = \mathbb{E}_p[w(X)h(X)], \qquad w(x) = \frac{g(x)}{p(x)}, \qquad \hat{I}_{IS} = \frac{1}{S}\sum_{s=1}^{S} w(x_s)h(x_s), \quad x_s \sim p.$$

$\hat{I}_{IS}$ is unbiased for any valid p, but its variance depends on p. A good proposal puts its mass where $|h(x)|g(x)$ is large and avoids huge weights. If $h \ge 0$, the choice $p \propto hg$ would give zero variance; it is infeasible, because its normalizing constant is I, but it shows the direction. Importance sampling is the natural tool for small probabilities and for densities that are hard to draw from.

Example 3.3 (A tail probability)**.** For $Y \sim N(0,1)$, $p_0 = P(Y \ge 3) = 0.00135$. With $S = 100$ direct draws, the estimate is zero in $(1 - p_0)^{100} \approx 0.874$ of all samples. With the proposal $N(\mu, 1)$ the weight is $w(y) = \phi(y)/\phi(y - \mu) = e^{-\mu y + \mu^2/2}$ and

$$\mathbb{E}_p\big[w(Y)^2 \mathbf{1}\{Y \ge 3\}\big] = e^{\mu^2}\big(1 - \Phi(3 + \mu)\big),$$

so the simulation standard deviation with $S = 100$ falls from 0.00367 ($\mu = 0$) to 3.09×10^{-4} ($\mu = 4$) and 2.48×10^{-4} ($\mu = 3$). Problem Set 3 derives this.

Antithetic draws. If g is symmetric around zero, pair each draw x_s with $-x_s$ (for uniforms: u_s with $1 - u_s$) and average $\frac{1}{2}[h(x_s) + h(-x_s)]$. With S pairs, i.e. $2S$ function evaluations,

$$\mathrm{Var}(\hat{I}_{anti}) = \frac{1}{2S}\Big[\sigma_h^2 + \mathrm{Cov}\,\big(h(X), h(-X)\big)\Big],$$

compared with $\sigma_h^2/(2S)$ for $2S$ independent draws. Antithetic draws help if the covariance is negative, which is the case for monotone h.

3.3.2 Quasi-random draws

Pseudo-random draws cluster by chance and leave gaps. **Quasi-random** (low-discrepancy) sequences are deterministic and fill $[0,1]$ evenly; they are transformed to the target distribution by the inverse cdf. The leading example is the **Halton sequence**. For a prime base b, write $i = \sum_k a_k b^k$ in base b and reflect the digits about the decimal point: $\varphi_b(i) = \sum_k a_k b^{-k-1}$. For $b = 3$ the sequence starts

$$\tfrac{1}{3},\ \tfrac{2}{3},\ \tfrac{1}{9},\ \tfrac{4}{9},\ \tfrac{7}{9},\ \tfrac{2}{9},\ \tfrac{5}{9},\ \tfrac{8}{9},\ \ldots$$

Each new block of points falls into the gaps left by the previous ones.

```
# Halton sequence in base b (equivalent to randtoolbox::halton for dim = 1, b =
↪  2)
halton_seq <- function(n, base = 2, skip = 0) {
  sapply(seq_len(n) + skip, function(i) {
    f <- 1; r <- 0
    while (i > 0) { f <- f / base; r <- r + f * (i %% base); i <- i %/% base }
    r
  })
}
round(halton_seq(8, base = 3), 3)
```

```
#> [1] 0.333 0.667 0.111 0.444 0.778 0.222 0.556 0.889
```

For smooth integrands in low dimensions the error of quasi-Monte Carlo is close to $1/S$ instead of $1/\sqrt{S}$. In d dimensions one uses a different prime for each coordinate; with many dimensions the coordinates of the early points become correlated, and scrambled or randomized versions are preferred. In estimation, one long sequence is generated and consecutive blocks are used for different observations; the first points are often discarded.

The figure shows the three rates: the Monte Carlo error falls by about one decimal for a hundred-fold increase in S, Halton draws roughly ten times faster, and Gauss–Hermite reaches machine precision with about a dozen nodes, because the integrand is smooth and one-dimensional.

3.4 Maximum simulated likelihood

Suppose the density of $\boldsymbol{y}_i$ given $\boldsymbol{x}_i$ is an integral over unobservables $\boldsymbol{u}$ with known density g,

$$f(\boldsymbol{y}_i \mid \boldsymbol{x}_i; \boldsymbol{\theta}) = \int \tilde{f}(\boldsymbol{y}_i \mid \boldsymbol{x}_i, \boldsymbol{u}; \boldsymbol{\theta})\, g(\boldsymbol{u})\, d\boldsymbol{u}.$$

Draw $\boldsymbol{u}_{i1}, \ldots, \boldsymbol{u}_{iS}$ from g for each i and replace f by the unbiased simulator $\hat{f}_i(\boldsymbol{\theta}) = S^{-1} \sum_s \tilde{f}(\boldsymbol{y}_i \mid \boldsymbol{x}_i, \boldsymbol{u}_{is}; \boldsymbol{\theta})$.

Definition 3.1 (Maximum simulated likelihood estimator)**.**

$$\hat{\boldsymbol{\theta}}_{MSL} = \arg\max_{\boldsymbol{\theta}} \ \frac{1}{n} \sum_{i=1}^{n} \ln \hat{f}_i(\boldsymbol{\theta}), \qquad \hat{f}_i(\boldsymbol{\theta}) = \frac{1}{S} \sum_{s=1}^{S} \tilde{f}(\boldsymbol{y}_i \mid \boldsymbol{x}_i, \boldsymbol{u}_{is}; \boldsymbol{\theta}).$$

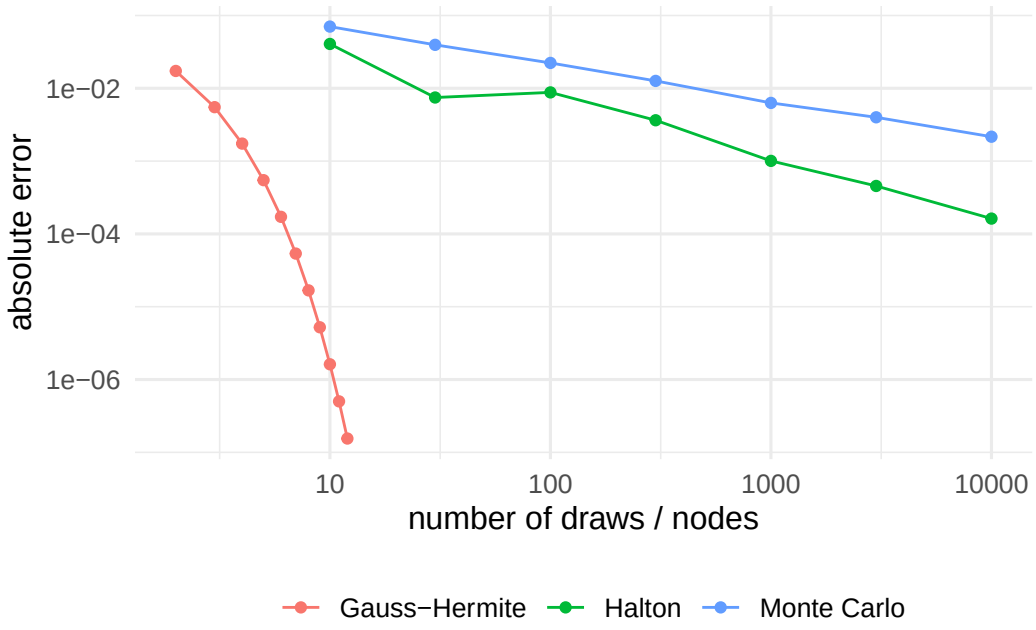


Figure 3.1: Absolute error in approximating $\mathbb{E}[\Phi(0.5+X)]$, $X \sim N(0,1)$: Monte Carlo (average over 200 replications), Halton draws and Gauss–Hermite quadrature. Log scales.

If the distribution of $\boldsymbol{u}$ depends on parameters (e.g. $c_i \sim N(0, \sigma_c^2)$), write $\boldsymbol{u} = \boldsymbol{u}(\boldsymbol{\varepsilon}; \boldsymbol{\theta})$ with draws $\boldsymbol{\varepsilon}$ from a fixed distribution ($c_{is} = \sigma_c \varepsilon_{is}$ with $\varepsilon_{is} \sim N(0,1)$).

Keep the draws fixed. The draws are generated once, before the optimization, and reused for every evaluation of the objective and its derivatives. New draws at every iteration would make the simulated likelihood jump between iterations and the optimizer fail to converge.

Simulation bias. $\hat{f}_i$ is unbiased for f_i, but the log is concave, so by Jensen's inequality $\mathbb{E}[\ln \hat{f}_i] < \ln f_i$. A second-order expansion gives

$$\mathbb{E}\big[\ln \hat{f}_i(\boldsymbol{\theta})\big] \approx \ln f_i(\boldsymbol{\theta}) - \frac{1}{2S} \frac{\operatorname{Var}\left(\tilde{f}(\boldsymbol{y}_i \mid \boldsymbol{x}_i, \boldsymbol{u}; \boldsymbol{\theta})\right)}{f_i(\boldsymbol{\theta})^2}.$$

The bias term depends on $\boldsymbol{\theta}$, so it shifts the maximizer: **with fixed S, MSL is inconsistent.**

Theorem 3.2 (Asymptotics of MSL (Cameron and Trivedi 2005, Ch. 12))**.** *Under the regularity conditions for ML and with independent draws across observations:*

1. *If $S \to \infty$ as $n \to \infty$, $\hat{\boldsymbol{\theta}}_{MSL}$ is consistent.*
2. *If $\sqrt{n}/S \to 0$, $\hat{\boldsymbol{\theta}}_{MSL}$ is asymptotically equivalent to the (infeasible) MLE: $\sqrt{n}(\hat{\boldsymbol{\theta}}_{MSL} - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathcal{I}(\boldsymbol{\theta}_0)^{-1})$.*

In practice S is chosen by experimentation: increase S (and use Halton draws) until the estimates stop changing. Standard errors are computed as for ML, from the Hessian or the sandwich of the simulated log-likelihood; with S large enough the simulation adds nothing to first order.

3.5 Method of simulated moments

If the moment function contains an integral, $\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}) = \int \tilde{\boldsymbol{\psi}}(\boldsymbol{w}_i, \boldsymbol{u}; \boldsymbol{\theta}) g(\boldsymbol{u})\, d\boldsymbol{u}$, replace it by the simulator $\hat{\boldsymbol{\psi}}_i(\boldsymbol{\theta}) = S^{-1} \sum_s \tilde{\boldsymbol{\psi}}(\boldsymbol{w}_i, \boldsymbol{u}_{is}; \boldsymbol{\theta})$ and apply (G)MM to $\frac{1}{n}\sum_i \hat{\boldsymbol{\psi}}_i(\boldsymbol{\theta})$ (McFadden 1989; Pakes and Pollard 1989).

The key difference to MSL: the simulator enters the objective **linearly**. Since $\mathbb{E}[\hat{\boldsymbol{\psi}}_i(\boldsymbol{\theta}_0)] = \mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)] = \mathbf{0}$, the simulated moment condition still holds at $\boldsymbol{\theta}_0$.

Theorem 3.3 (Properties of MSM)**.** *If the GMM estimator based on $\boldsymbol{\psi}$ is consistent and asymptotically normal and the draws are independent across i and of the data, then for **fixed** S the MSM estimator is consistent and asymptotically normal. Its asymptotic variance is that of GMM plus a term due to simulation noise, which vanishes as $S \to \infty$. With S independent draws per observation, simulation adds $\mathbb{E}\big[\operatorname{Var}(\tilde{\boldsymbol{\psi}} \mid \boldsymbol{w}_i)\big]/S$ to the variance $\boldsymbol{V}_0$ of the moment function.*

Table 3.2: MSL and MSM compared

	MSL	MSM
Consistent with fixed S	no	yes
Efficient as $S \to \infty$	yes, if $\sqrt{n}/S \to 0$	only if the moments are the efficient ones
Cost of small S	bias	variance

Example 3.4 (Unobserved heterogeneity (Cameron and Trivedi 2005, Ch. 12))**.** Let $y_i = \theta + u_i + \varepsilon_i$ with $\varepsilon_i \sim N(0,1)$ and u_i standard Gumbel (extreme value type I), $g(u) = e^{-u}\exp(-e^{-u})$, independent. The density of y_i,

$$f(y_i;\theta) = \int \phi(y_i - \theta - u)\, g(u)\, du,$$

has no closed form. With draws u_{is} from g:

- **MSL** maximizes $\frac{1}{n}\sum_i \ln\big[\frac{1}{S}\sum_s \phi(y_i - \theta - u_{is})\big]$.
- **MSM** uses $\mathbb{E}[y_i - \theta - u_i] = 0$ with $\mathbb{E}[u_i]$ replaced by the simulated mean, $\hat{\theta}_{MSM} = \bar{y} - \frac{1}{nS}\sum_i \sum_s u_{is}$.

(Here $\mathbb{E}[u_i]$ is in fact known, Euler's constant $\gamma \approx 0.5772$, so MM is feasible; the example is chosen so that we can compare.)

```
rgumbel <- function(n) -log(-log(runif(n)))
msl <- function(y, u) {                    # u: n x S matrix of draws
  obj <- function(th) mean(log(rowMeans(dnorm(y - th - u))))
  optimize(obj, c(-5, 5), maximum = TRUE)$maximum
}
msm <- function(y, u) mean(y) - mean(u)

set.seed(2027)
one_sample <- function(S, n = 100, theta = 1) {
  y <- theta + rgumbel(n) + rnorm(n)
  u <- matrix(rgumbel(n * S), n, S)        # draws fixed before optimizing
  c(msl = msl(y, u), msm = msm(y, u))
```

```
}
tab <- t(sapply(c(1, 2, 5, 20, 100, 1000), function(S) {
  r <- replicate(1000, one_sample(S))
  c(S = S, bias_MSL = mean(r["msl", ]) - 1, sd_MSL = sd(r["msl", ]),
    bias_MSM = mean(r["msm", ]) - 1, sd_MSM = sd(r["msm", ]))
}))
knitr::kable(round(tab, 3))
```

Table 3.3: MSL and MSM in the unobserved-heterogeneity model: bias and standard deviation over 1000 samples ($n = 100$, $\theta_0 = 1$).

S	bias_MSL	sd_MSL	bias_MSM	sd_MSM
1	-0.005	0.205	-0.005	0.205
2	0.145	0.191	-0.006	0.180
5	0.134	0.183	-0.006	0.170
20	0.065	0.165	-0.004	0.162
100	0.016	0.166	-0.007	0.173
1000	0.003	0.154	-0.001	0.162

The table illustrates Theorem 3.2 and Theorem 3.3:

- MSM is unbiased for every S; its standard deviation falls towards the MM value $\sqrt{(1+\pi^2/6)/n} = 0.163$. With independent draws, $\mathrm{Var}(\hat{\theta}_{MSM}) = [1 + \pi^2/6 + \pi^2/(6S)]/n$ (Problem Set 3, Problem 4, derives the analogous result for an exponential shock).
- MSL has a sizeable bias for small S that disappears only as S grows. With many draws it is slightly more precise than MSM, because it uses the whole likelihood.
- With $S = 1$ MSL and MSM coincide here: $\ln \phi$ is quadratic, so the first-order condition is linear in θ. This is special to the example.

3.6 The bootstrap

Simulation is also useful for inference. The asymptotic variance of many estimators is tedious to derive (two-step estimators, sample quantiles, functions of several estimates). The **nonparametric bootstrap** (Efron 1979) approximates the sampling distribution of an estimator $\hat{\theta} = T(\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n)$ by resampling the data:

1. Draw a bootstrap sample $\boldsymbol{w}_1^*, \ldots, \boldsymbol{w}_n^*$ of size n **with replacement** from the observed data (in a regression: resample whole observations $(y_i, \boldsymbol{x}_i)$, the *pairs* bootstrap).
2. Compute $\hat{\theta}_b^* = T(\boldsymbol{w}_1^*, \ldots, \boldsymbol{w}_n^*)$, using exactly the same procedure as for $\hat{\theta}$, including all estimation steps.
3. Repeat for $b = 1, \ldots, B$.

The **bootstrap standard error** is the standard deviation of $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$. The **percentile confidence interval** uses the $\alpha/2$ and $1-\alpha/2$ quantiles of the $\hat{\theta}_b^*$. B of a few hundred suffices for standard errors; percentile intervals need more ($B \geq 999$).

The logic: the empirical distribution of the data is to the sample what the population is to the data. If $\hat{\theta}$ is asymptotically normal and smooth in the empirical distribution, the bootstrap distribution of $\sqrt{n}(\hat{\theta}^* - \hat{\theta})$ approximates that of $\sqrt{n}(\hat{\theta} - \theta_0)$ (Cameron and Trivedi 2005, Ch. 11).

Example 3.5 (Standard error of a sample median)**.** For a sample of $n = 200$ from a lognormal distribution, the asymptotic variance of the median m is $1/(4f(m)^2 n)$, which needs an estimate of the density at the median. The bootstrap avoids this:

```
set.seed(2027)
y  <- exp(rnorm(200))
bt <- replicate(999, median(sample(y, replace = TRUE)))
c(median = median(y), se_boot = sd(bt),
  se_asymptotic = 1 / (2 * dlnorm(1) * sqrt(200)))   # uses the true density
```

```
#>        median       se_boot se_asymptotic
#>    0.99025928    0.09515879    0.08862269
```

```
quantile(bt, c(0.025, 0.975))                        # percentile 95% CI
```

```
#>     2.5%    97.5%
#> 0.853481 1.197444
```

The bootstrap standard error is close to the asymptotic one, which here uses the true density at the true median and would not be available in practice.

When it fails. The bootstrap is not a cure-all. It fails or needs modification for non-smooth or boundary problems (the maximum of a sample, parameters on the boundary of the parameter space), with weak instruments, and with dependent data, where whole blocks or clusters have to be resampled. In Computer Tutorial 2 we already used it for the two-step IPW estimator: there the bootstrap must re-estimate the propensity score in every replication.

3.7 Computer tutorial 3

Learning goals. Approximate an integral by Monte Carlo, quasi-random draws and Gaussian quadrature and compare their accuracy; implement maximum simulated likelihood and the method of simulated moments and see the simulation bias of MSL; compute bootstrap standard errors and confidence intervals and check their coverage.

Packages. `statmod` (Gaussian quadrature); optionally `randtoolbox` (Halton sequences). No data set: all data are simulated.

3.7.1 Problem 1: Three ways to compute an integral

Let $X \sim N(0,1)$ and $I = \mathbb{E}[\Phi(0.5 + X)]$, where Φ is the standard normal cdf.

a) Show that $I = \Phi(0.5/\sqrt{2})$. *Hint:* write $\Phi(0.5 + X) = P(Z \le 0.5 + X \mid X)$ for $Z \sim N(0,1)$ independent of X.

b) Write a function `mc_int(S)` that approximates I by direct Monte Carlo integration with S draws and returns the estimate together with its simulation standard error. Evaluate it for $S = 10, 100, 1000, 10000$.

c) Approximate I with S Halton draws (base 2), transformed to $N(0,1)$ by the inverse cdf. Use the function `halton_seq()` from Section 3.3.2 of the lecture notes or `randtoolbox::halton()`. Compare the errors with (b).

d) Approximate I by Gauss–Hermite quadrature with $m = 2, 3, 5, 10, 20$ nodes (`statmod::gauss.quad(m, kind = "hermite")`). Mind the change of variables in Example 3.1 of the lecture notes.

e) Use antithetic pairs $(x_s, -x_s)$ with $S/2$ pairs. Is the variance smaller than with S independent draws? Why?

3.7.2 Problem 2: MSL and MSM with unobserved heterogeneity

Consider the model of Example 3.4 of the lecture notes: $y_i = \theta + u_i + \varepsilon_i$ with $\varepsilon_i \sim N(0,1)$ and u_i standard Gumbel, independent, $\theta_0 = 1$.

a) Generate a sample of $n = 100$. Draw Gumbel variables by the inverse transform $u = -\ln(-\ln U)$, $U \sim U(0,1)$.

b) Write the simulated log-likelihood as a function of θ and of an $n \times S$ matrix of draws. Compute $\hat{\theta}_{MSL}$ for $S = 1, 10, 100, 1000$. Why must the draws be generated before the optimization?

c) Compute $\hat{\theta}_{MSM}$ for the same S, and the MM estimator $\bar{y} - \gamma$, which uses the known mean $\gamma = 0.5772$ of the Gumbel distribution.

d) Repeat (b) with Halton draws: take one long Halton sequence, discard the first 10 points, and use consecutive blocks of S points for the observations.

e) Monte Carlo: for $S = 2$ and $S = 50$, repeat (a)–(c) 300 times and report the mean and standard deviation of $\hat{\theta}_{MSL}$ and $\hat{\theta}_{MSM}$. Relate the results to Theorems 3.2 and 3.3 of the lecture notes.

3.7.3 Problem 3: Bootstrap standard errors and confidence intervals

a) Draw a sample of $n = 200$ from a lognormal distribution, $y_i = e^{z_i}$ with $z_i \sim N(0,1)$. The population median is 1.

b) Compute the bootstrap standard error of the sample median with $B = 999$ and the 95% percentile confidence interval.

c) Check the coverage of the percentile interval: repeat (a)–(b) 300 times with $B = 299$ and count how often the interval contains 1.

d) Do the same for the sample **maximum** as an estimator of the upper end of the support of a $U(0,1)$ distribution. What goes wrong?

3.8 Exercises

Problem Set 3 contains the analytical exercises for this chapter:

- **Problem 1**: Monte Carlo integration of $\int_0^1 xe^x\,dx$, the number of draws needed for a given precision, and antithetic draws.
- **Problem 2**: importance sampling for a normal tail probability.
- **Problem 3**: Gauss–Hermite quadrature and exactness for polynomials.
- **Problem 4**: MSL and MSM in the unobserved-heterogeneity model of Example 3.4, with an exponential instead of a Gumbel shock, for which the exact likelihood is available as a benchmark.

4 Nonparametric Methods

> **Readings**
>
> Cameron and Trivedi (2005), Ch. 9; B. E. Hansen (2022b), Ch. 17 (density estimation); B. E. Hansen (2022a), Ch. 19 (nonparametric regression); Li and Racine (2007), Ch. 1–2 and 7; Fan and Gijbels (1996), Ch. 2–3 (local polynomials).

The previous chapters assumed that the model is known up to a finite-dimensional parameter: a density $f(y \mid \boldsymbol{x}; \boldsymbol{\theta})$ for ML, a moment function $\boldsymbol{\psi}(\boldsymbol{w}, \boldsymbol{\theta})$ for GMM. This chapter drops that assumption for the two objects applied work cares about most, the density of a variable and the conditional mean of y given x. The price is a slower rate of convergence, a bandwidth that has to be chosen, and the curse of dimensionality. Semiparametric models, sketched at the end, keep some parametric structure to avoid the worst of these costs.

4.1 Parametric, nonparametric and semiparametric models

Much of applied econometrics estimates a conditional expectation function (CEF),

$$y_i = m(\boldsymbol{x}_i) + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid \boldsymbol{x}_i] = 0, \qquad m(\boldsymbol{x}) = \mathbb{E}[y_i \mid \boldsymbol{x}_i = \boldsymbol{x}].$$

There are three ways to model $m(\cdot)$.

- **Parametric:** $m(\boldsymbol{x}) = g(\boldsymbol{x}, \boldsymbol{\theta})$ with g known and $\boldsymbol{\theta}$ finite-dimensional, e.g. $\boldsymbol{x}'\boldsymbol{\beta}$ or $\Phi(\boldsymbol{x}'\boldsymbol{\beta})$. If g is correct, this is efficient, allows extrapolation outside the data and often has a behavioural interpretation. If g is wrong, the estimates converge to a pseudo-true value (Section 1.6) that may be far from anything of interest. Specification searches over many g's do not solve this: the correct model need not be among those tried, need not be selected, and the search invalidates the usual inference.
- **Nonparametric:** m is only assumed to be smooth. This minimizes the risk of misspecification and is useful for exploring data and for checking parametric models. The costs: slower convergence, results that are hard to display for more than two regressors, and precision that deteriorates quickly with the dimension of $\boldsymbol{x}$ (the **curse of dimensionality**, Section 4.3.4).
- **Semiparametric:** part of the model is parametric and part is not, for example $m(\boldsymbol{x}, z) = \boldsymbol{x}'\boldsymbol{\beta} + \lambda(z)$ with λ unknown (partially linear), or $m(\boldsymbol{x}) = g(\boldsymbol{x}'\boldsymbol{\beta})$ with g unknown (single index). Some authors also call a model semiparametric if the mean is parametric but the error distribution is left unspecified, which covers GMM.

4.2 Density estimation

Let $X_1, \ldots, X_n$ be i.i.d. with distribution function F and density f. A parametric estimator fits $f(x \mid \boldsymbol{\theta})$, e.g. $N(\mu, \sigma^2)$, which imposes symmetry, unimodality and thin tails; as Li and Racine

(2007) put it, the approach is somewhat circular, because we need to know the form of the density we want to estimate.

4.2.1 The histogram

Divide the support into bins $(b_j, b_{j+1}]$ of width $h = b_{j+1} - b_j$, $j = 1, \dots, k$. The empirical distribution function $\hat{F}(x) = n^{-1}\#\{i : X_i \le x\}$ estimates F consistently, and $f(x) = \lim_{h\to 0}\{F(b_{j+1}) - F(b_j)\}/h$ for $x \in (b_j, b_{j+1}]$. Replacing F by $\hat{F}$ and keeping h positive gives the histogram.

Definition 4.1 (Histogram estimator)**.** For $x \in (b_j, b_{j+1}]$,

$$\hat{f}(x) = \frac{\#\{i : X_i \in (b_j, b_{j+1}]\}}{nh} = \frac{1}{n}\sum_{i=1}^{n} W_{hi}(x), \qquad W_{hi}(x) = \begin{cases} 1/h & \text{if } X_i \text{ and } x \text{ are in the same bin,} \\ 0 & \text{otherwise.} \end{cases}$$

The histogram is a proper density: $\hat{f} \ge 0$, and $\int \hat{f}(x)\,dx = \sum_j h \cdot \#\{X_i \in \text{bin } j\}/(nh) = 1$.

Theorem 4.1 (Bias and variance of the histogram)**.** *Let f have a bounded second derivative and let $x \in (b_j, b_{j+1}]$. As $h \to 0$ and $nh \to \infty$,*

$$\mathbb{E}[\hat{f}(x)] - f(x) = \tfrac{1}{2}f'(x)\big[h - 2(x - b_j)\big] + O(h^2), \qquad \operatorname{Var}[\hat{f}(x)] = \frac{f(x)}{nh} + o\Big(\frac{1}{nh}\Big).$$

Proof

Bias. $\mathbb{E}[\hat{f}(x)] = \mathbb{E}[W_{hi}(x)] = h^{-1}\{F(b_{j+1}) - F(b_j)\}$. Expand $F(b) = F(x) + f(x)(b - x) + \frac{1}{2}f'(x)(b - x)^2 + R$ around x for $b = b_j$ and $b = b_{j+1}$. Both distances are at most h, so the remainders are $O(h^3)$. The terms $F(x)$ cancel, $f(x)(b_{j+1} - b_j)/h = f(x)$, and $(b_{j+1} - x)^2 - (b_j - x)^2 = h\{h - 2(x - b_j)\}$, which gives the result.
Variance. $nh\hat{f}(x)$ is binomial with success probability $p = F(b_{j+1}) - F(b_j) = hf(x) + O(h^2)$, so $\operatorname{Var}[\hat{f}(x)] = p(1 - p)/(nh^2) = f(x)/(nh) + O(1/n)$.

The bias is of order h, has the sign of $f'(x)$ in the left half of a bin and the opposite sign in the right half, and vanishes only at the bin centre. The variance is of order $1/(nh)$ and proportional to $f(x)$. A wider bin reduces the variance and increases the bias: this **bias-variance trade-off** is the central theme of the chapter. Since MSE $=$ Var $+$bias$^2 \to 0$ when $h \to 0$ and $nh \to \infty$, the histogram is consistent.

Table 4.1: Histogram of a Beta(2,4) density with $n = 1000$: Monte Carlo mean, bias and variance over 2000 samples, and the approximations of Theorem 4.1, at the deciles 0.1, 0.3, 0.5, 0.7 and 0.9 of the distribution. Bins on $[0, 1]$.

bins	x	$f(x)$	mean	bias	bias (Thm)	variance	$f/(nh)$
20	0.112	1.571	1.669	0.098	0.111	0.031	0.031
20	0.218	2.085	2.090	0.005	0.011	0.037	0.042
20	0.314	2.028	1.995	-0.032	-0.027	0.037	0.041
20	0.422	1.630	1.612	-0.018	-0.014	0.031	0.033
20	0.584	0.841	0.878	0.037	0.041	0.017	0.017
40	0.112	1.571	1.573	0.002	0.002	0.058	0.063

Table 4.1: Histogram of a Beta(2,4) density with $n = 1000$: Monte Carlo mean, bias and variance over 2000 samples, and the approximations of Theorem 4.1, at the deciles 0.1, 0.3, 0.5, 0.7 and 0.9 of the distribution. Bins on $[0, 1]$.

bins	x	$f(x)$	mean	bias	bias (Thm)	variance	$f/(nh)$
40	0.218	2.085	2.070	-0.016	-0.009	0.079	0.083
40	0.314	2.028	2.028	0.000	0.003	0.075	0.081
40	0.422	1.630	1.671	0.041	0.044	0.061	0.065
40	0.584	0.841	0.826	-0.016	-0.017	0.032	0.034

Table 4.1 shows the trade-off. Doubling the number of bins roughly doubles the variance everywhere. The bias does not fall at every point: at a given x it depends on where x happens to lie in its bin, and at $x = 0.422$ it even rises. What halves is the largest possible bias in a bin, $|f'(x)|h/2$. The approximations of Theorem 4.1 are accurate.

Optimal bin width. A global criterion is the integrated MSE. Integrating the variance gives $1/(nh)$; integrating the squared bias over a bin gives approximately $f'(x)^2 \int_0^h (h/2 - t)^2 dt/h = f'(x)^2 h^2/12$ per unit length. Hence

$$\mathrm{AIMSE}(h) = \frac{1}{nh} + \frac{h^2}{12} \int f'(x)^2 dx, \qquad h^* = \Big(\frac{6}{n \int f'(x)^2 dx} \Big)^{1/3} \propto n^{-1/3},$$

and the AIMSE at h^* is of order $n^{-2/3}$. For a normal reference density $\int f'^2 = 1/(4\sqrt{\pi}\sigma^3)$, so $h^* = (24\sqrt{\pi})^{1/3} \sigma n^{-1/3} \approx 3.49\, \sigma n^{-1/3}$; for an exponential with mean λ, $h^* \approx 2.29\, \lambda n^{-1/3}$.

Drawbacks. The histogram is a step function, depends on the location of the bin edges, has a bias of order h that is largest at the edges, and weights all observations in a bin equally, although those close to x carry more information about $f(x)$.

4.2.2 Kernel density estimation

The histogram uses the observations in a bin *containing* x. The kernel estimator centres the window *at* x. A histogram with a single bin $[x - h, x + h]$ is

$$\hat{f}(x) = \frac{1}{2nh} \#\{i : |X_i - x| \le h\} = \frac{1}{nh} \sum_{i=1}^{n} K\Big(\frac{X_i - x}{h} \Big), \qquad K(u) = \tfrac{1}{2} \mathbb{1}\{|u| \le 1\},$$

which suggests replacing the uniform weight K by one that declines smoothly with the distance from x.

Definition 4.2 (Kernel density estimator)**.**

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\Big(\frac{X_i - x}{h} \Big) = \frac{1}{n} \sum_{i=1}^{n} K_h(X_i - x), \qquad K_h(u) = \frac{1}{h} K\Big(\frac{u}{h} \Big),$$

where $h > 0$ is the **bandwidth** and the **kernel** K is a symmetric function with $\int K(u)\, du = 1$. A second-order kernel also satisfies $\mu_2(K) = \int u^2 K(u)\, du \in (0, \infty)$. We write $R_K = \int K(u)^2 du$ (the slides write $\|K\|_2^2$).

If $K \geq 0$, $\hat{f}$ is itself a density: an average of n small densities centred at the observations. Table 4.2 lists the common kernels.

Table 4.2: Second-order kernels. $\delta_0 = (R_K/\mu_2^2)^{1/5}$ is used in Equation 4.1.

Kernel	$K(u)$	$\mu_2(K)$	R_K	δ_0
Uniform	$\frac{1}{2}\mathbb{1}\{\lvert u\rvert \leq 1\}$	$1/3$	$1/2$	1.3510
Triangular	$(1 - \lvert u\rvert)\mathbb{1}\{\lvert u\rvert \leq 1\}$	$1/6$	$2/3$	1.8882
Epanechnikov	$\frac{3}{4}(1 - u^2)\mathbb{1}\{\lvert u\rvert \leq 1\}$	$1/5$	$3/5$	1.7188
Quartic (biweight)	$\frac{15}{16}(1 - u^2)^2\mathbb{1}\{\lvert u\rvert \leq 1\}$	$1/7$	$5/7$	2.0362
Gaussian	$\phi(u)$	1	$1/(2\sqrt{\pi})$	0.7764

Theorem 4.2 (Bias and variance of the kernel density estimator)**.** *Let f be twice continuously differentiable near x and K a second-order kernel. As $h \to 0$ and $nh \to \infty$,*

$$\mathbb{E}[\hat{f}(x)] - f(x) = \tfrac{1}{2}h^2\mu_2(K)f''(x) + o(h^2), \qquad \operatorname{Var}[\hat{f}(x)] = \frac{R_K f(x)}{nh} + o\Big(\frac{1}{nh}\Big).$$

Proof

Bias. Substituting $u = (t - x)/h$,

$$\mathbb{E}[\hat{f}(x)] = \int \frac{1}{h}K\Big(\frac{t-x}{h}\Big)f(t)\,dt = \int K(u)f(x + hu)\,du.$$

A second-order expansion $f(x + hu) = f(x) + huf'(x) + \frac{1}{2}h^2u^2f''(x) + o(h^2)$ and the kernel properties $\int K = 1$, $\int uK = 0$ (symmetry) and $\int u^2K = \mu_2$ give the result.
Variance. The summands are i.i.d., so $\operatorname{Var}[\hat{f}(x)] = n^{-1}\operatorname{Var}[K_h(X_i - x)] = n^{-1}\{\mathbb{E}[K_h(X_i - x)^2] - (\mathbb{E}[K_h(X_i - x)])^2\}$. The same substitution gives $\mathbb{E}[K_h^2] = h^{-1}\int K(u)^2 f(x + hu)\,du = h^{-1}R_K f(x) + O(1)$, and $(\mathbb{E}[K_h])^2 = O(1)$.

Compared with the histogram, the bias is of order h^2 instead of h and depends on the curvature f'', not the slope: the estimator underestimates peaks ($f'' < 0$) and overestimates troughs ($f'' > 0$). The variance has the same order $1/(nh)$ and increases with R_K.

Optimal bandwidth. Integrating the leading terms of the MSE gives the asymptotic mean integrated squared error

$$\operatorname{AMISE}(h) = \frac{R_K}{nh} + \frac{h^4}{4}\mu_2(K)^2 R(f''), \qquad R(f'') = \int f''(x)^2 dx.$$

Setting the derivative to zero,

$$h_{opt} = \Big(\frac{R_K}{\mu_2(K)^2 R(f'')\,n}\Big)^{1/5} \propto n^{-1/5}, \qquad \operatorname{AMISE}(h_{opt}) = \tfrac{5}{4}\big(\mu_2(K)^2 R(f'')\big)^{1/5} R_K^{4/5}\, n^{-4/5}.$$

The kernel estimator converges at rate $n^{-4/5}$ in MISE, faster than the histogram ($n^{-2/3}$), but slower than a correctly specified parametric model (n^{-1}). The optimal bandwidth depends on the kernel, the sample size and the unknown curvature $R(f'')$.

Rule of thumb. If f is $N(\mu, \sigma^2)$, then $R(f'') = 3/(8\sqrt{\pi}\sigma^5) \approx 0.212\sigma^{-5}$ and

$$h_{opt} = \delta_0 \Big(\frac{8\sqrt{\pi}}{3}\Big)^{1/5} \sigma n^{-1/5} \approx 1.3643\, \delta_0\, \hat{\sigma}\, n^{-1/5}, \qquad \hat{\sigma} = \min\Big\{s, \frac{\mathrm{IQR}}{1.349}\Big\}, \tag{4.1}$$

with δ_0 from Table 4.2; for the Gaussian kernel $h \approx 1.06\,\hat{\sigma} n^{-1/5}$. Using the smaller of the standard deviation and the scaled interquartile range protects against outliers and skewness (Silverman 1986). The rule works well for unimodal, roughly symmetric densities and oversmooths multimodal ones.

Least-squares cross-validation. Choose h to minimize an estimate of the integrated squared error $\int (\hat{f}_h - f)^2 = \int \hat{f}_h^2 - 2\int \hat{f}_h f + \int f^2$. The last term does not depend on h; $\int \hat{f}_h f = \mathbb{E}[\hat{f}_h(X)]$ is estimated by the leave-one-out average $n^{-1}\sum_i \hat{f}_{-i}(X_i)$, where $\hat{f}_{-i}$ omits observation i. This gives

$$\mathrm{CV}(h) = \frac{1}{n^2 h} \sum_{i=1}^{n}\sum_{j=1}^{n} \bar{K}\Big(\frac{X_i - X_j}{h}\Big) - \frac{2}{n}\sum_{i=1}^{n} \hat{f}_{-i}(X_i), \qquad \bar{K}(u) = \int K(u-t)K(t)\,dt.$$

For the Gaussian kernel $\bar{K}$ is the $N(0, 2)$ density. Cross-validation makes no reference-density assumption but is noisy in small samples.

Choice of kernel. It matters much less than the bandwidth. The Epanechnikov kernel minimizes the AMISE at the optimal bandwidth, but the efficiency loss of the others is a few percent. To give two kernels the same amount of smoothing, scale the bandwidths in proportion to δ_0: $h_B = (\delta_0^B/\delta_0^A)h_A$.

> ⚠ Bandwidths in R
>
> `density()` in R parametrizes every kernel by its *standard deviation* `bw`, not by the half-width of its support. For the Epanechnikov kernel on $[-1, 1]$, which has standard deviation $1/\sqrt{5}$, `bw` $= h/\sqrt{5}$. The default `bw.nrd0` is $0.9\hat{\sigma}n^{-1/5}$, slightly smaller than the Gaussian rule $1.06\hat{\sigma}n^{-1/5}$.

The rule-of-thumb bandwidth here is $h_{ROT} = 0.545$. Halving it produces spurious bumps; doubling it flattens the peak and spreads mass into the tails (Figure 4.1).

Inference. Under the conditions of Theorem 4.2, $\sqrt{nh}\,\{\hat{f}(x) - f(x) - \text{bias}\} \xrightarrow{d} N(0, R_K f(x))$. With the MSE-optimal $h \propto n^{-1/5}$, $\sqrt{nh}\cdot h^2$ converges to a constant, so the bias does not vanish relative to the standard error and the interval $\hat{f}(x) \pm 1.96\sqrt{R_K \hat{f}(x)/(nh)}$ is not centred at $f(x)$. In practice one either **undersmooths** (uses a bandwidth smaller than the MSE-optimal one, so that the bias is negligible) or bootstraps; the resulting bands are pointwise, not uniform in x.

4.3 Nonparametric regression

Let (y_i, x_i) be i.i.d. with $y_i = m(x_i) + \varepsilon_i$, $\mathbb{E}[\varepsilon_i \mid x_i] = 0$ and $\mathrm{Var}(\varepsilon_i \mid x_i) = \sigma^2(x_i)$; x is scalar until Section 4.3.4. With a discrete regressor, $m(x_0)$ is estimated by the average of y_i among observations with $x_i = x_0$. With many values or a continuous regressor, we average over observations with x_i *close to* x_0.

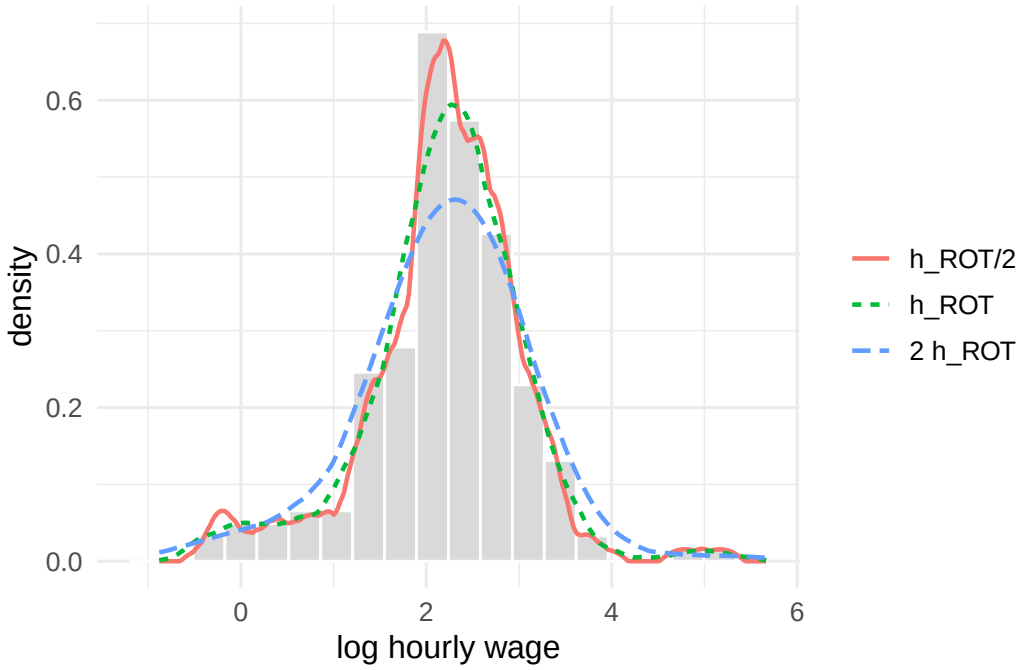


Figure 4.1: Log hourly wage of 177 women aged 36 (PSID; Cameron and Trivedi (2005), Ch. 9): histogram and Epanechnikov kernel estimates with the rule-of-thumb bandwidth h_{ROT} from Equation 4.1, half and twice of it.

4.3.1 The Nadaraya–Watson estimator

Definition 4.3 (Nadaraya–Watson estimator (Nadaraya 1964; Watson 1964))**.**

$$\hat{m}(x) = \frac{\sum_{i=1}^{n} K_h(x_i - x)\, y_i}{\sum_{i=1}^{n} K_h(x_i - x)} = \sum_{i=1}^{n} w_i(x)\, y_i, \qquad w_i(x) = \frac{K_h(x_i - x)}{\sum_{j=1}^{n} K_h(x_j - x)}.$$

With the uniform kernel this is the average of y_i over $|x_i - x| \le h$. The weights are nonnegative (for $K \ge 0$) and sum to one, so $\hat{m}(x)$ is a **local weighted average** and a **linear smoother**: linear in the y_i. Three equivalent views help:

1. **Plug-in:** $m(x) = \int y f(x, y)\, dy / f(x)$. Replacing the joint and the marginal density by kernel estimates with a product kernel gives exactly Definition 4.3: the numerator is an estimate of $\int y f(x, y)\, dy$ and the denominator, divided by n, is the kernel density estimate $\hat{f}(x)$.
2. **Local constant:** $\hat{m}(x) = \arg\min_\theta \sum_i K_h(x_i - x)(y_i - \theta)^2$, a weighted least squares fit of a constant.
3. **Local averaging with shrinking windows:** consistency requires $h \to 0$ (average ever closer to x, so the bias vanishes) and $nh \to \infty$ (the window contains ever more observations, so the variance vanishes).

Theorem 4.3 (Bias and variance of the NW estimator)**.** *Let x be an interior point of the support with $f(x) > 0$, let m and f be twice continuously differentiable and $\sigma^2(\cdot)$ continuous near x. As $h \to 0$ and $nh \to \infty$,*

$$\mathbb{E}[\hat{m}(x) \mid x_1, \ldots, x_n] - m(x) \approx h^2 \mu_2(K) \Big(\tfrac{1}{2} m''(x) + \frac{m'(x) f'(x)}{f(x)} \Big), \qquad \operatorname{Var}[\hat{m}(x) \mid x_1, \ldots, x_n] \approx \frac{R_K\, \sigma^2(x)}{nh\, f(x)}.$$

The variance is large where the data are sparse ($f(x)$ small) or noisy ($\sigma^2(x)$ large). The bias has two parts. The first, $\frac{1}{2}m''$, reflects curvature. The second, $m'f'/f$, is a **design bias**: if m is increasing and the density of x is increasing too, the window contains more observations to the right of x, where m is larger, and the local average is pulled up.

4.3.2 Local polynomial regression

NW fits a constant in a neighbourhood of x. A Taylor expansion $m(x_i) \approx m(x) + m'(x)(x_i - x) + \cdots + m^{(p)}(x)(x_i - x)^p/p!$ suggests fitting a polynomial instead.

Definition 4.4 (Local polynomial estimator)**.**

$$\hat{\boldsymbol{\theta}}(x) = \arg\min_{\theta_0,\ldots,\theta_p} \sum_{i=1}^{n} K_h(x_i - x)\Big[y_i - \theta_0 - \theta_1(x_i - x) - \cdots - \theta_p(x_i - x)^p\Big]^2,$$

and $\hat{m}(x) = \hat{\theta}_0(x)$; $k!\,\hat{\theta}_k(x)$ estimates the derivative $m^{(k)}(x)$. In matrix form, with $\boldsymbol{Z}_x$ the $n \times (p+1)$ matrix with rows $(1, x_i - x, \ldots, (x_i - x)^p)$ and $\boldsymbol{K}_x = \operatorname{diag}\{K_h(x_i - x)\}$, $\hat{\boldsymbol{\theta}}(x) = (\boldsymbol{Z}_x'\boldsymbol{K}_x\boldsymbol{Z}_x)^{-1}\boldsymbol{Z}_x'\boldsymbol{K}_x\boldsymbol{y}$.

$p = 0$ is NW and $p = 1$ the **local linear** (LL) estimator. For LL at an interior point,

$$\text{bias} \approx \tfrac{1}{2}h^2\mu_2(K)\,m''(x), \qquad \text{Var} \approx \frac{R_K\sigma^2(x)}{nhf(x)},$$

the same variance as NW but without the design bias (Fan and Gijbels 1996). More generally, the bias of odd-order fits does not depend on f, and going from p even to $p+1$ removes a bias term without increasing the asymptotic variance. This is why $p = 1$ (and sometimes $p = 3$) is recommended.

Boundaries. Near the edge of the support the kernel window is one-sided. NW then averages only observations on one side, and its bias is of order h rather than h^2. The local linear fit extrapolates the local slope and keeps a bias of order h^2 at the boundary.

Table 4.3: Bias and standard deviation of NW and local linear estimates at an interior point ($x = 0.5$) and at the boundary ($x = 0$) in the design of Figure 4.2, 1000 samples.

estimator	point	bias	sd
Nadaraya-Watson	interior x = 0.5	0.001	0.061
Nadaraya-Watson	boundary x = 0	0.499	0.090
local linear	interior x = 0.5	0.000	0.054
local linear	boundary x = 0	0.045	0.143

At the interior point both estimators are nearly unbiased, with similar standard deviations. At the boundary the NW bias is large, while the local linear estimator stays nearly unbiased at the cost of a larger variance (Table 4.3).

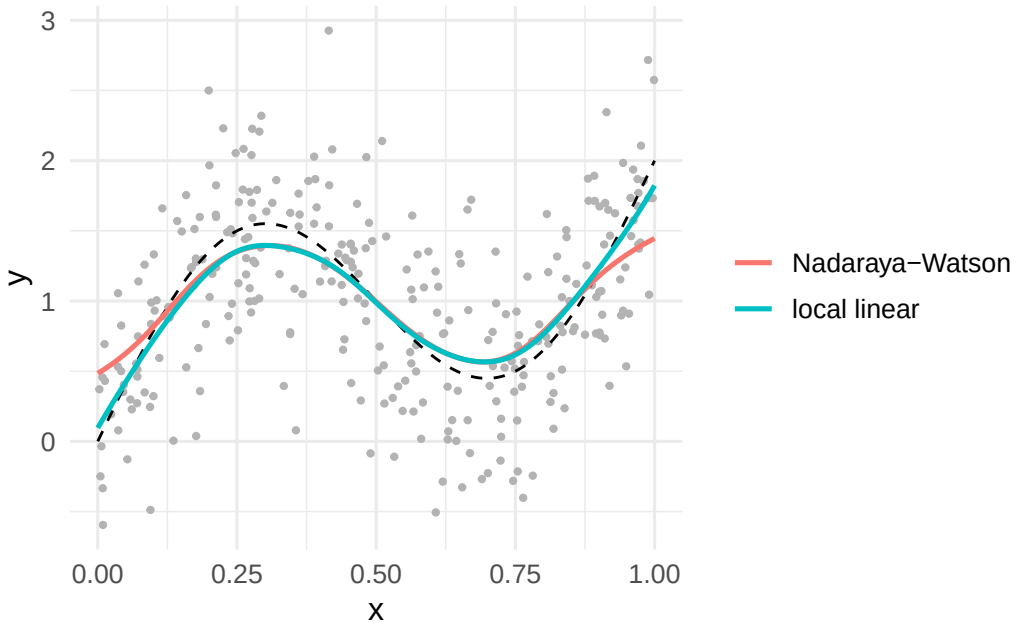


Figure 4.2: Nadaraya–Watson and local linear fits (Gaussian kernel, $h = 0.08$) to $n = 300$ observations from $y = \sin(2\pi x) + 2x + \varepsilon$, $x \sim U(0,1)$, $\varepsilon \sim N(0, 0.5^2)$. The true m is dashed. NW is biased towards the interior at both boundaries.

4.3.3 Bandwidth choice and inference

The integrated MSE of the local linear estimator over a region $[\xi_1, \xi_2]$ with weight $w(x) = \mathbb{1}\{\xi_1 \le x \le \xi_2\}$ is approximately

$$\operatorname{AIMSE}(h) = h^4 \mu_2(K)^2 \bar{B} + \frac{R_K \bar{\sigma}^2}{nh}, \qquad \bar{B} = \int \left(\tfrac{1}{2} m''(x)\right)^2 f(x) w(x)\, dx, \quad \bar{\sigma}^2 = \int \sigma^2(x) w(x)\, dx,$$

minimized by $h_0 = \left(R_K \bar{\sigma}^2 / (4 \mu_2(K)^2 \bar{B})\right)^{1/5} n^{-1/5}$.

Rule of thumb (Fan and Gijbels 1996, sec. 4.2; B. E. Hansen 2022a, sec. 19.9). Fit a global polynomial of order four by OLS, compute $\hat{m}''(x)$ from it, estimate $\bar{B}$ by $\frac{1}{n}\sum_i (\frac{1}{2}\hat{m}''(x_i))^2 w(x_i)$ and $\bar{\sigma}^2$ by $\hat{\sigma}^2(\xi_2 - \xi_1)$ with $\hat{\sigma}^2$ the residual variance of the polynomial fit (assuming homoskedasticity). For the Gaussian kernel $(R_K/4)^{1/5} \approx 0.58$, so

$$h_{ROT} = 0.58 \Big(\frac{\hat{\sigma}^2 (\xi_2 - \xi_1)}{n \hat{\bar{B}}} \Big)^{1/5}.$$

Cross-validation. Choose h to minimize the leave-one-out prediction error

$$\operatorname{CV}(h) = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \hat{m}_{-i}(x_i)\right)^2 w(x_i),$$

where $\hat{m}_{-i}$ is computed without observation i. Leaving i out is essential: with $\hat{m}(x_i)$ itself the criterion would be minimized by $h \to 0$, which interpolates the data.

Standard errors and bands. The local linear estimator is a weighted least squares coefficient, so a heteroskedasticity-robust variance is the sandwich

$$\widehat{\operatorname{Var}}[\hat{\boldsymbol{\theta}}(x)] = (\boldsymbol{Z}_x' \boldsymbol{K}_x \boldsymbol{Z}_x)^{-1} \Big(\sum_i K_h(x_i - x)^2 \, \boldsymbol{z}_i \boldsymbol{z}_i' \hat{\varepsilon}_i^2 \Big) (\boldsymbol{Z}_x' \boldsymbol{K}_x \boldsymbol{Z}_x)^{-1},$$

with $\boldsymbol{z}_i = (1, x_i - x)'$ and $\hat{\varepsilon}_i$ leave-one-out residuals, and $\hat{m}(x) \pm 1.96\,\mathrm{se}$ is a pointwise 95% interval. Alternatively, $\sigma^2(x)$ can be estimated by a nonparametric regression of $\hat{\varepsilon}_i^2$ on x_i and plugged into Theorem 4.3. As for densities, the interval is centred at $\mathbb{E}[\hat{m}(x)]$, not $m(x)$, unless one undersmooths; it is pointwise; and the bootstrap (resampling pairs (y_i, x_i)) is an alternative.

Nearest neighbours. The k-NN estimator averages y over the k observations with x_i closest to x: a uniform kernel with a bandwidth that adapts to the local density of the data, wide where data are sparse and narrow where they are dense. The number of neighbours k plays the role of nh.

4.3.4 The curse of dimensionality

All estimators extend to $\boldsymbol{x} \in \mathbb{R}^d$ with a multivariate (usually product) kernel, e.g. $\hat{m}(\boldsymbol{x}) = \sum_i \prod_{l=1}^{d} K_h(x_{il} - x_l) y_i / \sum_i \prod_l K_h(x_{il} - x_l)$. The window now has volume h^d, so the variance is of order $1/(nh^d)$ while the bias remains of order h^2. Balancing the two gives $h \propto n^{-1/(4+d)}$ and an optimal MSE of order $n^{-4/(4+d)}$.

Table 4.4: Sample size needed in d dimensions to reach the optimal MSE rate that $n = 100$ gives in one dimension, $n_d = 100^{(4+d)/5}$.

d	MSE rate	n_d
1	n^(-4/5)	100
2	n^(-4/6)	251
3	n^(-4/7)	631
5	n^(-4/9)	3,981
10	n^(-4/14)	398,107

With ten continuous regressors, about 400,000 observations deliver the precision that one hundred deliver with one regressor (Table 4.4). In addition, functions of more than two arguments are hard to display. This motivates restricting the model.

4.4 Semiparametric models

Semiparametric models keep a finite-dimensional parameter of interest and leave a function unspecified. This section gives the main ideas; Li and Racine (2007) (Ch. 7–8) and B. E. Hansen (2022a) (Sec. 19.24) treat them in detail.

4.4.1 Partially linear model

$$y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + \lambda(z_i) + u_i, \qquad \mathbb{E}[u_i \mid \boldsymbol{x}_i, z_i] = 0,$$

with λ unknown. Typically $\boldsymbol{x}_i$ contains the variables of interest and z_i a control whose effect is not of interest, e.g. a running variable, age or a location. Taking expectations conditional on z_i and subtracting,

$$\underbrace{y_i - \mathbb{E}[y_i \mid z_i]}_{\tilde{y}_i} = \underbrace{(\boldsymbol{x}_i - \mathbb{E}[\boldsymbol{x}_i \mid z_i])'}_{\tilde{\boldsymbol{x}}_i'}\boldsymbol{\beta} + u_i,$$

because $\lambda(z_i) - \mathbb{E}[\lambda(z_i) \mid z_i] = 0$. The nuisance function is gone.

Definition 4.5 (Robinson's estimator (Robinson 1988))**.**

1. Estimate $\mathbb{E}[y_i \mid z_i]$ and each element of $\mathbb{E}[\boldsymbol{x}_i \mid z_i]$ by nonparametric regressions on z_i; form residuals $\hat{\tilde{y}}_i$ and $\hat{\tilde{\boldsymbol{x}}}_i$.
2. Regress $\hat{\tilde{y}}_i$ on $\hat{\tilde{\boldsymbol{x}}}_i$ by OLS without a constant: $\hat{\boldsymbol{\beta}} = \big(\sum_i \hat{\tilde{\boldsymbol{x}}}_i \hat{\tilde{\boldsymbol{x}}}_i'\big)^{-1} \sum_i \hat{\tilde{\boldsymbol{x}}}_i \hat{\tilde{y}}_i$.
3. If needed, $\hat{\lambda}(z) = \hat{\mathbb{E}}[y \mid z] - \hat{\mathbb{E}}[\boldsymbol{x} \mid z]'\hat{\boldsymbol{\beta}}$.

Under regularity conditions (enough smoothness, suitable bandwidths, trimming where z is sparse), $\hat{\boldsymbol{\beta}}$ is $\sqrt{n}$-consistent and asymptotically normal with the variance it would have if the conditional means were known:

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} N\Big(\mathbf{0},\ \mathbb{E}[\tilde{\boldsymbol{x}}_i \tilde{\boldsymbol{x}}_i']^{-1} \mathbb{E}[u_i^2 \tilde{\boldsymbol{x}}_i \tilde{\boldsymbol{x}}_i'] \mathbb{E}[\tilde{\boldsymbol{x}}_i \tilde{\boldsymbol{x}}_i']^{-1}\Big).$$

The first-step nonparametric estimates converge slowly, but their errors enter the moment condition for $\boldsymbol{\beta}$ only as products, which vanish faster than $n^{-1/2}$. This is the same orthogonality idea that underlies double/debiased machine learning, which replaces the kernel regressions by machine-learning predictions with cross-fitting (Chapter 5). $\boldsymbol{\beta}$ is identified only if $\mathbb{E}[\tilde{\boldsymbol{x}}_i \tilde{\boldsymbol{x}}_i']$ is nonsingular: $\boldsymbol{x}$ must not be a function of z, and it cannot contain a constant (absorbed by λ).

Table 4.5: Partially linear model $y = \beta x + \lambda(z) + u$ with $\beta = 1$, $\lambda(z) = \sin(2z) + z^2$, $x = z^2 + v$: OLS of y on $(1, x, z)$ and Robinson's estimator (local linear first steps), $n = 500$, 500 samples.

estimator	bias	sd
OLS on (1, x, z)	0.661	0.057
Robinson	0.002	0.043

Controlling for z linearly leaves the nonlinear part of $\lambda(z)$ in the error. Because x depends nonlinearly on z too, this part is correlated with x, and OLS is severely biased. Robinson's estimator removes the bias at a small cost in precision (Table 4.5).

4.4.2 Single-index models

$$\mathbb{E}[y_i \mid \boldsymbol{x}_i] = g(\boldsymbol{x}_i'\boldsymbol{\beta}), \qquad g \text{ unknown.}$$

Probit and logit are single-index models with $g = \Phi$ or Λ; here g is estimated from the data. The index reduces the d-dimensional problem to a one-dimensional one.

Identification. For any a and $b \neq 0$, $g(\boldsymbol{x}'\boldsymbol{\beta}) = \tilde{g}(a + b\,\boldsymbol{x}'\boldsymbol{\beta})$ with $\tilde{g}(v) = g((v - a)/b)$, so $\boldsymbol{\beta}$ is identified only up to location and scale. One therefore excludes the constant from $\boldsymbol{x}$ and normalizes one coefficient to one. $\boldsymbol{x}$ must then contain at least two variables (otherwise nothing is left to estimate), and at least one with a continuous distribution (otherwise the index takes finitely many values and g can fit any function of them).

Semiparametric least squares (Ichimura 1993). If g were known, $\boldsymbol{\beta}$ would be estimated by nonlinear least squares. Ichimura replaces g by a leave-one-out kernel regression of y on the index:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \big(y_i - \hat{g}_{-i}(\boldsymbol{x}_i'\boldsymbol{\beta})\big)^2 \tau_i,$$

where τ_i trims observations where the density of the index is small. The objective is not smooth in $\boldsymbol{\beta}$ and often has several local minima, so good starting values matter. The estimator is $\sqrt{n}$-consistent and asymptotically normal.

Binary choice (Klein and Spady 1993). For binary y, $g(v) = P(y = 1 \mid \boldsymbol{x}'\boldsymbol{\beta} = v)$. By Bayes' rule,

$$g(v) = \frac{P(y=1)\, f(v \mid y=1)}{P(y=1)\, f(v \mid y=1) + P(y=0)\, f(v \mid y=0)},$$

which is estimated with kernel density estimates of the index in the two groups. Klein and Spady maximize the binary log-likelihood $\sum_i [y_i \ln \hat{g}(\boldsymbol{x}_i'\boldsymbol{\beta}) + (1-y_i)\ln(1-\hat{g}(\boldsymbol{x}_i'\boldsymbol{\beta}))]$ with $\hat{g}$ in place of Φ or Λ. With the same normalization as above, the estimator is $\sqrt{n}$-consistent, asymptotically normal, and attains the semiparametric efficiency bound when the error is independent of $\boldsymbol{x}$.

4.5 Computer tutorial 4

Learning goals. Program a kernel density estimator and a local polynomial regression from scratch and check them against built-in functions; compute rule-of-thumb and cross-validated bandwidths and see how sensitive the estimates are to them; build pointwise confidence bands; estimate a partially linear model with Robinson's estimator.

Packages. Base R and `ggplot2`; optionally `KernSmooth`. **Data.** `psidf3050.csv` (Cameron and Trivedi 2005, Ch. 9): 177 women aged 36 from the PSID, with the log hourly wage `lnhwage`. `cps09mar.csv` (B. E. Hansen 2022a): 50,742 respondents of the March 2009 Current Population Survey, with `age`, `female`, `education` (years), `earnings` (annual), `hours` (per week), `week` (weeks worked) and `race` (2 = Black).

4.5.1 Problem 1: Kernel density estimation

a) Write a function `kde(x0, x, h)` that returns the Epanechnikov kernel density estimate at the points `x0` (Definition 4.2 of the lecture notes). Estimate the density of `lnhwage` on a grid of 200 points with Silverman's rule-of-thumb bandwidth, $h = 1.3643\, \delta_0\, \hat{\sigma} n^{-1/5}$ with $\hat{\sigma} = \min\{s, \mathrm{IQR}/1.349\}$.

b) Reproduce your estimate with `density()`. Which value of `bw` do you need, and why is it not h?

c) Compute the least-squares cross-validation criterion for the Gaussian kernel on a grid of bandwidths and find its minimizer. Compare with `bw.ucv()` and with the Gaussian rule of thumb `bw.nrd()`. Plot the density estimates for the three bandwidths.

d) Add the normal density with the sample mean and standard deviation of `lnhwage` to the plot. What features of the data does it miss?

4.5.2 Problem 2: Wage–experience profiles

We study $m(x) = \mathbb{E}[\log \text{wage} \mid \text{experience} = x]$ with the hourly wage `earnings / (hours * week)` and potential experience `age - education - 6`.

a) Take the subsample of Black women with 12 years of education (high school graduates), and restrict attention to experience in $[0, 40]$, where about 90% of the observations lie.

   i) Compute the rule-of-thumb bandwidth for the local linear estimator with a Gaussian kernel (Section 4.3.3 of the lecture notes): fit a fourth-order polynomial, compute $\hat{\bar{B}}$ and $\hat{\sigma}^2$, and use $h_{ROT} = 0.58\{\hat{\sigma}^2(\xi_2 - \xi_1)/(n\hat{\bar{B}})\}^{1/5}$.

ii) Find the bandwidth that minimizes the leave-one-out cross-validation criterion of the local linear estimator on a grid from $h_{ROT}/3$ to $3h_{ROT}$, counting only observations with experience in $[0, 40]$.
iii) Plot the local linear estimates with both bandwidths.
iv) Add pointwise 95% confidence bands to the estimate with h_{ROT}, using the heteroskedasticity-robust sandwich with leave-one-out residuals.

b) Now take everyone with 20 years of education (professional degree or doctorate) and experience between 0 and 40.

i) Estimate the regression separately for men and women with the Nadaraya–Watson estimator and plot it with 95% bands. Use the rule-of-thumb bandwidth of each group.
ii) Repeat with the local linear estimator. How do the estimates and bands change, in particular near 0 and 40 years?
iii) Does the evidence suggest that expected wages fall with experience beyond 20 years in this group?

4.5.3 Problem 3: A partially linear wage equation

Take everyone with 16 years of education (bachelor's degree) and experience between 0 and 40, and consider

$$\log \text{wage}_i = \beta\, \texttt{female}_i + \lambda(\text{experience}_i) + u_i.$$

a) Estimate β by OLS with λ linear and with λ quadratic in experience.

b) Estimate β with Robinson's estimator (Section 4.4.1 of the lecture notes), using local linear regressions on experience with the rule-of-thumb bandwidth. Experience takes only integer values: use this to save computing time.

c) Compute a heteroskedasticity-robust standard error for $\hat{\beta}$ and compare the estimates. Why are they so similar here?

4.6 Exercises

Problem Set 4 contains the analytical exercises for this chapter:

- **Problem 1**: the histogram for the linear density $f(x) = 2x$: exact bias and variance, integrated MSE and the optimal bin width, compared with the normal rule of thumb.
- **Problem 2**: the kernel density estimator of a normal density: exact and approximate bias, where the bias vanishes, the optimal bandwidth and Silverman's rule.
- **Problem 3**: Nadaraya–Watson and local linear estimates by hand with a uniform kernel.
- **Problem 4**: the partially linear model: identification, Robinson's estimator with known conditional means, the binary-z case, and Yatchew's differencing estimator.

5 Introduction to Statistical Learning

Readings

James et al. (2023), Ch. 2, 5, 6 and 8; B. E. Hansen (2022a), Ch. 28–29; Belloni, Chernozhukov, and Hansen (2014); Chernozhukov et al. (2018).

Econometrics is mostly about estimating parameters with a causal or structural interpretation. Statistical (or machine) learning is mostly about **prediction**: given a new $\boldsymbol{x}_0$, predict y_0 as accurately as possible. The two goals call for different tools. For prediction, a biased estimator is welcome if it reduces the variance enough, and a model is judged by how it performs on data it has not seen. This chapter introduces the basic ideas: the bias–variance trade-off for prediction, model selection by penalties and cross-validation, and shrinkage estimators (ridge, lasso, elastic net). Two short outlook sections sketch tree-based methods and the use of the lasso for causal inference.

5.1 Prediction and the bias–variance trade-off

Terminology. In **supervised learning** we observe an outcome y and **features** (regressors) $\boldsymbol{x}$; it is called *regression* if y is continuous and *classification* if y is categorical. In **unsupervised learning** there is no outcome, and the aim is to find structure in $\boldsymbol{x}$ (e.g. clusters). We focus on regression. The **training sample** $\{(y_i, \boldsymbol{x}_i)\}_{i=1}^n$ is used to fit a prediction rule $\hat{f}(\boldsymbol{x})$; a **test sample** of new observations $(y_0, \boldsymbol{x}_0)$ from the same distribution is used to judge it.

Let $y = f(\boldsymbol{x}) + u$ with $\mathbb{E}[u \mid \boldsymbol{x}] = 0$ and $\operatorname{Var}(u \mid \boldsymbol{x}) = \sigma^2$, so that $f(\boldsymbol{x}) = \mathbb{E}[y \mid \boldsymbol{x}]$ is the best predictor under squared loss.

Theorem 5.1 (Bias–variance decomposition of the prediction error)**.** *For a test point $(y_0, \boldsymbol{x}_0)$ independent of the training sample,*

$$\mathbb{E}[(y_0 - \hat{f}(\boldsymbol{x}_0))^2 \mid \boldsymbol{x}_0] = \underbrace{\operatorname{Var}\left[\hat{f}(\boldsymbol{x}_0)\right]}_{\text{variance}} + \underbrace{\left\{\mathbb{E}[\hat{f}(\boldsymbol{x}_0)] - f(\boldsymbol{x}_0)\right\}^2}_{\text{bias}^2} + \underbrace{\sigma^2}_{\text{irreducible}},$$

where $\mathbb{E}$ and Var *are over training samples given $\boldsymbol{x}_0$.*

Proof

Write $y_0 - \hat{f} = (y_0 - f) + (f - \mathbb{E}\hat{f}) + (\mathbb{E}\hat{f} - \hat{f})$, with $f = f(\boldsymbol{x}_0)$ and $\hat{f} = \hat{f}(\boldsymbol{x}_0)$. The first term is u_0, which has mean zero and is independent of the training sample; the third has mean zero; the second is a constant. Squaring and taking expectations, all cross products vanish.

The irreducible error σ^2 cannot be reduced by any method. A more flexible method (more regressors, a smaller bandwidth, a smaller penalty) has smaller bias and larger variance. The best predictor balances the two, exactly as the MSE-optimal bandwidth in Chapter 4.

In-sample fit is optimistic. The training MSE $n^{-1}\sum_i (y_i - \hat{f}(\boldsymbol{x}_i))^2$ uses the same data to fit and to evaluate. It falls monotonically as the model becomes more flexible, while the test MSE eventually rises (Figure 5.1). For OLS with k regressors and fixed design, $\mathbb{E}[\text{training MSE}] = \sigma^2(n-k)/n$, whereas the expected squared error for new outcomes at the same $\boldsymbol{x}_i$ is $\sigma^2(n+k)/n$: the **optimism** is $2k\sigma^2/n$ (Problem Set 5, Problem 1). There are two remedies: penalize the in-sample fit for model size, or estimate the test error directly on data not used for fitting.

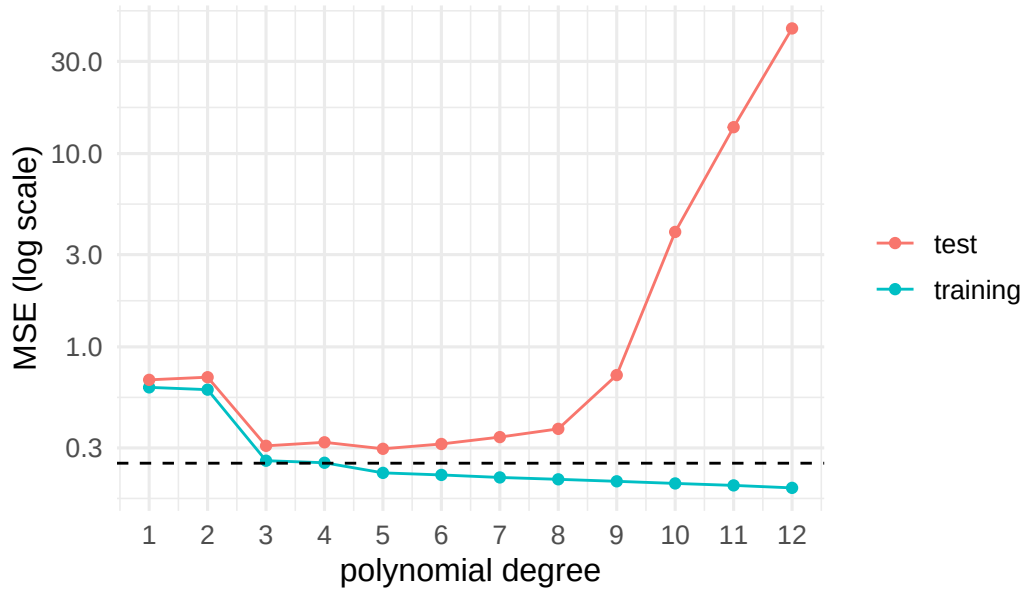


Figure 5.1: Training and test MSE of polynomial regressions of degree 1 to 12, $n = 50$, $y = \sin(2x) + x + u$, $x \sim U(-2, 2)$, $u \sim N(0, 0.5^2)$; averages over 200 samples (the test averages for high degrees are driven by a few samples with wild extrapolation near ± 2). The dashed line is the irreducible error $\sigma^2 = 0.25$.

5.2 Model selection

5.2.1 Penalized fit criteria

For a model with k parameters and maximized log-likelihood $\ln \hat{L}$,

$$\text{AIC} = -2\ln\hat{L} + 2k, \qquad \text{BIC} = -2\ln\hat{L} + k\ln n,$$

and smaller values are preferred. For the normal linear regression model with $\hat{\sigma}^2 = \text{MSE} = n^{-1}\sum_i \hat{u}_i^2$, $-2\ln\hat{L} = n\ln\text{MSE} + n(1 + \ln 2\pi)$, so up to a constant $\text{AIC} = n\ln\text{MSE} + 2k$ and $\text{BIC} = n\ln\text{MSE} + k\ln n$. Two related measures for OLS:

- **Adjusted** R^2, $\bar{R}^2 = 1 - \dfrac{\sum_i \hat{u}_i^2/(n-k)}{\sum_i (y_i - \bar{y})^2/(n-1)}$ (larger is better). In a comparison of nested models it prefers the larger model if the F statistic for the extra regressors exceeds one, a weak penalty.
- **Mallows'** C_p, $C_p = \dfrac{n\,\text{MSE}}{\tilde{\sigma}^2} - n + 2k$, where $\tilde{\sigma}^2$ is the unbiased error variance estimate from the *largest* model considered. C_p estimates the scaled prediction error $(n/\sigma^2)\mathbb{E}[\text{test MSE}] - n$ and is close to k for a correct model.

AIC approximates the expected out-of-sample log-likelihood (Kullback–Leibler) and tends to choose models that predict well, sometimes too large ones. BIC approximates the posterior

probability of a model; its penalty grows with n, and it selects the true model with probability approaching one if it is among the candidates (it is *consistent*), at the risk of underfitting in small samples. All three are derived for correctly specified, homoskedastic models.

Searching over models. With p candidate regressors there are 2^p subsets. *Best subset selection* fits all of them (feasible up to $p \approx 40$ with leaps-and-bounds algorithms); *forward selection* starts from the intercept and adds the variable that improves the fit most, and *backward selection* starts from the full model; both need at most $p(p+1)/2$ fits. Inference after any such search is problematic: the usual standard errors ignore that the model was chosen on the same data.

5.2.2 Cross-validation

Cross-validation estimates the test error directly by fitting on part of the data and predicting the rest.

- **Validation set.** Split the sample once into a training and a validation part. Simple, but the estimate depends strongly on the split and uses only part of the data for fitting.
- **K-fold cross-validation.** Split the sample into K folds of roughly equal size. For $j = 1, \dots, K$ fit on all folds except j and compute the MSE on fold j; then $\mathrm{CV}_K = K^{-1} \sum_{j=1}^{K} \mathrm{MSE}_j$. $K = 5$ or 10 is the usual compromise: each fit uses most of the data (small bias), and the K fits are not too similar (moderate variance).
- **Leave-one-out (LOOCV)**, $K = n$: $\mathrm{CV}_n = n^{-1} \sum_i \left(y_i - \hat{y}_{(-i)}\right)^2$, where $\hat{y}_{(-i)}$ is the prediction for observation i from a fit without it. This is the criterion used for bandwidths in Section 4.3.

Theorem 5.2 (LOOCV for linear smoothers)**.** *Let the fitted values be $\hat{\boldsymbol{y}} = \boldsymbol{H}\boldsymbol{y}$ for an $n \times n$ matrix $\boldsymbol{H}$ that does not depend on $\boldsymbol{y}$, and suppose that the prediction for observation i from the fit without it is $\hat{y}_{(-i)} = \sum_{j \neq i} H_{ij} y_j / (1 - H_{ii})$. Then*

$$y_i - \hat{y}_{(-i)} = \frac{y_i - \hat{y}_i}{1 - H_{ii}}, \qquad CV_n = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{y_i - \hat{y}_i}{1 - H_{ii}} \right)^2 .$$

The condition holds for OLS ($\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$, H_{ii} the leverage), for ridge regression with the total penalty $n\lambda$ held fixed, and for the Nadaraya–Watson estimator.

So for OLS a single fit suffices. The proof for OLS and NW is Problem Set 5, Problem 2.

One-standard-error rule. The K fold errors give a standard error $\mathrm{se}(\mathrm{CV}_K) = \{(K-1)^{-1} \sum_j (\mathrm{MSE}_j - \mathrm{CV}_K)^2\}^{1/2} / \sqrt{K}$. To guard against overfitting, one often chooses the simplest model whose CV error is within one standard error of the minimum.

Table 5.1: Model selection in a simulated sample ($n = 40$): $y = 2 + x_1 + u$, $u \sim N(0, 3^2)$, with x_1, x_2, x_3 standard normal and pairwise correlation 0.5. MSE is the training MSE; AIC and BIC are from R's `AIC()` and `BIC()`, which count σ^2 as a parameter (differences between models are unaffected); C_p uses $\tilde{\sigma}^2$ from the model with all three regressors. Best value in each column in bold.

	k	MSE	$\bar{R}^2$	AIC	BIC	C_p	LOOCV	5-fold CV
intercept only	1	9.80	0.000	208.81	212.19	6.17	10.31	10.01
x1	2	8.24	0.137	**203.89**	**208.96**	**1.15**	**9.05**	**9.51**

Table 5.1: Model selection in a simulated sample ($n = 40$): $y = 2 + x_1 + u$, $u \sim N(0, 3^2)$, with x_1, x_2, x_3 standard normal and pairwise correlation 0.5. MSE is the training MSE; AIC and BIC are from R's `AIC()` and `BIC()`, which count σ^2 as a parameter (differences between models are unaffected); C_p uses $\tilde{\sigma}^2$ from the model with all three regressors. Best value in each column in bold.

	k	MSE	$\bar{R}^2$	AIC	BIC	C_p	LOOCV	5-fold CV
x2	2	9.40	0.016	209.14	214.21	6.36	10.17	10.31
x3	2	9.74	-0.020	210.58	215.64	7.90	10.76	10.46
x1 + x2	3	8.24	0.114	205.89	212.64	3.14	9.37	9.83
x1 + x3	3	8.00	**0.140**	204.69	211.44	2.04	9.33	10.25
x2 + x3	3	9.40	-0.010	211.13	217.88	8.34	10.80	10.82
x1 + x2 + x3	4	**7.99**	0.117	206.64	215.08	4.00	9.69	10.79

In Table 5.1 the training MSE is smallest for the largest model, as it must be. All penalized criteria and both cross-validation criteria pick the true model with x_1 only, except $\bar{R}^2$, which prefers $x_1 + x_3$; with $n = 40$ this need not happen in every sample. (C_p is close to k for the correct model, 1.15 versus 2, and equals $k = 4$ by construction for the largest model.)

5.3 Shrinkage estimators

5.3.1 Why bias can help

Example 5.1 (Shrinking an unbiased estimator)**.** Let $\hat{\theta}$ be unbiased with variance v, and $\tilde{\theta} = a\hat{\theta}$ with $0 \le a \le 1$. Then $\text{bias}(\tilde{\theta}) = (a-1)\theta$, $\text{Var}(\tilde{\theta}) = a^2 v$ and

$$\text{MSE}(\tilde{\theta}) = a^2 v + (1-a)^2\theta^2 < v = \text{MSE}(\hat{\theta}) \iff \theta^2 < \frac{1+a}{1-a}\, v.$$

With $a = 0.9$, shrinking wins whenever $\theta^2 < 19v$, i.e. unless θ is large relative to its standard error. The MSE-optimal factor is $a^* = \theta^2/(\theta^2 + v)$, which depends on the unknown θ; the shrinkage methods below choose the amount of shrinkage from the data.

A famous extreme case is the **James–Stein** result: if $y_i \sim N(\mu_i, 1)$ independently, $i = 1, \dots, n \ge 4$, the estimator $\tilde{\mu}_i = \bar{y} + (1-c)(y_i - \bar{y})$ with $c = (n-3)/\sum_j (y_j - \bar{y})^2$ has a smaller total MSE $\sum_i \mathbb{E}(\tilde{\mu}_i - \mu_i)^2$ than the MLE $\hat{\mu}_i = y_i$ for every $(\mu_1, \dots, \mu_n)$ (B. E. Hansen 2022a, sec. 28.20). For prediction the same logic applies: with $\tilde{y}_0 = \boldsymbol{x}_0'\tilde{\boldsymbol{\beta}}$, $\mathbb{E}[(y_0 - \tilde{y}_0)^2 \mid \boldsymbol{x}_0] = \boldsymbol{x}_0' \,\text{MSE}(\tilde{\boldsymbol{\beta}})\, \boldsymbol{x}_0 + \sigma^2$, where $\text{MSE}(\tilde{\boldsymbol{\beta}}) = \mathbb{E}[(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta})(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta})']$, so a smaller MSE of the coefficients means a smaller prediction error.

Standardization. Shrinkage penalizes the size of the coefficients, which depends on the units of the regressors. Throughout, each regressor is standardized to mean zero and variance one, y is demeaned, and there is no intercept in $\boldsymbol{x}$ (it is recovered as $\bar{y}$). We write the model as $y_i = \boldsymbol{x}_i'\boldsymbol{\beta} + u_i$ with $\boldsymbol{x}_i$ of dimension p.

5.3.2 Ridge regression

Definition 5.1 (Ridge estimator)**.** For a tuning parameter $\lambda \geq 0$,

$$\hat{\boldsymbol{\beta}}_R(\lambda) = \arg\min_{\boldsymbol{\beta}} \ \frac{1}{n}\sum_{i=1}^{n}(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = (\boldsymbol{X}'\boldsymbol{X} + n\lambda\boldsymbol{I}_p)^{-1}\boldsymbol{X}'\boldsymbol{y}.$$

Equivalently, it minimizes the sum of squared residuals subject to $\sum_j \beta_j^2 \leq s$ for some s that corresponds to λ.

The first-order condition $-\frac{2}{n}\boldsymbol{X}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) + 2\lambda\boldsymbol{\beta} = \boldsymbol{0}$ gives the closed form, which exists even if $p > n$ or the regressors are collinear: $\boldsymbol{X}'\boldsymbol{X} + n\lambda\boldsymbol{I}$ is positive definite for $\lambda > 0$. Properties:

- $\hat{\boldsymbol{\beta}}_R(\lambda) \to \hat{\boldsymbol{\beta}}_{OLS}$ as $\lambda \to 0$ (if $\boldsymbol{X}'\boldsymbol{X}$ is invertible) and $\to \boldsymbol{0}$ as $\lambda \to \infty$.
- $\hat{\boldsymbol{\beta}}_R = \boldsymbol{A}_\lambda\hat{\boldsymbol{\beta}}_{OLS}$ with $\boldsymbol{A}_\lambda = (\boldsymbol{I} + n\lambda(\boldsymbol{X}'\boldsymbol{X})^{-1})^{-1}$: ridge shrinks OLS. If the regressors are orthogonal with $\boldsymbol{X}'\boldsymbol{X} = n\boldsymbol{I}$, $\hat{\beta}_{R,j} = \hat{\beta}_{OLS,j}/(1+\lambda)$, the same proportional shrinkage for every coefficient.
- Conditional on $\boldsymbol{X}$, $\mathbb{E}[\hat{\boldsymbol{\beta}}_R] = \boldsymbol{A}_\lambda\boldsymbol{\beta} \neq \boldsymbol{\beta}$ and $\mathrm{Var}[\hat{\boldsymbol{\beta}}_R] = \sigma^2(\boldsymbol{X}'\boldsymbol{X} + n\lambda\boldsymbol{I})^{-1}\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X} + n\lambda\boldsymbol{I})^{-1}$, which is smaller (in the matrix sense) than the OLS variance. There is always some $\lambda > 0$ with a smaller MSE than OLS (Problem Set 5, Problem 3 shows it for orthogonal regressors).
- Ridge works well when many regressors matter, each a little, and when OLS is imprecise because of collinearity. It never sets a coefficient exactly to zero.

5.3.3 The lasso

Definition 5.2 (Lasso (least absolute shrinkage and selection operator) (Tibshirani 1996))**.**

$$\hat{\boldsymbol{\beta}}_L(\lambda) = \arg\min_{\boldsymbol{\beta}} \ \frac{1}{n}\sum_{i=1}^{n}(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})^2 + \lambda\sum_{j=1}^{p}|\beta_j|,$$

or equivalently least squares subject to $\sum_j |\beta_j| \leq s$.

There is no closed form in general, but the problem is convex and fast algorithms (coordinate descent, LARS) compute the whole path of solutions over λ. With orthogonal regressors, $\boldsymbol{X}'\boldsymbol{X} = n\boldsymbol{I}$, the problem separates into p scalar problems and the solution is **soft thresholding**:

$$\hat{\beta}_{L,j} = \mathrm{sign}(\hat{\beta}_{OLS,j})\Big(|\hat{\beta}_{OLS,j}| - \frac{\lambda}{2}\Big)_+,$$

where $(a)_+ = \max(a, 0)$. Small OLS coefficients are set exactly to zero, large ones are shrunk by a constant $\lambda/2$ (Problem Set 5, Problem 3). This is why the lasso *selects* variables. Geometrically, the ℓ_1 constraint region $\sum_j |\beta_j| \leq s$ has corners on the axes, and the elliptical contours of the sum of squared residuals typically first touch it at a corner, where some coefficients are zero; the ℓ_2 region of ridge is round and has no corners (James et al. 2023, fig. 6.7).

The lasso works well when the true model is **sparse**: few regressors with nonzero coefficients. Its estimates of the selected coefficients are biased towards zero. The **post-lasso** estimator refits OLS on the selected regressors to remove this bias. A weighted version, $\lambda\sum_j \kappa_j|\beta_j|$, allows different penalties per coefficient; the *adaptive lasso* uses $\kappa_j = 1/|\hat{\beta}_j|$ from a first-step estimate.

5.3.4 Elastic net and the choice of λ

The **elastic net** combines both penalties,

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})^2 + \lambda\sum_{j=1}^{p}\left[\alpha|\beta_j| + (1-\alpha)\beta_j^2\right], \qquad 0 \le \alpha \le 1,$$

with ridge ($\alpha = 0$) and lasso ($\alpha = 1$) as special cases. The lasso tends to pick one variable out of a group of highly correlated ones, somewhat arbitrarily; the ridge part makes the elastic net select or drop such groups together. (The R package `glmnet` minimizes $\frac{1}{2n}\sum_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta})^2 + \lambda\sum_j[\alpha|\beta_j| + \frac{1-\alpha}{2}\beta_j^2]$, so its λ is on a different scale from the one used here.)

Choosing λ. λ is chosen by K-fold cross-validation over a decreasing grid: `lambda.min` minimizes the CV error, `lambda.1se` is the largest λ within one standard error of the minimum (a sparser model). Information criteria and theory-based penalties (Belloni, Chernozhukov, and Hansen 2014) are alternatives; the latter are used when the lasso serves as a first step for inference (Section 5.5).

Other models. Shrinkage applies to any M-estimator: for logit, probit or Poisson, replace the average squared residual by the average negative log-likelihood (half the deviance, up to a constant), e.g. $-\frac{1}{n}\sum_i[y_i \ln \Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}) + (1-y_i)\ln(1-\Lambda(\boldsymbol{x}_i'\boldsymbol{\beta}))] + \lambda\sum_j|\beta_j|$ for the lasso logit.

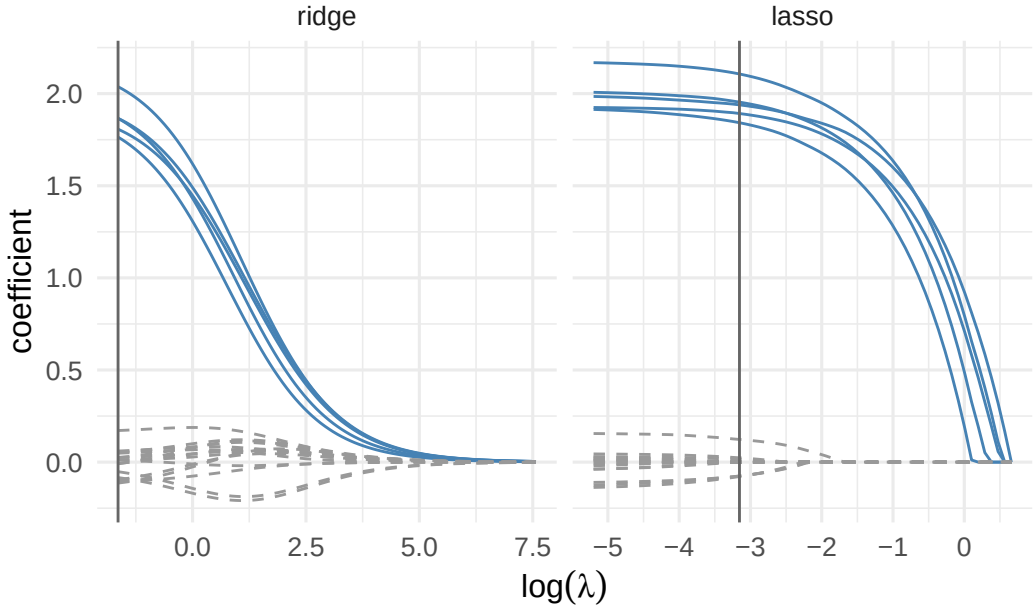


Figure 5.2: Coefficient paths of ridge and lasso (`glmnet`) for $n = 100$, $p = 20$ independent standard normal regressors, $\beta_j = 2$ for $j \le 5$ and 0 otherwise, $u \sim N(0,1)$. Solid: the five relevant regressors. Vertical line: λ chosen by 10-fold CV (`lambda.min`).

As λ falls (moving left in Figure 5.2), ridge coefficients grow smoothly and none is ever exactly zero; lasso coefficients enter one by one. At the cross-validated λ the lasso keeps the five relevant regressors and a few irrelevant ones with small coefficients.

Table 5.2: Test MSE (new observations, $\sigma^2 = 1$) of OLS, ridge, lasso and elastic net ($\alpha = 0.5$), λ by 10-fold CV; $n = 100$, $p = 50$ regressors with pairwise correlation 0.5; averages over 100 samples. Sparse: $\beta_j = 1$ for $j \le 5$, 0 otherwise. Dense: $\beta_j = 0.3$ for all j.

	OLS	ridge	lasso	elastic net
sparse	2.039	1.618	1.187	1.214
dense	2.107	1.195	1.919	1.885

With $p = 50$ and $n = 100$, OLS predicts poorly in both designs. In the sparse design the lasso and the elastic net are best; in the dense design, where every regressor matters a little, ridge is best and the lasso loses most of its advantage (Table 5.2). The elastic net is a robust compromise.

5.4 Outlook: trees and forests

Regression trees split the regressor space into rectangles and predict y by the mean in each rectangle, $\hat{f}(\boldsymbol{x}) = \sum_{m=1}^{M} c_m \mathbb{1}\{\boldsymbol{x} \in R_m\}$. The splits are chosen greedily: at each step, for every regressor j and split point s, compute the sum of squared residuals of the two half-spaces $\{x_j < s\}$ and $\{x_j \ge s\}$, and take the best split; then repeat within each part. A large tree is grown and then pruned back with a penalty on the number of leaves chosen by CV. Trees are easy to interpret and capture interactions, but they predict poorly because they have high variance: small changes in the data change the first splits and therefore the whole tree.

- **Bagging** (bootstrap aggregation) averages the predictions of trees grown on B bootstrap samples, $\hat{f}_{bag}(\boldsymbol{x}) = B^{-1} \sum_b \hat{f}_b(\boldsymbol{x})$. Averaging reduces variance. Each bootstrap sample leaves out about $(1 - 1/n)^n \approx e^{-1} \approx 37\%$ of the observations, which serve as a built-in test sample (out-of-bag error).
- **Random forests** additionally consider only a random subset of $m < p$ regressors at each split (often $m \approx \sqrt{p}$ or $p/3$). This decorrelates the trees, so averaging reduces the variance further.
- **Boosting** fits a sequence of small trees, each to the residuals of the current fit, and adds them with a small learning rate: $\hat{f} \leftarrow \hat{f} + \eta \hat{f}_b$.

Random forests are close relatives of the nearest-neighbour and kernel estimators of Chapter 4: the prediction is a weighted average of y_i's of observations that fall in the same leaves as $\boldsymbol{x}$, with data-driven weights. Computer Tutorial 5 compares them with the shrinkage estimators.

5.5 Outlook: machine learning for causal inference

Prediction methods do not deliver causal effects by themselves, but they can estimate **nuisance functions** in a causal model. Consider the partially linear model of Section 4.4 with a treatment d and many controls $\boldsymbol{x}$,

$$y = \alpha d + g(\boldsymbol{x}) + u, \qquad \mathbb{E}[u \mid d, \boldsymbol{x}] = 0,$$

where α is of interest (selection on observables) and g is approximated by a linear combination of many terms (polynomials, interactions, dummies), $g(\boldsymbol{x}) \approx \boldsymbol{x}'\boldsymbol{\gamma}$, of which few matter (approximate sparsity).

Naive post-selection fails. Running a lasso of y on d and $\boldsymbol{x}$ (with d unpenalized) and then OLS on d and the selected controls is biased: a control that is strongly correlated with d but

has a moderate effect on y is easily dropped, because d already explains part of its effect. The omitted control then biases $\hat{\alpha}$.

Double selection (Belloni, Chernozhukov, and Hansen 2014). (1) Lasso of y on $\boldsymbol{x}$, selected set $\boldsymbol{x}_y$. (2) Lasso of d on $\boldsymbol{x}$, selected set $\boldsymbol{x}_d$. (3) OLS of y on d and the union of $\boldsymbol{x}_y$ and $\boldsymbol{x}_d$. A control that matters for d is kept through step 2 even if its coefficient in the y equation is small. Under sparsity conditions, $\hat{\alpha}$ is $\sqrt{n}$-consistent and asymptotically normal, and the usual robust standard errors from step 3 are valid.

Orthogonality and double/debiased machine learning (DML) (Chernozhukov et al. 2018). The partialling-out estimator of Robinson solves the sample analogue of

$$\mathbb{E}\Big[\big(d-\eta_1(\boldsymbol{x})\big)\big\{y-\eta_2(\boldsymbol{x})-\alpha\big(d-\eta_1(\boldsymbol{x})\big)\big\}\Big]=0, \qquad \eta_1(\boldsymbol{x})=\mathbb{E}[d\mid\boldsymbol{x}],\ \eta_2(\boldsymbol{x})=\mathbb{E}[y\mid\boldsymbol{x}].$$

This moment is **Neyman orthogonal**: its derivative with respect to the nuisance functions is zero in expectation at the true values (Problem Set 5, Problem 4). Small errors in $\hat{\eta}_1, \hat{\eta}_2$ therefore affect $\hat{\alpha}$ only through their *product*, which vanishes faster than $n^{-1/2}$ if each converges faster than $n^{-1/4}$. This is the condition $\boldsymbol{A}_{21}=\boldsymbol{0}$ of the two-step GMM theory in Section 2.9: the first step does not change the asymptotic distribution of the second. DML estimates η_1, η_2 with any machine learner (lasso, random forests, boosting) and uses **cross-fitting**: split the sample into K folds, estimate the nuisance functions without fold k, compute the residuals $d_i-\hat{\eta}_1(\boldsymbol{x}_i)$ and $y_i-\hat{\eta}_2(\boldsymbol{x}_i)$ in fold k, and regress the pooled y-residuals on the pooled d-residuals. Cross-fitting prevents the overfitting of the machine learner from contaminating the second step.

Table 5.3: Estimates of $\alpha=0.5$ in $y=\alpha d+\boldsymbol{x}'\boldsymbol{\gamma}+u$, $d=\boldsymbol{x}'\boldsymbol{\theta}+v$, $n=400$, $p=150$ independent standard normal controls; five confounders with $\theta_j=0.6$ and $\gamma_j=0.25$, all other coefficients zero. Lasso penalties by 5-fold CV; 100 samples.

estimator	bias	sd
OLS of y on d only	0.271	0.028
naive post-lasso	0.264	0.052
double selection	-0.003	0.055
DML (cross-fitted lasso)	0.028	0.052
OLS on d and all 150 controls	0.002	0.066

Ignoring the controls, or selecting them only by their effect on y, gives a large bias. Double selection and DML are close to unbiased. OLS with all 150 controls is unbiased too, but noisier; with more controls than observations it would not be available at all (Table 5.3).

5.6 Computer tutorial 5

Learning goals. Fit ridge, lasso and elastic net with `glmnet`, read coefficient paths and cross-validation curves, and compare the methods by their test error; compare shrinkage with OLS and a random forest in a real prediction problem; use the lasso for inference on a treatment coefficient with many controls (double selection and cross-fitted partialling out).

Packages. `glmnet`, `randomForest`, `ggplot2`. **Data.** Simulated data and `cps09mar.csv` (B. E. Hansen 2022a), as in Computer Tutorial 4.

5.6.1 Problem 1: Ridge, lasso and elastic net on simulated data

a) Simulate $n = 100$ observations with $p = 30$ regressors. The regressors come in three groups of ten: within each group the pairwise correlation is 0.8, across groups they are independent. Let $\beta_j = 1.5$ for the first three regressors of group 1, $\beta_j = -1$ for the first three regressors of group 2, and $\beta_j = 0$ otherwise; $y = \boldsymbol{X}\boldsymbol{\beta} + u$, $u \sim N(0, 1)$.

b) Fit ridge ($\alpha = 0$), lasso ($\alpha = 1$) and elastic net ($\alpha = 0.5$) with `glmnet` and plot the coefficient paths against $\log \lambda$. Which coefficients become exactly zero, and in which order do the regressors enter the lasso?

c) Choose λ by 10-fold cross-validation with `cv.glmnet`. Plot the CV curve for the lasso and report `lambda.min` and `lambda.1se`. How many regressors does the lasso select with each? Which of the truly relevant ones are missed, and which irrelevant ones are included?

d) Generate a test sample of 10,000 observations from the same model and compare the test MSE of OLS, ridge, lasso, elastic net and the post-lasso estimator (OLS on the regressors selected with `lambda.min`).

5.6.2 Problem 2: Predicting wages

We predict the log hourly wage in `cps09mar.csv` from age, education, sex, ethnicity, race, marital status, union membership and region.

a) Draw a random subsample of 6000 observations. Recode race into white, Black and other, and marital status into married (codes 1–3), never married (7) and previously married (4–6). Build a design matrix with fourth-order polynomials in potential experience, a quadratic in education, the dummies, and all their pairwise interactions. How many regressors are there?

b) Use the first 2000 observations as training sample and the other 4000 as test sample. Compute the test MSE of: OLS on all regressors; a simple Mincer regression of log wage on experience, experience squared, education and `female`; ridge, lasso and elastic net with λ chosen by CV; a random forest on the original variables.

c) Which variables does the lasso keep? Compare the test MSEs with the variance of log wages in the test sample. What do you conclude?

5.6.3 Problem 3: The gender wage gap with many controls

Using the 6000 observations of Problem 2, we estimate α in

$$\log \text{wage}_i = \alpha\, \texttt{female}_i + g(\boldsymbol{x}_i) + u_i,$$

where $\boldsymbol{x}_i$ contains all regressors of Problem 2(a) that do not involve `female`.

a) Estimate α by OLS of log wage on `female` alone and on `female` and all controls.

b) Estimate α by double selection: lasso of log wage on the controls, lasso of `female` on the controls (use `lambda.1se`), then OLS on `female` and the union of the selected controls with a robust standard error.

c) Estimate α by cross-fitted partialling out (DML) with five folds and the lasso as learner, and compute its standard error from the residual regression.

d) Compare the estimates. Why does a naive lasso that always keeps `female` and selects among the controls by their effect on wages not suffice in general?

5.7 Exercises

Problem Set 5 contains the analytical exercises for this chapter:

- **Problem 1**: the optimism of the training MSE for OLS and its relation to C_p and AIC.
- **Problem 2**: the leave-one-out shortcut for OLS and the Nadaraya–Watson estimator.
- **Problem 3**: ridge, lasso, elastic net and best subset selection with orthogonal regressors: closed forms, MSE and the optimal ridge penalty.
- **Problem 4**: Neyman orthogonality of the partialling-out moment, and why the naive moment is not orthogonal.

Part III

Optional reading

6 Causal Inference

Optional reading

This chapter is **optional** and **not part of the exam**. It summarises the two causal inference lectures (Causal Inference I and II) for students who want a written reference. There is no problem set or computer tutorial for it; the short exercises at the end are for self-study.

Readings

Imbens and Wooldridge (2009); Angrist and Pischke (2009), Ch. 3, 4 and 6; Wooldridge (2010), Ch. 21; Imbens and Rubin (2015) for a book-length treatment. Instrumental variables and LATE: Imbens and Angrist (1994), Angrist, Imbens, and Rubin (1996), Abadie (2003), Słoczyński, Uysal, and Wooldridge (2025). Regression discontinuity: Imbens and Lemieux (2008), Lee and Lemieux (2010), Cattaneo, Idrobo, and Titiunik (2020).

Section 2.9.1 introduced potential outcomes, the average treatment effect and inverse probability weighting as the leading example of a two-step GMM estimator. This chapter places those results in the wider toolbox of program evaluation. It covers three identification strategies: **selection on observables** (randomized experiments, matching, weighting, regression and doubly robust estimators), **instrumental variables** with heterogeneous effects (the local average treatment effect), and **regression discontinuity** designs. The estimators reuse the tools of the course: logit for the propensity score (Chapter 1), two-step GMM for inference (Section 2.9), and local linear regression at a boundary (Section 4.3).

6.1 Potential outcomes and causal parameters

The notation follows Section 2.9.1: $D_i \in \{0,1\}$ is the treatment, (Y_{0i}, Y_{1i}) are the potential outcomes, $Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}$ is observed, and the no-interference assumption (SUTVA) is maintained. Besides the ATE we use the average effects on the treated and on the untreated,

$$\tau = \mathbb{E}[Y_{1i} - Y_{0i}], \quad \tau_T = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 1], \quad \tau_U = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_i = 0],$$

which are linked by $\tau = \pi\tau_T + (1 - \pi)\tau_U$ with $\pi = P(D_i = 1)$. The three differ whenever effects are heterogeneous and units select into treatment on the basis of their gains.

What the naive comparison estimates. Let $\Delta = \mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] = \mathbb{E}[Y_{1i} \mid$

$D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]$. Adding and subtracting terms gives three equivalent decompositions,

$$\begin{aligned}\Delta &= \tau_T + (\mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]) \\ &= \tau + (\mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]) \\ &\quad + (1-\pi)(\tau_T - \tau_U) \\ &= \pi\tau_T + (1-\pi)\tau_U \\ &\quad + (1-\pi)(\mathbb{E}[Y_{1i} \mid D_i = 1] - \mathbb{E}[Y_{1i} \mid D_i = 0]) \\ &\quad + \pi(\mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]).\end{aligned} \tag{6.1}$$

The first equality is the familiar one from Section 2.9.1; the term in brackets is the selection bias. The second shows that, relative to the ATE, there are two sources of bias: different baseline outcomes, and different gains ($\tau_T \neq \tau_U$). The third (Exercise 6.1) writes Δ as the ATE plus a weighted sum of the differences between treated and untreated in *both* potential outcomes; it shows that $\Delta = \tau$ if the treated and untreated have the same mean of Y_0 and the same mean of Y_1, which is weaker than full independence.

6.2 Randomized experiments

If D_i is randomly assigned, $(Y_{0i}, Y_{1i}) \perp D_i$, so $\mathbb{E}[Y_{di} \mid D_i = 1] = \mathbb{E}[Y_{di} \mid D_i = 0]$ for $d = 0, 1$. Both bias terms in Equation 6.1 vanish and $\Delta = \tau = \tau_T = \tau_U$. The difference in sample means $\bar{Y}_1 - \bar{Y}_0$ is unbiased for τ. It is algebraically identical to the OLS slope in the regression of Y_i on a constant and D_i, so a regression is just a convenient way to compute it. Two points matter for inference. First, with heterogeneous effects the variances of Y differ between the arms, so heteroskedasticity-robust standard errors should be used; the conventional OLS standard error is valid only if the two arms have equal variances or equal sizes. Second, adding pre-treatment covariates $\boldsymbol{x}_i$ does not change what is estimated (they are independent of D_i) but can reduce the variance. With heterogeneous effects, the regression should include D_i interacted with the demeaned covariates, Y_i on $(1, D_i, \boldsymbol{x}_i - \bar{\boldsymbol{x}}, D_i(\boldsymbol{x}_i - \bar{\boldsymbol{x}}))$, for the coefficient on D_i to be consistent for τ and, in large samples, never less precise than the simple difference in means.

6.3 Selection on observables

Without randomization we rely on the two assumptions of Section 2.9.1:

- **Unconfoundedness (conditional independence, CIA):** $(Y_{0i}, Y_{1i}) \perp D_i \mid \boldsymbol{x}_i$.
- **Overlap (common support):** $0 < p(\boldsymbol{x}) < 1$ for all $\boldsymbol{x}$, with $p(\boldsymbol{x}) = P(D_i = 1 \mid \boldsymbol{x}_i = \boldsymbol{x})$.

For τ_T it suffices that $Y_{0i} \perp D_i \mid \boldsymbol{x}_i$ and $p(\boldsymbol{x}) < 1$. Under the CIA, $\mu_d(\boldsymbol{x}) \equiv \mathbb{E}[Y_i \mid D_i = d, \boldsymbol{x}_i = \boldsymbol{x}] = \mathbb{E}[Y_{di} \mid \boldsymbol{x}_i = \boldsymbol{x}]$, so the conditional effect $\tau(\boldsymbol{x}) = \mu_1(\boldsymbol{x}) - \mu_0(\boldsymbol{x})$ is identified wherever there is overlap, and

$$\tau = \mathbb{E}[\mu_1(\boldsymbol{x}_i) - \mu_0(\boldsymbol{x}_i)], \qquad \tau_T = \mathbb{E}[\mu_1(\boldsymbol{x}_i) - \mu_0(\boldsymbol{x}_i) \mid D_i = 1]. \tag{6.2}$$

All estimators below are different ways of estimating these averages. They differ in what they model: the outcome regressions μ_d (regression adjustment), the propensity score p (weighting, propensity score matching), both (doubly robust), or neither (covariate matching).

6.3.1 Subclassification and matching

Subclassification. If $\boldsymbol{x}_i$ takes K values $\boldsymbol{x}^1, \dots, \boldsymbol{x}^K$, Equation 6.2 is a weighted average of within-cell differences in means,

$$\hat{\tau} = \sum_{k=1}^{K} (\bar{Y}_{1k} - \bar{Y}_{0k}) \frac{N_k}{N}, \qquad \hat{\tau}_T = \sum_{k=1}^{K} (\bar{Y}_{1k} - \bar{Y}_{0k}) \frac{N_{1k}}{N_1},$$

where N_k and N_{1k} are the numbers of units and of treated units in cell k. Overlap requires every cell (for τ_T, every cell with treated units) to contain controls. With several or continuous covariates the cells become empty quickly: the curse of dimensionality of Section 4.3.4 in another guise.

Matching replaces exact cells by nearest neighbours. Let $\mathcal{J}_M(i)$ be the set of the M units in the opposite group closest to unit i in covariate space. The missing potential outcome is imputed by the average outcome of the matches,

$$\hat{Y}_{0i} = \begin{cases} Y_i & D_i = 0 \\ M^{-1} \sum_{j \in \mathcal{J}_M(i)} Y_j & D_i = 1 \end{cases}, \qquad \hat{Y}_{1i} = \begin{cases} M^{-1} \sum_{j \in \mathcal{J}_M(i)} Y_j & D_i = 0 \\ Y_i & D_i = 1 \end{cases},$$

so the treated are imputed Y_0 from untreated matches and the untreated are imputed Y_1 from treated matches. Then $\hat{\tau} = N^{-1} \sum_i (\hat{Y}_{1i} - \hat{Y}_{0i})$ and $\hat{\tau}_T = N_1^{-1} \sum_{i:D_i=1} (Y_i - \hat{Y}_{0i})$. Matching is usually done with replacement (a control can serve as a match several times), which reduces bias at the cost of variance.

Distance. Covariates on different scales are combined with a weighting matrix, $\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_{\boldsymbol{V}} = \{(\boldsymbol{x}_i - \boldsymbol{x}_j)' \boldsymbol{V} (\boldsymbol{x}_i - \boldsymbol{x}_j)\}^{1/2}$. The **normalized Euclidean** distance uses $\boldsymbol{V} = \operatorname{diag}(\hat{\sigma}_1^{-2}, \dots, \hat{\sigma}_k^{-2})$, i.e. $\sum_l (x_{li} - x_{lj})^2 / \hat{\sigma}_l^2$ under the root, and is invariant to rescaling each covariate. The **Mahalanobis** distance uses $\boldsymbol{V} = \hat{\boldsymbol{\Sigma}}_x^{-1}$ and also accounts for correlation between covariates.

Bias. When matches are not exact, $\hat{\tau}_T$ contains the term $N_1^{-1} \sum_{i:D_i=1} M^{-1} \sum_{j \in \mathcal{J}_M(i)} \{\mu_0(\boldsymbol{x}_i) - \mu_0(\boldsymbol{x}_j)\}$. Abadie and Imbens (2006) show that this conditional bias is of order $N^{-1/k}$ with k continuous covariates. For $k \geq 2$ it is not negligible relative to the standard error, which is of order $N^{-1/2}$, and for $k \geq 3$ it dominates the distribution of the estimator. The **bias-corrected** matching estimator (Abadie and Imbens 2011) estimates μ_0 (and μ_1) by a regression and subtracts $\hat{\mu}_0(\boldsymbol{x}_i) - \hat{\mu}_0(\boldsymbol{x}_j)$ from each matched difference. The standard bootstrap is not valid for matching with a fixed number of matches (Abadie and Imbens 2008); analytic variance formulas are available in Abadie and Imbens (2006).

6.3.2 The propensity score

Rosenbaum and Rubin (1983) showed that the propensity score is a **balancing score**: $D_i \perp \boldsymbol{x}_i \mid p(\boldsymbol{x}_i)$, and if the CIA holds given $\boldsymbol{x}_i$ it also holds given $p(\boldsymbol{x}_i)$ alone,

$$(Y_{0i}, Y_{1i}) \perp D_i \mid \boldsymbol{x}_i \quad \Longrightarrow \quad (Y_{0i}, Y_{1i}) \perp D_i \mid p(\boldsymbol{x}_i)$$

(Exercise 6.2). Matching or subclassifying on the scalar $p(\boldsymbol{x}_i)$ therefore removes the dimensionality problem of covariate matching, at the cost of having to model p, in practice by a logit or probit with a flexible specification. Common implementations are nearest-neighbour matching on $\hat{p}$ (optionally within a **caliper** $|\hat{p}_i - \hat{p}_j| < \delta$), radius matching, stratification on $\hat{p}$, and kernel or local linear matching, which use a weighted average of all controls with weights that decline with $|\hat{p}_i - \hat{p}_j|$. Estimating p affects the variance of the matching estimator, and treating $\hat{p}$ as known is not valid (Abadie and Imbens 2016).

Overlap in practice. Plotting the distribution of $\hat{p}$ for treated and controls shows whether overlap is plausible. If some treated units have $\hat{p}$ above the largest control value, they have no comparable controls. A common response is to drop them (and report that the estimand now refers to the treated *on the common support*, so averages are over the treated units remaining after trimming) or to trim units with $\hat{p} \notin [\alpha, 1-\alpha]$, e.g. $\alpha = 0.1$, which changes the estimand to the ATE for the trimmed population.

Balance. Because p is only a device to balance $\boldsymbol{x}$, the specification is judged by whether the weighted or matched samples are balanced, e.g. by standardized differences in covariate means, not by the fit of the logit.

6.3.3 Weighting

The IPW identification result and the unnormalized and normalized ATE estimators are in Section 2.9.1 (Equation 2.1). For the ATT only Y_0 of the treated is missing, and controls are reweighted to look like the treated:

$$\tau_T = \frac{1}{\pi}\mathbb{E}\Big[\frac{D_i - p(\boldsymbol{x}_i)}{1 - p(\boldsymbol{x}_i)} Y_i\Big],$$

$$\hat{\tau}_{T,norm} = \frac{\sum_i D_i Y_i}{\sum_i D_i} - \frac{\sum_i (1-D_i)\omega_i Y_i}{\sum_i (1-D_i)\omega_i}, \quad \omega_i = \frac{\hat{p}(\boldsymbol{x}_i)}{1-\hat{p}(\boldsymbol{x}_i)}.$$

Controls with a high propensity score, those that look like treated units, receive large weights (the odds of treatment). Problem Set 2, Problem 4 derives the moments and the two-step variance. As stressed in Section 2.9.1, the normalized versions, whose weights sum to one within each group, are preferred (Busso, DiNardo, and McCrary 2014). One reason is a simple invariance property discussed in Section 6.4.1: an unnormalized estimator changes if a constant is added to Y.

6.3.4 Regression adjustment and what OLS estimates

Regression adjustment (RA) estimates μ_0 and μ_1 by separate regressions in the control and treated samples, predicts both for every unit and averages according to Equation 6.2:

$$\hat{\tau}_{RA} = \frac{1}{N}\sum_{i=1}^{N} \big(\hat{\mu}_1(\boldsymbol{x}_i) - \hat{\mu}_0(\boldsymbol{x}_i)\big), \qquad \hat{\tau}_{T,RA} = \frac{1}{N_1}\sum_{i:D_i=1} \big(\hat{\mu}_1(\boldsymbol{x}_i) - \hat{\mu}_0(\boldsymbol{x}_i)\big).$$

With linear $\hat{\mu}_d$ this is the same as the fully interacted regression of Section 6.2.

OLS with additive controls. Most applied work instead runs $Y_i = \alpha + \tau D_i + \boldsymbol{x}_i'\boldsymbol{\beta} + u_i$. What does τ estimate when effects are heterogeneous?

Saturated controls. If $\boldsymbol{x}_i$ is discrete and the regression includes a dummy for each cell, the OLS coefficient converges to

$$\tau_{OLS} = \sum_{k=1}^{K} \tau(\boldsymbol{x}^k)\, w_k, \qquad w_k = \frac{p_k(1-p_k)P(\boldsymbol{x}_i = \boldsymbol{x}^k)}{\sum_{l=1}^{K} p_l(1-p_l)P(\boldsymbol{x}_i = \boldsymbol{x}^l)}, \quad p_k = p(\boldsymbol{x}^k).$$

The ATE weights the cells by $P(\boldsymbol{x}_i = \boldsymbol{x}^k)$ and the ATT by $p_k P(\boldsymbol{x}_i = \boldsymbol{x}^k)/\pi$. OLS instead weights by the conditional variance of treatment, $p_k(1-p_k)$: cells where treated and controls are balanced (p_k near $1/2$) count most, cells where almost everyone or almost no one is treated

count little. OLS is a variance-weighted average of cell effects, which coincides with the ATE only if $\tau(\boldsymbol{x})$ is constant or uncorrelated with $p(1-p)$.

Linear controls. Słoczyński (2022) shows that, for any $\boldsymbol{x}_i$, the OLS coefficient is exactly

$$\begin{aligned} \tau_{OLS} &= w_1\,\tau_T^{\ell} + w_0\,\tau_U^{\ell}, \qquad w_0 = 1 - w_1, \\ w_1 &= \frac{(1-\pi)\operatorname{Var}[p^{\ell}(\boldsymbol{x}_i) \mid D_i = 0]}{\pi \operatorname{Var}[p^{\ell}(\boldsymbol{x}_i) \mid D_i = 1] + (1-\pi)\operatorname{Var}[p^{\ell}(\boldsymbol{x}_i) \mid D_i = 0]}, \end{aligned} \tag{6.3}$$

where $p^{\ell}(\boldsymbol{x}_i)$ is the linear projection of D_i on $(1, \boldsymbol{x}_i)$, and τ_T^{ℓ}, τ_U^{ℓ} are the effects on the treated and untreated implied by separate linear regressions of Y on $p^{\ell}(\boldsymbol{x}_i)$ in each group. They equal τ_T and τ_U if $\mathbb{E}[Y_{di} \mid \boldsymbol{x}_i]$ is linear in $p^{\ell}(\boldsymbol{x}_i)$. If the two conditional variances are similar, $w_1 \approx 1 - \pi$: **the smaller group gets the larger weight**. When few units are treated, OLS is close to the ATT, not the ATE; when most are treated, it is close to the ATU. The ATE puts weight π on τ_T, so OLS can be far from the ATE when π is far from 1/2 and $\tau_T \neq \tau_U$.

6.3.5 Doubly robust estimation

The **augmented IPW** (AIPW) estimator combines both models (Robins, Rotnitzky, and Zhao 1994):

$$\hat{\tau}_{AIPW} = \frac{1}{N}\sum_{i=1}^{N}\Big[\hat{\mu}_1(\boldsymbol{x}_i) - \hat{\mu}_0(\boldsymbol{x}_i) + \frac{D_i(Y_i - \hat{\mu}_1(\boldsymbol{x}_i))}{\hat{p}(\boldsymbol{x}_i)} - \frac{(1-D_i)(Y_i - \hat{\mu}_0(\boldsymbol{x}_i))}{1-\hat{p}(\boldsymbol{x}_i)}\Big].$$

It starts from RA and adds an IPW-weighted average of the residuals. If the outcome models are correct, the residual terms have mean zero; if the propensity score is correct, they remove the bias of misspecified outcome models. AIPW is therefore consistent if *either* model is correctly specified (**double robustness**), and it reaches the semiparametric efficiency bound when both are. The closely related **IPWRA** estimator runs the RA regressions weighted by $1/\hat{p}$ (treated) and $1/(1-\hat{p})$ (controls) and has the same double robustness. The AIPW moment is also Neyman orthogonal, which is why it is the basis of the DML estimator of the ATE (Section 5.5 discusses the partially linear version).

6.3.6 Illustration: smoking and birth weight

We return to the data of Computer Tutorial 2 (Cattaneo 2010): the effect of smoking during pregnancy (`mbsmoke`) on birth weight in grams, with controls for marital status, age (and its square), education and first birth. The propensity score is a logit, the outcome models are linear, and covariate matching uses one nearest neighbour with the Mahalanobis distance (without the squared age).

Table 6.1: Effect of smoking on birth weight (grams), $N = 4642$, of whom 19% smoked. Matching: one nearest neighbour (ties averaged), Mahalanobis distance, with replacement; bias correction by linear regression.

estimator	ATE	ATT
Difference in means	-275.3	
OLS with additive controls	-225.0	
Regression adjustment	-231.4	-225.0

Table 6.1: Effect of smoking on birth weight (grams), $N = 4642$, of whom 19% smoked. Matching: one nearest neighbour (ties averaged), Mahalanobis distance, with replacement; bias correction by linear regression.

estimator	ATE	ATT
IPW, unnormalized	-331.1	-298.5
IPW, normalized	-231.7	-225.7
AIPW (doubly robust)	-231.1	-229.1
NN matching	-204.8	-239.2
NN matching, bias-corrected	-203.9	-239.1

The raw difference of about -275 g overstates the harm of smoking, because smokers are less often married and less educated. All estimators that model the outcome or the normalized propensity score agree on an ATE of about -230 g and an ATT of about -225 g (Table 6.1). Two results stand out. First, the **unnormalized IPW** estimate of the ATE is far off. Birth weights are around 3000 g, and the IPW weights $D_i/\hat{p}_i$ average to one only in expectation, so a small imbalance in the weights is multiplied by a large outcome level; the normalized estimator removes this. The unnormalized IPW estimate of the ATT is off for the same reason. Second, OLS with additive controls is close to the ATT rather than the ATE. The Słoczyński decomposition Equation 6.3 gives $w_1 = 0.79$, $\tau_T^\ell = -224.9$, $\tau_U^\ell = -225.1$, so $w_1\tau_T^\ell + w_0\tau_U^\ell = -225.0$, which reproduces the OLS coefficient exactly. With only 19% smokers, OLS puts about 80% of its weight on the effect for smokers. Here τ_T and τ_U happen to be close, so the distinction is minor; in applications with more heterogeneity it is not. Matching gives a smaller ATE and a larger ATT than the regression-based estimators. Since all matching covariates are discrete, about 90% of smokers and 72% of non-smokers have an exact match, so the bias correction changes almost nothing. The difference reflects that matching does not impose linear outcome models, together with its larger sampling variance; it is a reminder that estimates under the same identifying assumption can differ by more than rounding.

6.4 Instrumental variables and the LATE

When unconfoundedness is implausible, an instrument can identify a causal effect, but with heterogeneous effects not the ATE. Let $Z_i \in \{0, 1\}$ be a binary instrument (an offer, an eligibility rule, a lottery) and D_i the treatment actually taken. Each unit has two **potential treatments** D_{0i}, D_{1i}, the treatment it would take with $Z_i = 0$ and $Z_i = 1$, so $D_i = Z_i D_{1i} + (1 - Z_i)D_{0i}$, and potential outcomes $Y_i(d, z)$.

The pair (D_{0i}, D_{1i}) defines four **types**:

	$D_{1i} = 0$	$D_{1i} = 1$
$D_{0i} = 0$	never-taker	**complier**
$D_{0i} = 1$	defier	always-taker

Types are not observed: a unit with $Z_i = 1, D_i = 1$ is a complier or an always-taker. Imbens and Angrist (1994) and Angrist, Imbens, and Rubin (1996) impose:

1. **Independence:** $\{Y_i(d, z), D_{0i}, D_{1i}\} \perp Z_i$ (the instrument is as good as randomly assigned).
2. **Exclusion:** $Y_i(d, 0) = Y_i(d, 1) \equiv Y_{di}$ (the instrument affects Y only through D).

3. **First stage:** $\mathbb{E}[D_i \mid Z_i = 1] \neq \mathbb{E}[D_i \mid Z_i = 0]$.
4. **Monotonicity:** $D_{1i} \geq D_{0i}$ for all i (no defiers).

Theorem 6.1 (LATE theorem)**.** *Under assumptions 1–4, the Wald estimand equals the average effect for compliers:*

$$\frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]} = \mathbb{E}[Y_{1i} - Y_{0i} \mid D_{1i} > D_{0i}] \equiv \tau_{LATE},$$

and the denominator is the share of compliers, $P(D_{1i} > D_{0i})$.

The proof is Exercise 6.3. The numerator is the **intention-to-treat** effect: always- and never-takers contribute nothing to it, because Z does not change their treatment. With a binary instrument and no covariates, the Wald ratio is also the IV and 2SLS estimator. Compliers are the units whose treatment is changed by *this* instrument, so different instruments identify different LATEs, and τ_{LATE} can differ substantially from τ and τ_T. The size and characteristics of the complier population are identified even though compliers cannot be individually identified: for example, $P(D_{1i} = D_{0i} = 1) = \mathbb{E}[D_i \mid Z_i = 0]$ is the share of always-takers.

6.4.1 Covariates and kappa weighting

Often the instrument is only as good as random conditional on covariates, $\{Y_i(d, z), D_{0i}, D_{1i}\} \perp Z_i \mid \boldsymbol{x}_i$, with **instrument propensity score** $q(\boldsymbol{x}) = P(Z_i = 1 \mid \boldsymbol{x}_i = \boldsymbol{x}) \in (0, 1)$. The common practice of adding $\boldsymbol{x}_i$ linearly to 2SLS does not in general estimate the LATE when effects vary with $\boldsymbol{x}$; Angrist and Imbens (1995) show that even a saturated specification yields a weighted average of covariate-specific LATEs with weights that differ from the complier shares. Weighting estimators avoid this.

Abadie's kappa (Abadie 2003). Define

$$\kappa_i = 1 - \frac{D_i(1 - Z_i)}{1 - q(\boldsymbol{x}_i)} - \frac{(1 - D_i)Z_i}{q(\boldsymbol{x}_i)},$$
$$\kappa_{1i} = D_i \frac{Z_i - q(\boldsymbol{x}_i)}{q(\boldsymbol{x}_i)(1 - q(\boldsymbol{x}_i))}, \qquad \kappa_{0i} = (1 - D_i)\frac{q(\boldsymbol{x}_i) - Z_i}{q(\boldsymbol{x}_i)(1 - q(\boldsymbol{x}_i))}.$$

For any function g,

$$\mathbb{E}[g(Y_i, D_i, \boldsymbol{x}_i) \mid D_{1i} > D_{0i}] = \frac{\mathbb{E}[\kappa_i g(Y_i, D_i, \boldsymbol{x}_i)]}{P(D_{1i} > D_{0i})},$$
$$\mathbb{E}[g(Y_{1i}, \boldsymbol{x}_i) \mid D_{1i} > D_{0i}] = \frac{\mathbb{E}[\kappa_{1i} g(Y_i, \boldsymbol{x}_i)]}{P(D_{1i} > D_{0i})},$$

and the same with Y_{0i} and κ_{0i}. Intuitively, κ_i removes the always-takers (observed with $D = 1, Z = 0$) and never-takers (observed with $D = 0, Z = 1$) by giving the corresponding cells negative weight. With $g = 1$, $\mathbb{E}[\kappa_i] = \mathbb{E}[\kappa_{1i}] = \mathbb{E}[\kappa_{0i}] = P(D_{1i} > D_{0i})$. Taking $g(\boldsymbol{x}_i) = \boldsymbol{x}_i$ gives the **mean characteristics of compliers**, $\mathbb{E}[\boldsymbol{x}_i \mid D_{1i} > D_{0i}] = \mathbb{E}[\kappa_i \boldsymbol{x}_i]/\mathbb{E}[\kappa_i]$, which helps to interpret a LATE.

Weighting estimators of the LATE. Since $\kappa_{1i} - \kappa_{0i} = (Z_i - q_i)/\{q_i(1 - q_i)\}$ with $q_i = q(\boldsymbol{x}_i)$,

$$\tau_{LATE} = \frac{\mathbb{E}[(\kappa_{1i} - \kappa_{0i})Y_i]}{P(D_{1i} > D_{0i})},$$

and each of the three sample estimators of $P(D_{1i} > D_{0i})$, the sample means of $\hat{\kappa}_i$, $\hat{\kappa}_{1i}$ and $\hat{\kappa}_{0i}$, gives an estimator: $\hat{\tau}_a$, $\hat{\tau}_{a,1}$ and $\hat{\tau}_{a,0}$. The estimator $\hat{\tau}_{a,1}$ is identical to the ratio of two

unnormalized IPW estimators, the effects of Z on Y and on D (Tan 2006), which is the most popular weighting estimator of the LATE.

Słoczyński, Uysal, and Wooldridge (2025) show that none of these is **normalized**: written as a difference between a weighted mean of Y for compliers under treatment and under control, the weights do not sum to one. As a consequence they are not **translation invariant**: adding a constant c to every Y_i changes the estimate (Exercise 6.4). They recommend two normalized estimators,

$$\hat{\tau}_{a,10} = \frac{\sum_i \hat{\kappa}_{1i} Y_i}{\sum_i \hat{\kappa}_{1i}} - \frac{\sum_i \hat{\kappa}_{0i} Y_i}{\sum_i \hat{\kappa}_{0i}},$$

$$\hat{\tau}_u = \frac{\hat{\mathbb{E}}_w[Y \mid Z=1] - \hat{\mathbb{E}}_w[Y \mid Z=0]}{\hat{\mathbb{E}}_w[D \mid Z=1] - \hat{\mathbb{E}}_w[D \mid Z=0]},$$

where $\hat{\mathbb{E}}_w[V \mid Z=1] = \sum_i Z_i V_i/\hat{q}_i / \sum_i Z_i/\hat{q}_i$ and $\hat{\mathbb{E}}_w[V \mid Z=0] = \sum_i (1-Z_i) V_i/(1-\hat{q}_i) / \sum_i (1-Z_i)/(1-\hat{q}_i)$ are normalized IPW means. $\hat{\tau}_u$ is the ratio of two normalized IPW estimators of the effects of Z on Y and on D. Both are translation invariant and scale equivariant; the authors recommend $\hat{\tau}_u$. The same lesson as for the ATE applies: normalize the weights.

Table 6.3: Estimators of the LATE (= 1; ATE = 1.25) with a covariate-dependent instrument. $x \sim N(0,1)$, $q(x) = \Lambda(0.3 + 0.8x)$; 20% always-takers (effect 3), 30% never-takers (effect 0.5), 50% compliers (effect $1+x$); $n = 1000$, 1000 samples. Column 4: standard deviation when 10 is added to every Y. Column 5: change in the estimate when 100 is added to Y in one sample.

estimator	bias	sd	sd if Y + 10	change if Y + 100
2SLS with linear control	-0.091	0.229	0.229	0.000
$\hat{\tau}_a$	0.012	0.257	0.449	6.746
$\hat{\tau}_{a,1}$ (Tan)	0.011	0.255	0.441	6.552
$\hat{\tau}_{a,0}$	0.016	0.265	0.464	7.011
$\hat{\tau}_{a,10}$ (normalized)	0.007	0.243	0.243	0.000
$\hat{\tau}_u$ (normalized)	0.005	0.242	0.242	0.000

In this design 2SLS with x as a linear control is biased for the LATE, because the complier effect varies with x, 2SLS weights the covariate-specific effects differently from the complier distribution, and the linear control only approximates the role of x. All kappa estimators are close to unbiased. The unnormalized ones are only slightly less precise in the original design, but their precision depends on the arbitrary level of Y: after adding 10 to the outcome their standard deviation rises by about 75%, and adding 100 moves the estimate in a single sample by several units. The normalized estimators (and 2SLS) are unaffected (Table 6.3).

6.5 Regression discontinuity designs

In a regression discontinuity (RD) design, treatment is determined, fully or partly, by whether a **running variable** (score) X_i exceeds a known **cutoff** c. Examples are test-score thresholds for scholarships, age or income limits for programs, and vote-share thresholds in elections. The idea is that units just above and just below the cutoff are comparable, so the jump in the outcome at c reveals the effect of treatment, at least for units at the cutoff. We discuss the continuity-based framework; the alternative local randomization framework treats units in a small window around c as if randomly assigned (Cattaneo, Idrobo, and Titiunik 2020).

6.5.1 Sharp and fuzzy designs

Let $T_i = \mathbb{1}\{X_i \geq c\}$ be treatment *assignment*.

Sharp RD. Treatment is a deterministic function of the score, $D_i = T_i$. Then $\mathbb{E}[Y_i \mid X_i = x] = \mathbb{E}[Y_{1i} \mid X_i = x]$ for $x \geq c$ and $\mathbb{E}[Y_{0i} \mid X_i = x]$ for $x < c$. There is no overlap at all: at any value of X we see only treated or only controls, and unconfoundedness given X holds trivially. Identification comes from continuity instead (Hahn, Todd, and Van der Klaauw 2001): if $\mathbb{E}[Y_{0i} \mid X_i = x]$ and $\mathbb{E}[Y_{1i} \mid X_i = x]$ are continuous at $x = c$,

$$\tau_{SRD} \equiv \mathbb{E}[Y_{1i} - Y_{0i} \mid X_i = c] = \lim_{x \downarrow c} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i \mid X_i = x].$$

The effect is **local**: it is the average effect for units at the cutoff. Extrapolating it away from c requires additional assumptions on how effects vary with the score.

Fuzzy RD. Crossing the cutoff changes the probability of treatment, but not from 0 to 1, e.g. because eligibility does not force participation. Then T_i is an instrument for D_i at the cutoff. With potential treatments D_{1i}, D_{0i} (under $T_i = 1$ and $T_i = 0$), continuity and monotonicity,

$$\begin{aligned} \tau_{FRD} &= \frac{\lim_{x \downarrow c} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i \mid X_i = x]}{\lim_{x \downarrow c} \mathbb{E}[D_i \mid X_i = x] - \lim_{x \uparrow c} \mathbb{E}[D_i \mid X_i = x]} \\ &= \mathbb{E}[Y_{1i} - Y_{0i} \mid X_i = c,\, D_{1i} > D_{0i}], \end{aligned}$$

a LATE for compliers at the cutoff: the ratio of the jump in the outcome (the intention-to-treat effect) to the jump in the treatment probability (the first stage).

Extensions. In a **kink** design the slope rather than the level of the treatment intensity changes at c, and the effect is identified from the change in the slope of $\mathbb{E}[Y \mid X]$ (Card et al. 2015). With **multiple cutoffs** the score is usually centred at each unit's cutoff and pooled; the pooled estimand is then a weighted average of cutoff-specific effects. With **multiple scores**, e.g. two exam thresholds or a geographic boundary, the effect is a function along the boundary.

6.5.2 Estimation by local polynomials

The RD estimand is a difference of two regression functions evaluated at a **boundary point**: c is the left boundary of the treated sample and the right boundary of the control sample. This is exactly the situation of Table 4.3, where Nadaraya–Watson has a bias of order h and local linear regression a bias of order h^2. The standard estimator is therefore local linear regression on each side, with a kernel K and bandwidth h:

$$(\hat{\alpha}_+, \hat{\beta}_+) = \arg\min_{a,b} \sum_{i: X_i \geq c} K\Big(\frac{X_i - c}{h}\Big)\big(Y_i - a - b(X_i - c)\big)^2,$$

$(\hat{\alpha}_-, \hat{\beta}_-)$ analogously for $X_i < c$, and $\hat{\tau}_{SRD} = \hat{\alpha}_+ - \hat{\alpha}_-$. Practical recommendations (Cattaneo, Idrobo, and Titiunik 2020):

- **Order:** $p = 1$ (or $p = 2$). Global high-order polynomials (fitting e.g. a quartic in X to the whole sample) give the boundary points erratic weights and are not recommended.
- **Kernel:** the triangular kernel $K(u) = (1 - |u|)\mathbb{1}\{|u| \leq 1\}$ is optimal for boundary estimation; the uniform kernel amounts to OLS of Y on $(1, T_i, X_i - c, T_i(X_i - c))$ within $[c - h, c + h]$ (Exercise 6.5). Results are usually not sensitive to the kernel.
- **Bandwidth:** the bias-variance trade-off of Section 4.3 applies. The standard choice minimizes the asymptotic MSE of $\hat{\tau}_{SRD}$, possibly with different bandwidths on the two sides.

- **Inference:** the MSE-optimal bandwidth is too large for conventional confidence intervals, whose centre is then off by a bias of the same order as the standard error (as for the density bands of Section 4.2). **Robust bias-corrected** intervals (Calonico, Cattaneo, and Titiunik 2014) subtract an estimate of the bias and inflate the standard error to account for estimating it. They are implemented, with the bandwidth selectors, in the R and Stata packages `rdrobust` (which includes `rdplot`) and `rddensity`.
- **Covariates:** pre-determined covariates can be added linearly to the local regression. They should not change the estimate, but they can reduce the variance. Covariates affected by treatment must not be included.

The fuzzy RD is estimated as the ratio of two such jumps, which is numerically the same as local 2SLS with T_i as instrument for D_i.

RD plots. A scatter plot of Y against X is often uninformative. The standard RD plot shows means of Y in bins of X (evenly spaced or quantile-spaced) together with a smooth fit on each side, which displays the jump and the shape of the regression functions.

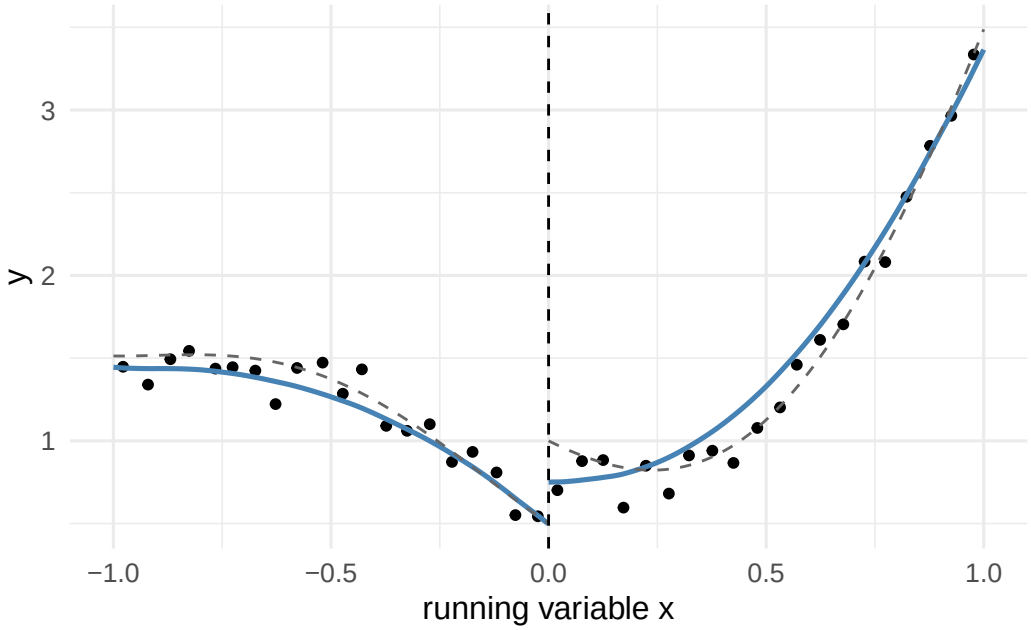


Figure 6.1: Sharp RD with $n = 1000$, $x \sim U(-1, 1)$, cutoff 0, $\mathbb{E}[Y_0 \mid x] = 0.5 + 0.6x + 1.5x^2 - 0.8\sin(3x)$ and effect $0.5 + 0.5x$, so $\tau_{SRD} = 0.5$. Dots: means in 20 bins per side; solid lines: local linear fits (triangular kernel, $h = 0.5$); dashed: true regression functions.

Table 6.4: Bias, standard deviation and RMSE of sharp RD estimators of $\tau_{SRD} = 0.5$ in the design of Figure 6.1, 1000 samples of $n = 1000$. Local linear: triangular kernel. Global: polynomials of order 1 and 4 in x, interacted with T, on the full sample.

estimator	bias	sd	rmse
local linear, h = 0.1	0.006	0.222	0.222
local linear, h = 0.25	-0.006	0.142	0.142
local linear, h = 0.5	-0.077	0.100	0.126
local linear, h = 1	-0.517	0.073	0.522
global linear	-0.924	0.074	0.927
global quartic	0.001	0.162	0.161

Table 6.4 shows the bandwidth trade-off. A small bandwidth gives little bias but a large variance; a large one fits a straight line through a curved function and is badly biased; $h = 1$ uses all data on each side and is as bad as a global linear fit. Figure 6.1 shows the same bias in a single sample: on the right of the cutoff, where the regression function is strongly curved, the local linear fit with $h = 0.5$ starts below the true intercept. The global quartic happens to work reasonably here, because the true function is smooth on the whole support, but it is noisier than local linear regression with a moderate bandwidth.

6.5.3 Validation and falsification

The continuity assumptions are about unobserved potential outcomes and cannot be tested. Their implications can:

- **Manipulation of the score.** If units can precisely sort to one side of the cutoff (students retaking a test until they pass, officials adjusting a poverty index), units just above and below c are no longer comparable. Sorting usually shows up as a discontinuity in the density of X at c, which is tested by comparing density estimates on both sides (McCrary 2008; Cattaneo, Jansson, and Ma 2020).
- **Pre-determined covariates and placebo outcomes.** Characteristics fixed before treatment should not jump at the cutoff. Running the RD estimator with each covariate as outcome is a balance test.
- **Placebo cutoffs.** Estimating "effects" at values of X where treatment does not change should give zero (using only treated or only control observations so that the true effect does not contaminate the estimate).
- **Sensitivity.** Results should be robust to the bandwidth and, as a **donut-hole** check, to dropping the observations closest to c, which are the most likely to have manipulated their score.

Example. Meyersson (2014) studies whether the election of an Islamic party mayor in Turkish municipalities in 1994 affected the share of young women who completed high school by 2000. The score is the Islamic party's margin of victory, with cutoff zero. The naive comparison of municipalities with Islamic and secular mayors suggests a *negative* association, because Islamic parties won in poorer and more conservative places. The local linear RD estimate (triangular kernel, MSE-optimal bandwidth of about 17 percentage points) is a *positive* effect of about 3 percentage points, with a robust bias-corrected p-value of about 0.08 (Cattaneo, Idrobo, and Titiunik 2020). The density test and the covariate balance tests do not indicate manipulation.

6.6 Further reading

Imbens and Rubin (2015) is the standard reference for the potential outcomes framework; Angrist and Pischke (2009) is an accessible applied treatment of all three strategies. For matching and weighting see Imbens and Wooldridge (2009) and Busso, DiNardo, and McCrary (2014), for LATE and kappa weighting Abadie (2003) and Słoczyński, Uysal, and Wooldridge (2025), and for RD Cattaneo, Idrobo, and Titiunik (2020) and Lee and Lemieux (2010). Difference-in-differences, the fourth main design, is not covered in this course.

6.7 Exercises

These exercises are optional and for self-study.

Exercise 6.1 (Decomposing the naive comparison)**.**

(a) Show that $\Delta = \tau + (\mathbb{E}[Y_{0i} \mid D_i = 1] - \mathbb{E}[Y_{0i} \mid D_i = 0]) + (1 - \pi)(\tau_T - \tau_U)$.
(b) Verify the third equality in Equation 6.1.
(c) Suppose $\mathbb{E}[Y_{0i} \mid D_i = 1] = \mathbb{E}[Y_{0i} \mid D_i = 0]$ but individuals with larger gains select into treatment. Which of τ, τ_T and τ_U does Δ equal?

Exercise 6.2 (The propensity score theorem)**.** Let $p(\boldsymbol{x}) = P(D = 1 \mid \boldsymbol{x})$ with $0 < p(\boldsymbol{x}) < 1$. (a) Show that $P(D = 1 \mid \boldsymbol{x}, p(\boldsymbol{x})) = P(D = 1 \mid p(\boldsymbol{x}))$, i.e. $D \perp \boldsymbol{x} \mid p(\boldsymbol{x})$. (b) Show that if $(Y_0, Y_1) \perp D \mid \boldsymbol{x}$, then $(Y_0, Y_1) \perp D \mid p(\boldsymbol{x})$.

Exercise 6.3 (The LATE theorem)**.** Prove Theorem 6.1. Hint: write $Y_i = Y_{0i} + D_i(Y_{1i} - Y_{0i})$ and use independence to express $\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]$ in terms of $D_{1i} - D_{0i}$.

Exercise 6.4 (Normalization and translation invariance)**.**

(a) Show that $\hat{\tau}_{a,1}$ equals the ratio of the unnormalized IPW estimators of the effect of Z on Y and of Z on D, $\sum_i (Z_i/\hat{q}_i - (1 - Z_i)/(1 - \hat{q}_i))Y_i / \sum_i (Z_i/\hat{q}_i - (1 - Z_i)/(1 - \hat{q}_i))D_i$.
(b) Show that replacing Y_i by $Y_i + c$ changes $\hat{\tau}_a$ by $c \sum_i (\hat{\kappa}_{1i} - \hat{\kappa}_{0i}) / \sum_i \hat{\kappa}_i$ and explain why this is generally non-zero in finite samples but zero in expectation.
(c) Show that $\hat{\tau}_{a,10}$ and $\hat{\tau}_u$ are translation invariant.

Exercise 6.5 (RD as a regression)**.**

(a) Show that the sharp RD local linear estimator with the uniform kernel equals the OLS coefficient on T_i in the regression of Y_i on $(1, T_i, X_i - c, T_i(X_i - c))$ using observations with $|X_i - c| \leq h$.
(b) In a fuzzy design, show that the ratio of the two local linear jumps (in Y and in D), with the same kernel and bandwidth, equals the IV estimator of the coefficient on D_i in $Y_i = \alpha + \tau D_i + \beta_-(X_i - c) + \beta_+ T_i(X_i - c) + u_i$ with instruments $(1, T_i, X_i - c, T_i(X_i - c))$, using the same window.

Exercise 6.6 (Computing: ATT estimators)**.** Using `cattaneo2.csv`, estimate the effect of smoking on the treated by (i) regression adjustment, (ii) normalized IPW with logit propensity score, (iii) one-to-one nearest-neighbour matching on the estimated propensity score with replacement. Plot the distribution of $\hat{p}$ for smokers and non-smokers and comment on overlap.

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A Large-Sample Asymptotics

This appendix collects the asymptotic tools used throughout the notes. They are assumed known from Research Metrics 1; the summary is for reference. Wooldridge (2010), Ch. 3, B. E. Hansen (2022a), Ch. 6 and B. E. Hansen (2022b), Ch. 7–8 give proofs.

A.1 Modes of convergence

Let $\boldsymbol{z}_n$ be a sequence of random vectors.

Definition A.1 (Convergence in probability)**.** $\boldsymbol{z}_n \xrightarrow{p} \boldsymbol{c}$ if for every $\varepsilon > 0$, $P(\|\boldsymbol{z}_n - \boldsymbol{c}\| > \varepsilon) \to 0$ as $n \to \infty$. We also write $\operatorname{plim} \boldsymbol{z}_n = \boldsymbol{c}$. An estimator $\hat{\boldsymbol{\theta}}_n$ is **consistent** if $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$.

Definition A.2 (Convergence in distribution)**.** $\boldsymbol{z}_n \xrightarrow{d} \boldsymbol{z}$ if $P(\boldsymbol{z}_n \le \boldsymbol{t}) \to P(\boldsymbol{z} \le \boldsymbol{t})$ at every continuity point $\boldsymbol{t}$ of the cdf of $\boldsymbol{z}$. The distribution of $\boldsymbol{z}$ is the **limiting distribution**.

Convergence in probability implies convergence in distribution; the converse holds only if the limit is a constant.

Order notation. $\boldsymbol{z}_n = o_p(1)$ means $\boldsymbol{z}_n \xrightarrow{p} \mathbf{0}$; $\boldsymbol{z}_n = O_p(1)$ means $\boldsymbol{z}_n$ is bounded in probability (for every ε there is an M with $P(\|\boldsymbol{z}_n\| > M) < \varepsilon$ for all n). Any $\boldsymbol{z}_n$ that converges in distribution is $O_p(1)$, and $o_p(1) \cdot O_p(1) = o_p(1)$.

A.2 Laws of large numbers and central limit theorems

Theorem A.1 (Weak law of large numbers)**.** *If $\boldsymbol{y}_1, \dots, \boldsymbol{y}_n$ are i.i.d. with $\mathbb{E}\|\boldsymbol{y}_i\| < \infty$, then $\bar{\boldsymbol{y}}_n = \frac{1}{n}\sum_i \boldsymbol{y}_i \xrightarrow{p} \mathbb{E}[\boldsymbol{y}_i]$.*

Theorem A.2 (Central limit theorem (Lindeberg–Lévy))**.** *If $\boldsymbol{y}_1, \dots, \boldsymbol{y}_n$ are i.i.d. with mean $\boldsymbol{\mu}$ and finite variance matrix $\boldsymbol{\Sigma}$, then*

$$\sqrt{n}\,(\bar{\boldsymbol{y}}_n - \boldsymbol{\mu}) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} (\boldsymbol{y}_i - \boldsymbol{\mu}) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Sigma}).$$

Both apply to functions of the data: in the notes $\boldsymbol{y}_i$ is replaced by a score $S_i(\boldsymbol{\theta}_0)$, a moment function $\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta}_0)$ or a simulated quantity $h(x_s)$.

A.3 Continuous mapping, Slutsky and the delta method

Theorem A.3 (Continuous mapping theorem)**.** *Let g be continuous.*

(i) If $\boldsymbol{z}_n \xrightarrow{p} \boldsymbol{c}$, then $g(\boldsymbol{z}_n) \xrightarrow{p} g(\boldsymbol{c})$.

(ii) If $\boldsymbol{z}_n \xrightarrow{d} \boldsymbol{z}$, then $g(\boldsymbol{z}_n) \xrightarrow{d} g(\boldsymbol{z})$.

For (i), continuity at $\boldsymbol{c}$ suffices.

Theorem A.4 (Slutsky's theorem)**.** *If $z_n \xrightarrow{d} z$ and $c_n \xrightarrow{p} c$ (a constant), then $z_n + c_n \xrightarrow{d} z + c$, $c_n z_n \xrightarrow{d} cz$, and $z_n / c_n \xrightarrow{d} z/c$ if $c \neq 0$.*

In matrix form: if $\boldsymbol{H}_n \xrightarrow{p} \boldsymbol{H}$ and $\boldsymbol{b}_n \xrightarrow{d} N(\boldsymbol{\mu}, \boldsymbol{B})$, then $\boldsymbol{H}_n \boldsymbol{b}_n \xrightarrow{d} N(\boldsymbol{H}\boldsymbol{\mu}, \boldsymbol{H}\boldsymbol{B}\boldsymbol{H}')$.

The matrix form is the workhorse of the notes. Every asymptotic normality proof writes $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = \boldsymbol{H}_n \boldsymbol{b}_n$, where $\boldsymbol{H}_n$ is an (inverse) average of derivatives that converges in probability by the WLLN, and $\boldsymbol{b}_n$ is $\sqrt{n}$ times an average of mean-zero terms that is asymptotically normal by the CLT.

Theorem A.5 (Delta method)**.** *If $\sqrt{n}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{V})$ and $\boldsymbol{g} : \mathbb{R}^k \to \mathbb{R}^q$ is continuously differentiable at $\boldsymbol{\beta}$ with Jacobian $\boldsymbol{G} = \partial \boldsymbol{g}(\boldsymbol{\beta}) / \partial \boldsymbol{\beta}'$, then*

$$\sqrt{n}(\boldsymbol{g}(\hat{\boldsymbol{\beta}}_n) - \boldsymbol{g}(\boldsymbol{\beta})) \xrightarrow{d} N(\mathbf{0},\ \boldsymbol{G}\boldsymbol{V}\boldsymbol{G}').$$

Why: by the mean value theorem, $\boldsymbol{g}(\hat{\boldsymbol{\beta}}_n) = \boldsymbol{g}(\boldsymbol{\beta}) + \boldsymbol{G}(\boldsymbol{\beta}^*)(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta})$ with $\boldsymbol{\beta}^*$ between $\hat{\boldsymbol{\beta}}_n$ and $\boldsymbol{\beta}$. Since $\boldsymbol{\beta}^* \xrightarrow{p} \boldsymbol{\beta}$, $\boldsymbol{G}(\boldsymbol{\beta}^*) \xrightarrow{p} \boldsymbol{G}$, and Slutsky gives the result.

Example A.1 (Delta method)**.** Let y_i be i.i.d. with mean μ and variance σ^2, and $g(\mu) = e^{\mu}$. Then $\sqrt{n}(e^{\bar{y}} - e^{\mu}) \xrightarrow{d} N(0, e^{2\mu}\sigma^2)$, and a standard error of $e^{\bar{y}}$ is $e^{\bar{y}}\hat{\sigma}/\sqrt{n}$.

A.4 Mean value expansions

Most proofs expand a first-order condition around the true value. For a differentiable vector function $\bar{\boldsymbol{g}}_n(\boldsymbol{\theta})$,

$$\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) = \bar{\boldsymbol{g}}_n(\boldsymbol{\theta}_0) + \frac{\partial \bar{\boldsymbol{g}}_n(\boldsymbol{\theta}^*)}{\partial \boldsymbol{\theta}'} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0),$$

where $\boldsymbol{\theta}^* = \boldsymbol{\theta}_0 + \lambda(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$, $0 \leq \lambda \leq 1$, lies on the segment between $\hat{\boldsymbol{\theta}}$ and $\boldsymbol{\theta}_0$ (strictly, a different $\boldsymbol{\theta}^*$ for each row). If $\hat{\boldsymbol{\theta}}$ is consistent, so is $\boldsymbol{\theta}^*$. If $\bar{\boldsymbol{g}}_n(\hat{\boldsymbol{\theta}}) = \mathbf{0}$ by definition of the estimator, solving for $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0$ gives the representation used for the MLE (Theorem 1.5), MM (Theorem 2.1) and GMM (Theorem 2.2).

A.5 Consistency of extremum estimators

ML, GMM and MSL estimators maximize or minimize a sample objective $Q_n(\boldsymbol{\theta})$. The standard argument has three ingredients:

1. **Identification:** the limit objective $Q_0(\boldsymbol{\theta}) = \operatorname{plim} Q_n(\boldsymbol{\theta})$ has a unique maximum at $\boldsymbol{\theta}_0$ (for ML this follows from the KLIC inequality; for GMM from $\mathbb{E}[\boldsymbol{\psi}(\boldsymbol{w}_i, \boldsymbol{\theta})] = \mathbf{0}$ only at $\boldsymbol{\theta}_0$).
2. **Uniform convergence:** $\sup_{\boldsymbol{\theta} \in \Theta} |Q_n(\boldsymbol{\theta}) - Q_0(\boldsymbol{\theta})| \xrightarrow{p} 0$, which holds under compactness of Θ, continuity and a dominance condition (a uniform law of large numbers).
3. **Continuity** of Q_0.

Then $\hat{\boldsymbol{\theta}}_n \xrightarrow{p} \boldsymbol{\theta}_0$ (Wooldridge 2010, Thm. 12.2; B. E. Hansen 2022a, Ch. 22.4). Pointwise convergence of Q_n is not enough: the maximizer of Q_n could drift away from $\boldsymbol{\theta}_0$.

B A Short R Primer

> **In preparation**
>
> This chapter will be added before the corresponding lectures. Planned content:
>
> - Projects, scripts and working directories
> - Data import, data frames and matrices
> - Matrix algebra and OLS by hand
> - Writing functions; optimisation with optim
> - Computer tutorial 0